Pure-MSE gamma short | Val IC 00

T_0=1262.8 K P_0=4.913 atm ϕ_0=0.746 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 00

Repaired | Cantera vs NN NRMSE | T0=1262.8K, P0=4.913atm, phi0=0.746

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.142	0.722	8.08	0.983	0.445	4.33	0.208	0.922	1.84	0	Complete
1 ANN10	0.1	0.237	2.16	7.6	0.825	0.431	6.1	0.362	0.86	2.17	0	Complete
1 ANN10	0.2	0.152	1.35	0.985	0.934	0.303	4.4	0.249	0.906	0.935	0	Complete
1 ANN10	0.3	9.12e-03	0.508	1.02	0.38	0.0341	0.0511	0.0212	0.626	0.952	0	Complete
1 ANN10	0.5	0.0407	0.744	1.57	0.72	0.0684	0.939	0.0682	0.957	0.988	0	Complete
1 ANN10	1	0.0106	0.771	4.54	0.485	0.113	0.941	0.0502	0.937	8.94	0	Complete
2 P-only DeepONet10	0.05	0.0506	0.0939	0.954	2.94	0.268	1.02	0.0407	0.41	0.204	0	Complete
2 P-only DeepONet10	0.1	8.74e-03	0.0714	0.934	1.18	0.0399	0.33	0.022	0.031	1.56	0	Complete
2 P-only DeepONet10	0.2	0.0883	0.956	2.78	0.0672	0.0425	1.2	0.117	6.05	47.2	0	Complete
2 P-only DeepONet10	0.3	0.0444	0.441	0.981	0.274	0.0153	0.696	0.058	1.26	0.496	0	Complete
2 P-only DeepONet10	0.5	0.539	17	2.64	0.963	2.36	0.561	0.904	0.97	28.5	0	Complete
2 P-only DeepONet10	1	0.494	17.1	1.09	2.05	2.34	0.223	0.858	0.804	40.6	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0401	0.0556	0.935	2.95	0.248	0.753	0.025	0.443	0.0877	0	Complete
3 ANN10 + P-only DeepONet10	0.1	4.21e-03	0.0554	0.9	1.43	0.0382	0.335	0.0177	0.0339	1.13	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0799	0.705	2.57	0.471	0.01	1.36	0.109	2.58	27.9	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0386	0.392	0.973	0.0295	5.11e-03	0.64	0.053	1.15	0.652	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.152	0.336	1.93	1.05	0.589	4.45	0.185	0.797	118	0	Complete
3 ANN10 + P-only DeepONet10	1	0.469	16.8	1.42	1.37	2.36	0.462	0.831	0.996	92.8	0	Complete
4 T/P DeepONet10	0.05	0.137	0.589	2.34	1.19	0.291	2.96	0.168	2.54	2.35	0	Complete
4 T/P DeepONet10	0.1	0.0431	0.292	0.503	0.622	0.113	0.941	0.0529	0.97	0.991	0	Complete
4 T/P DeepONet10	0.2	5.47e-03	0.279	0.237	0.1	0.043	0.0292	0.0124	0.212	0.147	0	Complete
4 T/P DeepONet10	0.3	4.90e-03	0.257	0.179	0.0998	0.0413	0.0284	0.0114	0.238	0.154	0	Complete
4 T/P DeepONet10	0.5	0.069	2.1	3.17	0.839	0.323	0.313	0.105	2.78	2.61	0	Complete
4 T/P DeepONet10	1	0.48	17.1	1	2.17	2.41	0.721	0.839	0.999	84.1	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.365	0.208	15.1	0.257	0.603	7.31	0.401	3.95	3.52	0	Complete
5 ANN10 + T/P DeepONet10	0.1	3.44e-03	0.21	0.195	0.0841	0.0328	0.0408	8.88e-03	0.219	0.192	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.06e-03	0.234	0.177	0.105	0.0364	0.0324	0.0107	0.22	0.16	0	Complete
5 ANN10 + T/P DeepONet10	0.3	5.25e-03	0.253	0.209	0.0963	0.0386	0.0295	0.0114	0.228	0.153	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0477	1.73	2.21	0.449	0.295	0.377	0.0795	0.877	0.912	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0454	0.815	5.17	0.252	0.0976	0.441	0.0242	0.718	11.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0608	0.279	0.301	0.549	0.214	1.75	0.0825	1.45	1.92	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0528	0.439	0.857	0.67	0.0878	0.962	0.0655	0.832	0.88	0	Complete
6 T/P/ϕ DeepONet10	0.2	5.20e-03	0.274	0.345	0.112	0.0427	0.0769	0.0125	0.214	0.169	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0733	1.09	0.663	1.85	0.0872	1.09	0.0968	0.484	0.378	0	Complete
6 T/P/ϕ DeepONet10	0.5	7.17e-03	0.336	0.351	0.128	0.0495	0.0719	0.0153	0.276	0.189	0	Complete
6 T/P/ϕ DeepONet10	1	0.356	12.5	1.01	1.79	1.7	0.174	0.624	1	40.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0658	0.22	0.34	0.375	0.251	2.04	0.0907	1.57	2.28	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0461	0.357	0.71	0.424	0.084	0.937	0.0594	0.844	0.915	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	5.43e-03	0.244	0.351	0.135	0.0372	0.0913	0.0119	0.219	0.166	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0448	0.796	0.91	1.05	0.0576	0.744	0.0657	0.506	0.409	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0108	0.301	0.239	0.168	0.0384	0.121	0.0174	0.303	0.242	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.298	9.53	4.21	1.28	1.35	0.799	0.51	6.76	20.8	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 00

T_0=1262.8 K P_0=4.913 atm ϕ_0=0.746 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 00

Repaired | Cantera vs NN NRMSE | T0=1262.8K, P0=4.913atm, phi0=0.746

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.277	1.65	0.82	0.868	0.703	8.11	0.421	0.246	0.994	0	Complete
1 ANN10	0.1	0.0502	0.692	0.847	5.01	0.125	0.481	0.0224	0.162	0.567	0	Complete
1 ANN10	0.2	0.0813	0.263	0.533	0.818	0.4	3.36	0.127	0.663	0.232	0	Complete
1 ANN10	0.3	0.0101	0.211	0.29	0.113	0.0135	0.259	0.0195	0.138	0.331	0	Complete
1 ANN10	0.5	0.0259	0.452	1.21	0.4	0.0948	0.805	0.0449	0.58	0.963	0	Complete
1 ANN10	1	0.0208	0.726	3.18	0.544	0.087	0.866	0.0538	0.915	1.6	0	Complete
2 P-only DeepONet10	0.05	0.0492	0.234	9.07	0.633	0.25	0.823	9.88e-03	1.37	1.79	0	Complete
2 P-only DeepONet10	0.1	0.139	0.68	3.87	0.92	0.503	4.36	0.173	0.751	0.771	0	Complete
2 P-only DeepONet10	0.2	1.17e-03	0.0296	0.119	0.0432	3.60e-03	0.0309	1.38e-03	0.0171	0.0552	0	Complete
2 P-only DeepONet10	0.3	0.0833	0.572	6.9	0.271	0.094	1.22	0.0652	0.604	0.541	0	Complete
2 P-only DeepONet10	0.5	0.0242	1.31	0.935	0.403	0.179	0.239	0.0509	0.507	0.238	0	Complete
2 P-only DeepONet10	1	0.625	1.2	28.6	19	0.717	3.92	0.406	0.986	798	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0551	0.299	10.2	0.78	0.271	0.863	0.0114	0.831	1.21	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.125	0.624	1.88	0.85	0.55	4.49	0.169	0.745	0.696	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.69e-04	0.0368	0.128	0.0627	4.15e-03	0.0345	8.33e-04	9.30e-03	0.0638	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0322	0.594	3.34	0.496	0.0899	0.733	0.0215	0.55	0.179	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0237	1.27	0.925	0.399	0.172	0.221	0.0499	0.471	0.176	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0701	0.456	12.7	0.986	0.314	0.953	0.0152	0.899	1.78	0	Complete
4 T/P DeepONet10	0.05	0.359	1.43	15.2	2.52	0.449	5.47	0.366	8.49	34.5	0	Complete
4 T/P DeepONet10	0.1	3.78e-03	0.128	0.17	0.046	0.0186	0.0644	6.78e-03	0.137	0.112	0	Complete
4 T/P DeepONet10	0.2	2.67e-03	0.128	0.15	0.0565	0.0193	0.014	5.74e-03	0.172	0.122	0	Complete
4 T/P DeepONet10	0.3	4.44e-03	0.205	0.166	0.0846	0.0333	0.0175	9.46e-03	0.218	0.155	0	Complete
4 T/P DeepONet10	0.5	0.0581	1.52	3.36	0.693	0.158	0.319	0.0854	0.506	0.771	0	Complete
4 T/P DeepONet10	1	0.322	10.8	4.53	1.67	1.54	0.932	0.558	10.4	56.4	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0564	0.441	0.949	0.674	0.0713	0.991	0.0702	0.926	0.92	0	Complete
5 ANN10 + T/P DeepONet10	0.1	2.14e-03	0.0769	0.119	0.0281	0.011	0.0439	4.01e-03	0.0792	0.0808	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.18e-03	0.166	0.218	0.0639	0.0235	0.0133	6.79e-03	0.139	0.112	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.23e-03	0.177	0.182	0.0838	0.0274	0.0163	7.71e-03	0.192	0.138	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0381	0.704	4.16	0.41	0.0781	0.23	0.0402	0.582	0.588	0	Complete
5 ANN10 + T/P DeepONet10	1	0.432	9.6	2.81	0.211	0.908	4.26	0.689	27.9	92.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0616	0.58	0.954	0.827	0.0596	0.989	0.0771	0.95	0.967	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0196	0.186	0.521	0.144	0.028	0.348	0.025	0.372	0.393	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.021	0.307	3.65	0.115	0.0597	0.194	7.23e-03	0.707	0.216	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0234	1.09	0.992	0.433	0.124	0.0501	0.0472	0.474	0.344	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0409	0.722	0.302	0.706	0.05	0.598	0.059	0.872	0.695	0	Complete
6 T/P/ϕ DeepONet10	1	0.404	5.13	1.71	6.65	0.284	5.87	0.519	1.02	92.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0572	0.511	0.925	0.572	0.0565	0.989	0.0735	0.936	0.967	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0388	0.326	0.65	0.193	0.0409	0.746	0.0521	0.7	0.728	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0121	0.348	3.23	0.259	0.05	0.233	8.16e-03	0.589	0.195	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0271	1.13	0.998	0.568	0.119	0.0141	0.0515	0.388	0.293	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	5.13e-03	0.253	0.302	0.116	0.0372	0.0332	0.0112	0.237	0.173	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.403	12.1	1.76	2.21	1.55	1.29	0.668	5.28	110	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 09

T_0=901.6 K P_0=2.084 atm ϕ_0=1.343 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 09

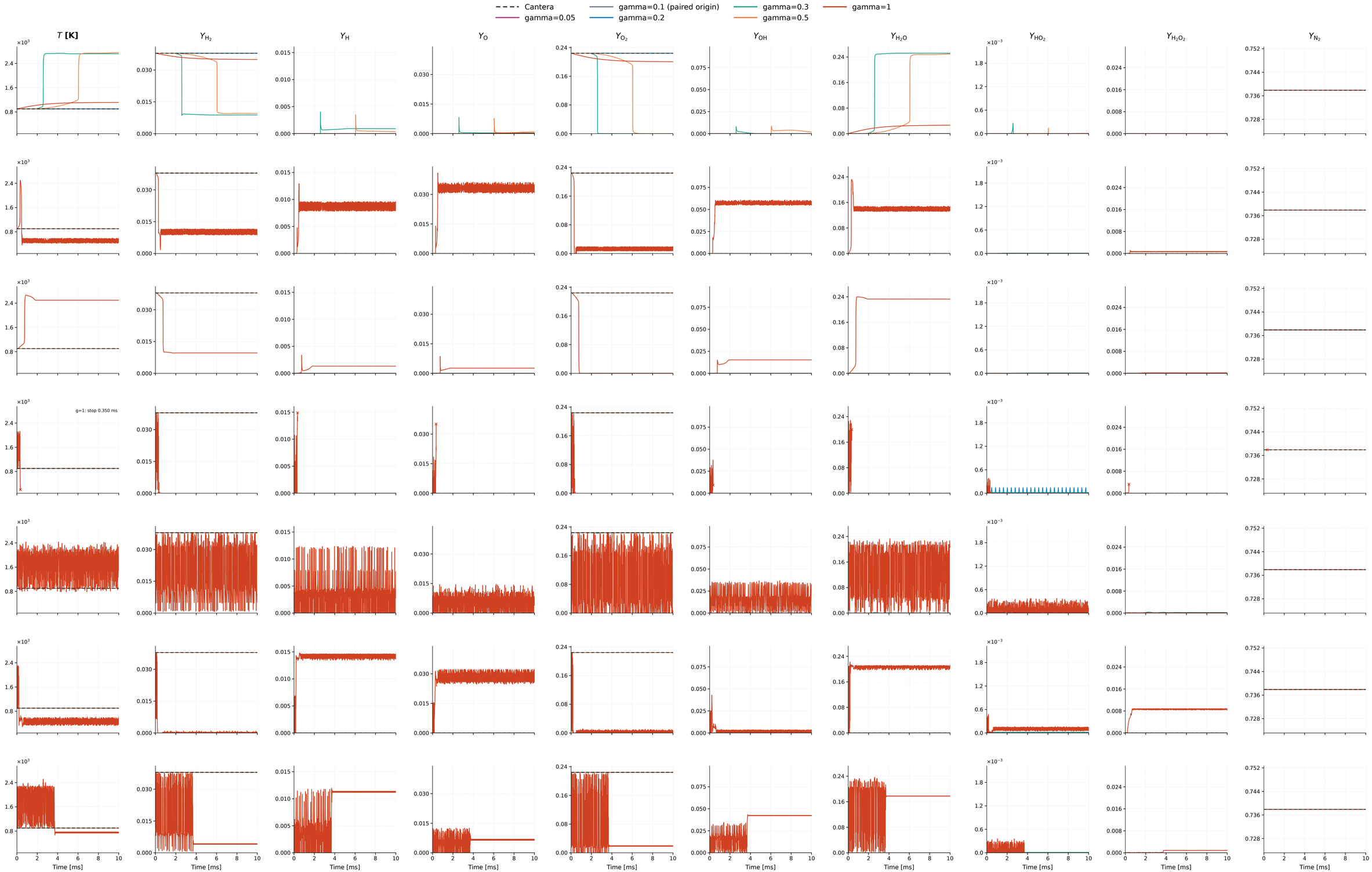
Repaired | Cantera vs NN NRMSE | T0=901.6K, P0=2.084atm, phi0=1.343

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	3.36e-07	2.40e-07	0.998	104	7.91e-07	1	0.998	0.964	0.993	0	Complete
1 ANN10	0.1	3.23e-07	2.35e-07	3.72	1	7.92e-07	1	0.897	0.98	0.999	0	Complete
1 ANN10	0.2	1.60e-07	1.44e-07	1	1	6.66e-07	1	1.04	0.995	1	0	Complete
1 ANN10	0.3	3.37e-07	2.41e-07	1	1.01	8.00e-07	1.04	0.988	0.979	1	0	Complete
1 ANN10	0.5	0.131	0.0424	4.19e+05	2.81e+05	0.0569	2.73e+05	6.89e+05	1	0.982	0	Complete
1 ANN10	1	0.0869	0.028	3.68e+04	1.82e+06	0.0376	2.62e+06	4.54e+05	0.594	147	0	Complete
2 P-only DeepONet10	0.05	0.24	0.217	2.49e+09	3.75e+05	0.224	6.43e+06	1.22e+06	1.75e+04	4.54e+05	0	Complete
2 P-only DeepONet10	0.1	3.93e-06	2.14e-06	0.999	0.38	5.74e-06	0.996	4.69	1.59	18.3	0	Complete
2 P-only DeepONet10	0.2	2.18e-07	1.47e-07	0.999	0.526	4.78e-07	0.873	0.628	0.546	0.687	0	Complete
2 P-only DeepONet10	0.3	1.04e-04	3.41e-05	35.7	5.45e+04	4.73e-05	3.32e+05	545	1.45	2	0	Complete
2 P-only DeepONet10	0.5	2.09e-04	6.82e-05	6.24e+03	9.69e+04	9.18e-05	1.55e+05	1.1e+03	0.947	0.873	0	Complete
2 P-only DeepONet10	1	0.022	0.0199	1.26e+08	4.48e+07	0.0239	6.22e+08	2.12e+05	27.2	2.08e+04	0	Complete
3 ANN10 + P-only DeepONet10	0.05	7.57e-07	5.01e-07	1.71	0.454	1.56e-06	0.996	1.29	1.38	3.44	0	Complete
3 ANN10 + P-only DeepONet10	0.1	3.38e-07	2.43e-07	0.999	0.7	8.04e-07	1	0.999	0.988	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.12e-07	2.24e-07	0.995	0.797	7.44e-07	0.906	0.872	0.896	0.975	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.14e-03	3.74e-04	1.87e+04	3.25e+05	5.09e-04	4.77e+06	6e+03	0.741	0.99	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.97e-04	6.33e-05	1.97e+03	4.10e+04	8.49e-05	4.49e+04	1.03e+03	0.985	0.918	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0539	0.0187	1.04e+07	2.42e+07	0.0248	5.62e+07	2.95e+05	8.43	794	0	Complete
4 T/P DeepONet10	0.05	2.05	0.71	1.17e+06	4.68e+06	0.954	8.19e+07	1.16e+07	93.4	7.93e+03	0	Complete
4 T/P DeepONet10	0.1	7.51e-05	3.45e-05	26.1	44.3	8.12e-05	6.28	237	33.6	181	0	Complete
4 T/P DeepONet10	0.2	3.39e-07	2.44e-07	1	1	8.10e-07	0.997	1	1	1	0	Complete
4 T/P DeepONet10	0.3	6.13e-06	2.24e-06	71.5	2.98e+04	3.08e-06	1.91e+05	36	0.997	1	0	Complete
4 T/P DeepONet10	0.5	1.82e-04	8.95e-05	8.55e+03	2.49e+06	1.55e-04	1.50e+07	1.24e+03	0.646	1	0	Complete
4 T/P DeepONet10	1	0.82	0.93	5.69e+09	1.20e+09	0.991	2.78e+10	1.11e+07	484	425	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.817	0.622	1.57e+09	8.66e+08	0.938	1.54e+11	6.94e+06	237	1.98e+04	0	Complete
5 ANN10 + T/P DeepONet10	0.1	5.08e-06	3.41e-06	8.95	6.67	1.06e-05	1.39	3.38	7.02	27.6	0	Complete
5 ANN10 + T/P DeepONet10	0.2	3.39e-07	2.44e-07	0.998	1	8.10e-07	0.993	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.47e-05	4.86e-06	286	6.66e+03	6.39e-06	1.81e+05	78.6	0.998	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	1.51e-03	5.19e-04	2.18e+05	1.73e+05	6.96e-04	8.89e+06	8.18e+03	0.975	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0387	0.0189	5.84e+07	1.24e+08	0.0248	6.48e+08	2.68e+05	4.4e+03	323	0	Complete
6 T/P/ϕ DeepONet10	0.05	2.04	0.708	2.76e+06	1.19e+07	0.951	1.47e+08	1.15e+07	106	2.46e+03	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.507	0.744	3.97e+09	1.53e+08	0.939	1.26e+11	7.79e+06	1.03e+03	6.03e+03	0	Complete
6 T/P/ϕ DeepONet10	0.2	3.39e-07	2.44e-07	1	1	8.10e-07	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	1.29e-04	4.24e-05	1.03e+03	9.6e+03	5.77e-05	8.94e+05	679	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.79e-03	5.96e-04	1.70e+05	5.96e+05	8.11e-04	1.04e+07	9.49e+03	0.995	1	0	Complete
6 T/P/ϕ DeepONet10	1	1.48	0.779	2.06e+09	3.26e+09	0.997	3.05e+10	1.09e+07	154	42	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	1.90e-05	9.13e-06	0.994	57.8	2.24e-05	6.9	52.3	8.15	58	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	1.00e-05	5.02e-06	0.985	63.9	1.29e-05	8.25	30.7	7.33	29.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	3.39e-07	2.44e-07	0.998	1	8.10e-07	0.993	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	3.04e-05	9.88e-06	106	2.78e+03	1.31e-05	1.97e+05	160	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.31e-03	4.28e-04	7.58e+04	3.18e+05	5.81e-04	4.98e+06	6.88e+03	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	1.49	0.78	2.03e+09	2.91e+09	0.996	2.96e+10	1.10e+07	190	84.1	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 09

T_0=901.6 K P_0=2.084 atm ϕ_0=1.343 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 09

Repaired | Cantera vs NN NRMSE | T0=901.6K, P0=2.084atm, phi0=1.343

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	3.34e-07	2.30e-07	0.92	113	7.45e-07	0.909	0.999	0.879	0.966	0	Complete
1 ANN10	0.1	5.48e-08	1.19e-07	0.999	3.11e+03	5.78e-07	1	1	0.654	1.28	0	Complete
1 ANN10	0.2	3.39e-07	2.43e-07	0.848	1	8.04e-07	1	1	0.988	0.999	0	Complete
1 ANN10	0.3	1.74	0.659	5.19e+08	2.87e+08	0.861	1.44e+09	1.04e+07	145	30.6	0	Complete
1 ANN10	0.5	1.28	0.471	2.12e+08	3.15e+08	0.629	3.95e+09	7.48e+06	45.4	69.1	0	Complete
1 ANN10	1	0.196	0.064	2.65e+05	9.97e+07	0.0866	1.15e+06	1.04e+06	0.767	242	0	Complete
2 P-only DeepONet10	0.05	3.38e-07	2.42e-07	1	0.956	8.03e-07	0.989	0.999	0.985	0.999	0	Complete
2 P-only DeepONet10	0.1	3.39e-07	2.43e-07	0.754	0.936	8.06e-07	0.948	0.998	0.992	1	0	Complete
2 P-only DeepONet10	0.2	3.39e-07	2.44e-07	0.999	0.996	8.10e-07	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.3	1.65e-04	6.77e-05	2.63e+03	160	1.41e-04	1.29e+03	716	76.1	179	0	Complete
2 P-only DeepONet10	0.5	2.69e-05	9.88e-06	1.89	1.30e+04	1.75e-05	7.77e+03	131	6.92	12.9	0	Complete
2 P-only DeepONet10	1	0.464	0.729	5.75e+09	2.01e+10	0.939	1.00e+11	6.80e+06	3.42	1.08e+04	0	Complete
3 ANN10 + P-only DeepONet10	0.05	3.39e-07	2.43e-07	0.999	0.91	8.06e-07	0.991	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	3.39e-07	2.42e-07	0.998	0.781	8.04e-07	0.997	0.999	0.987	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.39e-07	2.44e-07	0.999	1	8.09e-07	1	1	0.998	1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.42e-04	9.27e-05	499	53.1	1.76e-04	850	1.1e+03	76.7	193	0	Complete
3 ANN10 + P-only DeepONet10	0.5	2.00e-05	6.95e-06	40	4.71e+03	1.12e-05	6.05e+03	99.7	3.03	5.97	0	Complete
3 ANN10 + P-only DeepONet10	1	1.71	0.716	8.31e+08	1.55e+09	0.96	2.58e+10	1.08e+07	8.48	2.75e+03	0	Complete
4 T/P DeepONet10	0.05	2.62e-07	1.75e-07	0.434	0.572	5.56e-07	0.628	0.769	0.63	0.762	0	Complete
4 T/P DeepONet10	0.1	4.86e-07	2.19e-07	814	0.995	8.10e-07	8.95	0.996	0.999	1	0	Complete
4 T/P DeepONet10	0.2	1.07e-04	7.74e-05	478	2.91e+05	2.62e-04	1.66e+05	609	432	33.3	0	Complete
4 T/P DeepONet10	0.3	8.83e-05	4.44e-05	67.2	3.40e+05	1.14e-04	3.88e+05	507	144	0.808	0	Complete
4 T/P DeepONet10	0.5	3.39e-04	1.19e-04	3.36e+04	2.73e+05	1.66e-04	4.88e+06	1.85e+03	0.666	1	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.350 ms
5 ANN10 + T/P DeepONet10	0.05	2.46e-07	1.61e-07	0.392	0.517	5.11e-07	0.573	0.725	0.572	0.707	0	Complete
5 ANN10 + T/P DeepONet10	0.1	9.58e-07	1.60e-07	3.14e+03	0.992	8.02e-07	74.1	0.939	0.994	1	0	Complete
5 ANN10 + T/P DeepONet10	0.2	4.74e-06	1.52e-06	825	461	1.69e-06	1.15e+04	26.2	0.963	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	5.18e-03	1.81e-03	1.46e+04	4.13e+05	2.92e-03	5.58e+05	2.45e+04	290	3.29e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.5	2.37e-04	8.04e-05	2.06e+04	8.32e+04	1.09e-04	1.57e+06	1.27e+03	0.56	0.998	0	Complete
5 ANN10 + T/P DeepONet10	1	0.967	0.62	2.89e+09	3.41e+09	0.765	3.66e+10	8.02e+06	1.11e+03	1.02e+03	0	Complete
6 T/P/ϕ DeepONet10	0.05	3.42e-07	2.39e-07	51.9	0.671	8.03e-07	9.62	0.974	0.989	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	9.00e-06	1.39e-06	4.00e+04	0.915	7.94e-07	330	0.868	0.992	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	3.39e-07	2.07e-07	346	895	7.58e-07	5.94e+03	0.698	0.991	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	3.08e-04	1.22e-04	274	4.86e+05	2.35e-04	3.72e+06	1.68e+03	175	1.5	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.28e-03	4.39e-04	1.41e+05	3.10e+05	5.96e-04	1.00e+07	6.91e+03	0.905	0.98	0	Complete
6 T/P/ϕ DeepONet10	1	0.511	0.992	9.46e+09	1.74e+10	0.983	7.34e+09	9.87e+06	936	1.40e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	3.38e-07	2.38e-07	44.1	0.754	7.97e-07	8.24	0.959	0.979	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	4.79e-07	2.13e-07	795	0.89	7.99e-07	22.8	0.942	0.987	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	3.04e-07	2.26e-07	54	51.3	7.85e-07	330	0.781	0.995	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	3.44e-04	1.23e-04	343	2.66e+05	2.05e-04	1.32e+06	1.83e+03	102	0.757	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	7.51e-04	2.54e-04	7.00e+04	1.58e+05	3.44e-04	4.49e+06	4.03e+03	0.856	0.973	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.611	0.801	6.29e+09	3.76e+09	0.861	6.37e+10	8.37e+06	743	1.15e+04	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 10

T_0=1317.2 K P_0=1.262 atm ϕ_0=0.769 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 10

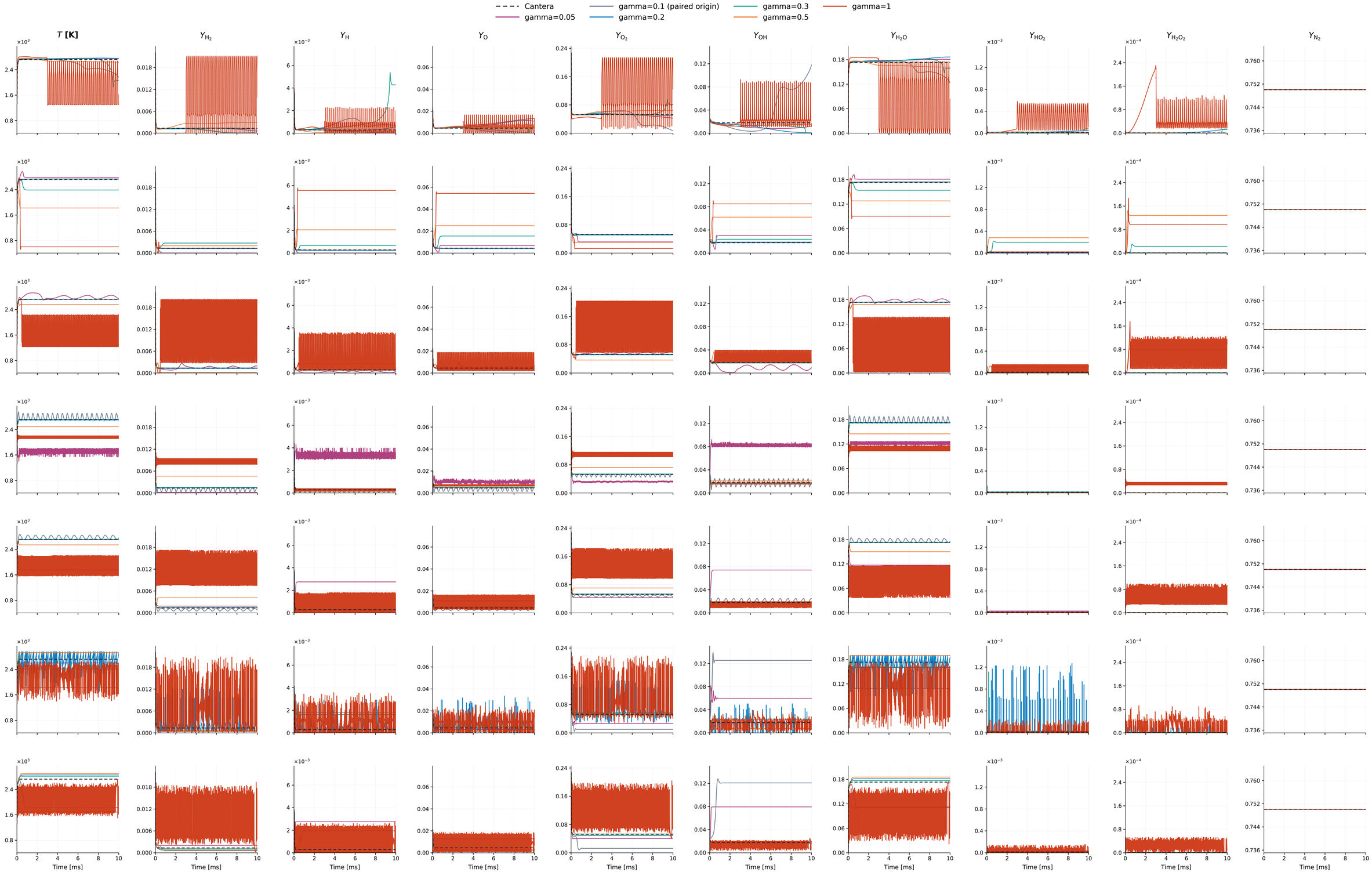
Repaired | Cantera vs NN NRMSE | T0=1317.2K, P0=1.262atm, phi0=0.769

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.134	0.278	0.752	0.748	0.46	3.63	0.217	0.611	0.952	0	Complete
1 ANN10	0.1	0.112	0.215	0.68	0.668	0.426	3.22	0.186	0.69	1.28	0	Complete
1 ANN10	0.2	0.0133	0.529	0.825	0.786	0.358	1.66	0.0365	0.899	0.936	0	Complete
1 ANN10	0.3	3.15e-03	0.309	0.201	0.421	0.108	0.451	3.48e-03	0.202	0.956	0	Complete
1 ANN10	0.5	0.0765	0.767	1.17	0.318	0.101	0.795	0.115	6.71	29.4	0	Complete
1 ANN10	1	0.284	4.94	4.02	1.11	1.23	1.13	0.458	15.8	55.9	0	Complete
2 P-only DeepONet10	0.05	0.11	0.871	1.06	0.907	0.0879	0.952	0.138	0.887	0.895	0	Complete
2 P-only DeepONet10	0.1	0.11	0.8	0.91	0.953	0.037	0.955	0.135	0.966	0.925	0	Complete
2 P-only DeepONet10	0.2	0.0971	0.79	0.897	0.956	0.0605	0.619	0.117	0.97	0.974	0	Complete
2 P-only DeepONet10	0.3	9.41e-03	0.794	0.207	0.771	0.313	1.14	9.79e-03	0.976	0.66	0	Complete
2 P-only DeepONet10	0.5	0.0752	0.916	4.21	0.171	0.129	0.219	0.0369	0.97	0.938	0	Complete
2 P-only DeepONet10	1	0.0317	0.393	0.408	0.21	0.0806	0.139	0.0434	0.926	3.64	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.108	0.779	0.882	0.927	0.0319	0.972	0.134	0.925	0.915	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.103	0.759	0.869	0.825	0.0352	0.913	0.129	0.927	0.946	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0761	0.762	0.872	0.803	0.134	0.185	0.0896	0.958	0.974	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0271	0.773	0.569	0.758	0.298	0.928	0.0254	0.961	0.85	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0455	0.795	2.82	0.139	0.121	0.399	0.01	0.97	0.947	0	Complete
3 ANN10 + P-only DeepONet10	1	0.27	5.22	2.33	1.37	1.33	0.752	0.46	5.49	33.9	0	Complete
4 T/P DeepONet10	0.05	0.486	0.202	11.8	2.54	0.184	3.51	0.399	1.26	0.727	0	Complete
4 T/P DeepONet10	0.1	0.0909	0.534	0.841	0.813	0.0404	0.879	0.108	0.718	0.678	0	Complete
4 T/P DeepONet10	0.2	0.154	1.62	0.743	2.94	0.132	0.302	0.168	4.36	2.62	0	Complete
4 T/P DeepONet10	0.3	0.216	1.58	0.797	4.64	0.161	1.15	0.214	28	20.6	0	Complete
4 T/P DeepONet10	0.5	0.574	0.678	14.7	8.27	0.569	2.29	0.309	51.1	79.3	0	Complete
4 T/P DeepONet10	1	0.184	4.01	0.358	0.521	0.988	0.262	0.331	0.793	30.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0629	0.29	0.904	0.975	0.268	0.996	0.0464	0.718	0.473	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.082	0.364	0.794	0.693	0.0804	0.924	0.0948	0.825	0.804	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0611	0.585	0.448	0.834	0.0538	0.249	0.0714	1.83	1.72	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0123	0.0857	0.257	0.0933	0.0153	0.0589	0.013	0.387	0.379	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.374	0.759	7.37	6.93	0.897	2.8	0.208	49.1	34.4	0	Complete
5 ANN10 + T/P DeepONet10	1	0.199	4.2	0.422	0.555	1.05	0.291	0.351	0.818	23.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0815	0.377	0.929	0.982	0.197	0.997	0.0839	0.77	0.686	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0787	0.279	0.888	0.56	0.0958	0.962	0.0899	0.658	0.393	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0759	0.356	1.57	0.546	0.0638	0.316	0.0734	0.839	0.556	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.178	0.67	4.79	1.19	0.218	1.28	0.144	16.8	11.1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.715	2.58	7.49	14.7	0.895	3.54	0.532	18.3	5.37	0	Complete
6 T/P/ϕ DeepONet10	1	0.142	2.28	3.27	1.32	0.62	0.336	0.194	0.984	10.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0771	0.105	0.913	0.978	0.193	0.997	0.0726	0.877	0.788	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0606	0.0869	0.821	0.233	0.125	0.947	0.0678	0.703	0.529	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0511	0.168	1.1	0.204	0.0204	0.348	0.0511	0.898	0.834	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0216	0.097	0.623	0.144	0.0347	0.18	0.0196	1.25	1.45	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.057	2.55	0.876	0.718	0.775	0.765	0.165	0.975	0.991	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.153	2.44	2.41	1.2	0.634	0.277	0.215	0.695	11.4	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 10

T_0=1317.2 K P_0=1.262 atm ϕ_0=0.769 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 10

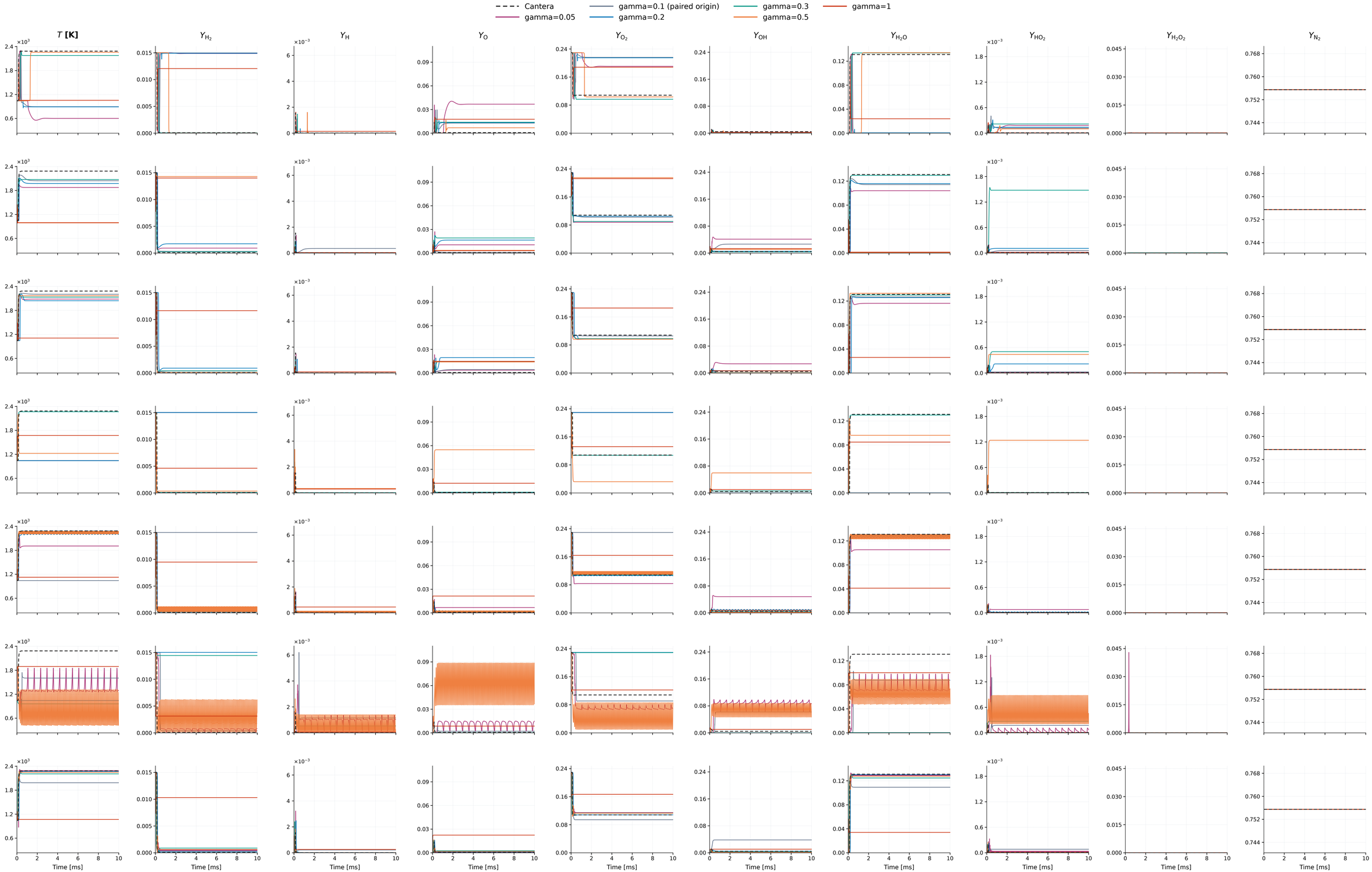
Repaired | Cantera vs NN NRMSE | T0=1317.2K, P0=1.262atm, phi0=0.769

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.011	0.051	0.482	1.02	0.0659	0.426	0.0312	0.317	0.394	0	Complete
1 ANN10	0.1	0.0771	0.475	1.71	0.632	0.377	2.24	0.0992	0.556	0.644	0	Complete
1 ANN10	0.2	0.0119	0.158	0.224	0.846	0.024	0.589	0.0414	0.775	5.28	0	Complete
1 ANN10	0.3	0.0687	0.511	3.95	0.395	0.162	0.371	0.0379	1.03	3.58	0	Complete
1 ANN10	0.5	0.0205	0.748	0.523	0.25	0.175	0.223	0.0475	0.296	0.699	0	Complete
1 ANN10	1	0.329	6.79	1.48	0.79	1.68	0.859	0.599	17.7	92.2	0	Complete
2 P-only DeepONet10	0.05	0.0277	0.779	0.895	0.485	0.376	0.673	0.047	0.961	0.977	0	Complete
2 P-only DeepONet10	0.1	7.13e-03	0.0762	0.109	0.105	0.0138	0.0124	7.78e-03	0.0568	0.0796	0	Complete
2 P-only DeepONet10	0.2	5.78e-03	0.159	0.203	0.0826	0.0441	0.0588	0.011	0.284	0.376	0	Complete
2 P-only DeepONet10	0.3	0.118	0.819	1.22	2.31	0.0289	0.323	0.108	11.3	26.7	0	Complete
2 P-only DeepONet10	0.5	0.327	0.413	5.65	4.37	0.41	2.44	0.26	16.8	155	0	Complete
2 P-only DeepONet10	1	0.767	0.352	16.6	10.6	0.733	3.68	0.472	0.525	117	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0375	0.238	0.673	0.239	0.0858	0.568	0.0413	0.42	0.48	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.59e-03	0.0281	0.0439	0.0331	6.67e-03	0.0121	2.05e-03	0.0402	0.0722	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.31e-03	0.0943	0.114	0.0505	0.0274	0.0469	6.20e-03	0.19	0.273	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.70e-03	0.0481	0.111	0.0183	0.0129	0.0374	2.59e-03	0.0506	0.0536	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0592	0.72	1.83	0.233	0.291	1.14	0.0341	5.88	95.2	0	Complete
3 ANN10 + P-only DeepONet10	1	0.458	8.69	3.38	1.2	2.2	0.544	0.779	6.09	103	0	Complete
4 T/P DeepONet10	0.05	0.376	0.7	9.99	1.26	0.384	3.58	0.297	0.976	0.987	0	Complete
4 T/P DeepONet10	0.1	0.0367	0.315	0.472	0.368	0.0672	0.26	0.0418	0.291	0.288	0	Complete
4 T/P DeepONet10	0.2	7.84e-03	0.113	0.0857	0.128	0.0224	0.0177	9.54e-03	0.242	0.264	0	Complete
4 T/P DeepONet10	0.3	1.60e-03	0.0671	0.0827	0.0624	0.0117	0.0256	3.26e-03	0.127	0.11	0	Complete
4 T/P DeepONet10	0.5	0.0847	1.94	0.91	0.584	0.366	0.161	0.163	0.675	0.247	0	Complete
4 T/P DeepONet10	1	0.205	4.33	0.509	0.556	1.04	0.12	0.367	0.692	37.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.353	0.296	7.85	1.38	0.174	3.11	0.323	1.19	0.665	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.028	0.311	0.434	0.313	0.0895	0.218	0.0311	0.434	0.462	0	Complete
5 ANN10 + T/P DeepONet10	0.2	4.69e-03	0.0897	0.0329	0.101	0.018	0.0296	6.13e-03	0.232	0.272	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.36e-03	0.0734	0.0966	0.0672	0.0129	0.0294	3.33e-03	0.117	0.101	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0638	1.66	0.898	0.339	0.333	0.0656	0.133	0.605	0.269	0	Complete
5 ANN10 + T/P DeepONet10	1	0.283	4.95	2.2	1.43	1.21	0.175	0.444	0.546	62.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0504	0.491	0.845	0.388	0.486	2.34	0.0782	0.963	0.98	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.326	0.435	4.86	0.295	0.79	5.92	0.366	0.974	0.984	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0673	0.941	1.14	0.798	0.257	0.317	0.0853	6.34	2.72	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.113	0.21	4.25	0.333	0.0822	0.312	0.0707	1.37	1.92	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0763	0.636	0.761	0.661	0.0766	0.589	0.0948	0.913	0.922	0	Complete
6 T/P/ϕ DeepONet10	1	0.182	2.95	3.44	1.61	0.756	0.357	0.261	2.66	26.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.378	0.404	7.87	1.77	0.226	3.4	0.349	1.04	0.363	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.318	0.447	5.14	0.284	0.725	5.56	0.35	0.951	0.963	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.048	0.332	0.84	0.555	0.0605	0.083	0.0468	0.233	0.438	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0334	0.158	0.894	0.553	0.0486	0.105	0.0214	0.792	0.575	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0632	0.403	0.82	0.558	0.062	0.532	0.0717	0.936	0.962	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.198	3.08	3.08	1.49	0.791	0.203	0.28	1.85	20.6	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 14

T_0=1043.0 K P_0=1.518 atm ϕ_0=0.520 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 14

Repaired | Cantera vs NN NRMSE | T0=1043.0K, P0=1.518atm, phi0=0.520

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.715	8.04	1.12	37.9	0.766	0.66	0.989	11.6	16.5	0	Complete
1 ANN10	0.1	0.6	8.1	1.19	13	0.963	0.911	0.99	8.13	25	0	Complete
1 ANN10	0.2	0.606	8.13	1.02	13.7	0.962	0.83	0.996	9.32	31	0	Complete
1 ANN10	0.3	0.0822	1.07	1.35	14.4	0.178	0.995	0.13	14.6	64	0	Complete
1 ANN10	0.5	0.184	2.8	1.38	6.42	0.374	0.989	0.341	6.36	27.2	0	Complete
1 ANN10	1	0.536	6.55	2.89	18.8	0.715	0.465	0.815	0.745	200	0	Complete
2 P-only DeepONet10	0.05	0.182	0.79	1.17	10.8	0.202	7.99	0.219	1.37	1.11	0	Complete
2 P-only DeepONet10	0.1	0.103	0.274	6.95	0.63	0.0604	4.65	0.128	3.61	4.89	0	Complete
2 P-only DeepONet10	0.2	0.138	1.06	1.08	17.2	0.0835	0.23	0.132	7.25	3.78	0	Complete
2 P-only DeepONet10	0.3	0.0952	0.465	1.13	20.5	0.172	0.294	0.054	104	225	0	Complete
2 P-only DeepONet10	0.5	0.566	7.79	0.977	3.22	0.957	1.41	0.995	0.939	16.5	0	Complete
2 P-only DeepONet10	1	0.563	7.64	1.14	2.74	0.934	1.94	0.988	0.816	15.4	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0918	0.539	1.2	3.32	0.123	4.97	0.127	1.07	1.47	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0396	0.416	1.08	3.98	0.0564	0.561	0.0531	1.47	2.35	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.122	1.13	1.2	20.4	0.168	0.898	0.13	14.4	15	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0657	0.21	0.761	15.3	0.0906	0.787	0.0145	34.9	60.9	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0599	0.518	1.02	14.8	0.129	0.913	0.0656	30.6	50.6	0	Complete
3 ANN10 + P-only DeepONet10	1	0.514	6.34	1.88	15.7	0.693	0.443	0.802	1.02	160	0	Complete
4 T/P DeepONet10	0.05	0.542	8.19	1	1	1.09	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.1	0.542	8.19	1	1	1.09	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.2	0.542	8.19	1	1	1.09	1	1	0.997	1	0	Complete
4 T/P DeepONet10	0.3	0.0336	0.58	1.62	0.856	0.0804	0.378	0.0673	1.02	1.26	0	Complete
4 T/P DeepONet10	0.5	0.464	0.678	7.25	59.8	0.687	11.9	0.276	87.6	111	0	Complete
4 T/P DeepONet10	1	0.269	2.57	6.21	13	0.238	1.28	0.36	0.966	73.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.165	0.41	1.66	6.75	0.223	9.44	0.201	5.17	4.14	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.542	8.19	1	1	1.09	1	1	0.996	1	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0327	0.313	1.8	1.59	0.0456	0.955	0.0456	0.977	1.36	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0326	0.538	1.29	1.51	0.075	0.175	0.0632	1.01	1.16	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0416	0.675	2.05	1.49	0.0957	0.739	0.0771	1.25	1.92	0	Complete
5 ANN10 + T/P DeepONet10	1	0.506	5.17	9.3	23.1	0.505	0.727	0.686	0.785	147	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.421	1.11	20.9	13.1	0.374	18	0.46	10.4	2.26e+03	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.309	1.49	23.7	1.68	0.252	12.3	0.371	12.1	10.7	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.542	8.19	1	1	1.09	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.579	7.87	10.9	0.966	1.08	0.997	0.998	18.8	48.9	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.671	1.92	15.3	72.2	0.645	14.2	0.495	37.4	8.89	0	Complete
6 T/P/ϕ DeepONet10	1	0.174	1.83	1.72	8.85	0.17	1.27	0.251	0.787	1.55	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0207	0.328	2.22	0.644	0.0631	0.981	0.0318	1.42	4.56	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.13	0.262	4.53	1.61	0.127	7.28	0.168	5.39	5.39	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0352	0.559	1.83	1.08	0.0785	0.192	0.066	1.12	1.48	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0488	0.772	1.74	2.28	0.0998	0.349	0.0883	1.08	1.17	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0245	0.452	1.2	1.46	0.0628	0.667	0.0459	0.98	1.29	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.532	5.62	5.26	24.2	0.528	1.37	0.741	0.875	69.3	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 14

T_0=1043.0 K P_0=1.518 atm ϕ_0=0.520 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 14

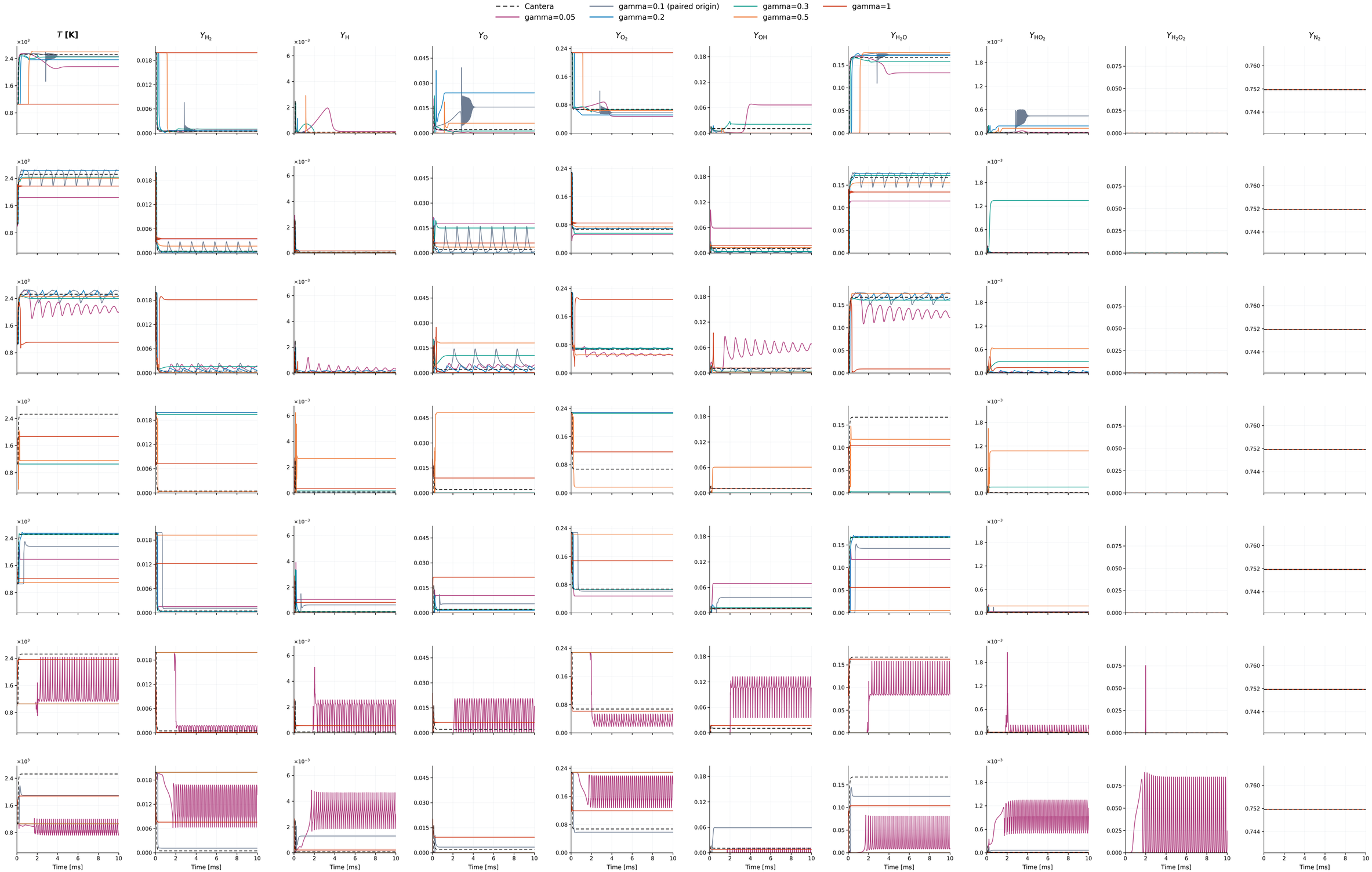
Repaired | Cantera vs NN NRMSE | T0=1043.0K, P0=1.518atm, phi0=0.520

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.485	6.93	1.09	8.15	0.876	0.951	0.846	4.76	24.9	0	Complete
1 ANN10	0.1	0.115	0.527	0.862	23.3	0.136	0.966	0.0484	23.5	50.4	0	Complete
1 ANN10	0.2	0.0963	0.899	1.14	19.3	0.166	0.98	0.104	8.36	31.8	0	Complete
1 ANN10	0.3	0.127	0.496	22.5	6.71	0.0778	0.968	0.0769	10.7	27.6	0	Complete
1 ANN10	0.5	0.0235	0.429	1.29	4.04	0.0627	0.97	0.0533	3.57	4.95	0	Complete
1 ANN10	1	0.542	8.19	0.997	1	1.09	1	1	0.999	42.1	0	Complete
2 P-only DeepONet10	0.05	0.0826	0.595	3.59	1.02	0.0893	3.8	0.12	3.56	8.7	0	Complete
2 P-only DeepONet10	0.1	0.0404	0.658	1.66	0.932	0.0915	0.623	0.0781	1.18	1.31	0	Complete
2 P-only DeepONet10	0.2	0.542	8.19	1	1	1.09	1	1	1	1	0	Complete
2 P-only DeepONet10	0.3	0.0669	1.06	1.28	1.02	0.144	0.295	0.127	1.97	5.06	0	Complete
2 P-only DeepONet10	0.5	0.0231	0.395	1.18	0.927	0.0553	0.169	0.0458	0.941	1.54	0	Complete
2 P-only DeepONet10	1	0.714	3.89	43.6	20.7	0.308	9.67	0.801	1.53	245	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.182	0.571	10.8	1.16	0.131	8.48	0.227	4.83	4.03	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.204	0.486	37.3	0.78	0.147	0.828	0.164	10.6	13.4	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.162	2.44	3.01	1.16	0.325	0.618	0.298	3.81	7.17	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0309	0.523	1.8	0.903	0.0732	0.184	0.0607	1.02	2.16	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0424	0.613	5.79	0.93	0.0879	0.456	0.0719	1.19	3.65	0	Complete
3 ANN10 + P-only DeepONet10	1	0.54	8.13	1.68	1.43	1.08	1.08	0.994	15.5	45.7	0	Complete
4 T/P DeepONet10	0.05	0.746	0.0831	22	75.1	0.959	21.9	0.474	0.658	0.627	0	Complete
4 T/P DeepONet10	0.1	0.0818	1	3.56	3.38	0.137	1.35	0.128	5.9	4.49	0	Complete
4 T/P DeepONet10	0.2	0.524	7.89	1	0.928	1.04	0.997	0.967	111	431	0	Complete
4 T/P DeepONet10	0.3	0.0264	0.42	1.01	1.56	0.056	0.273	0.0491	1.74	5.64	0	Complete
4 T/P DeepONet10	0.5	0.11	1.12	3.32	5.02	0.0988	1.53	0.162	7.25	44.6	0	Complete
4 T/P DeepONet10	1	0.231	2.54	2.54	7.79	0.251	1.69	0.351	0.787	56.9	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0269	0.469	0.833	2.47	0.0607	0.97	0.0411	2.08	1.04	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0284	0.427	1.28	1.26	0.0613	0.747	0.0511	1.78	2.16	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0685	1.02	1.65	2.07	0.139	0.649	0.124	3.13	8.15	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0158	0.222	0.808	1.24	0.0278	0.186	0.0263	0.466	0.818	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.078	0.869	1.15	3.15	0.0787	0.853	0.121	1.18	1.99	0	Complete
5 ANN10 + T/P DeepONet10	1	0.315	3.45	5.36	12.5	0.359	0.599	0.455	1.38	82.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.648	0.153	10.6	14.3	0.832	39.6	0.77	25.7	1.66	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.705	0.285	4.01	0.892	0.965	50.5	0.946	0.771	0.742	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.035	0.597	1.33	0.765	0.0834	0.416	0.0698	1.14	1.09	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.503	7.55	1.18	0.872	1	0.96	0.925	13.6	9.09	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.267	1.18	6.54	19.9	0.209	6.87	0.273	96.8	12.5	0	Complete
6 T/P/ϕ DeepONet10	1	0.832	0.964	39.2	112	0.991	8.88	0.325	1.01	9.27e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0604	0.698	0.666	6.24	0.0606	0.968	0.0698	5.08	1.46	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.151	0.205	3.27	0.605	0.202	10.8	0.209	0.415	1.69	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0538	0.851	1.31	1.11	0.117	0.568	0.102	1.24	1.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.104	0.441	16.4	2.82	0.0657	0.715	0.0848	1.21	1.58	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.106	0.835	3.58	7.86	0.124	2.09	0.127	40.4	6.38	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.731	1.35	11.7	103	0.848	9.5	0.371	0.956	2.73e+03	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 17

T_0=1057.2 K P_0=1.276 atm ϕ_0=0.692 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 17

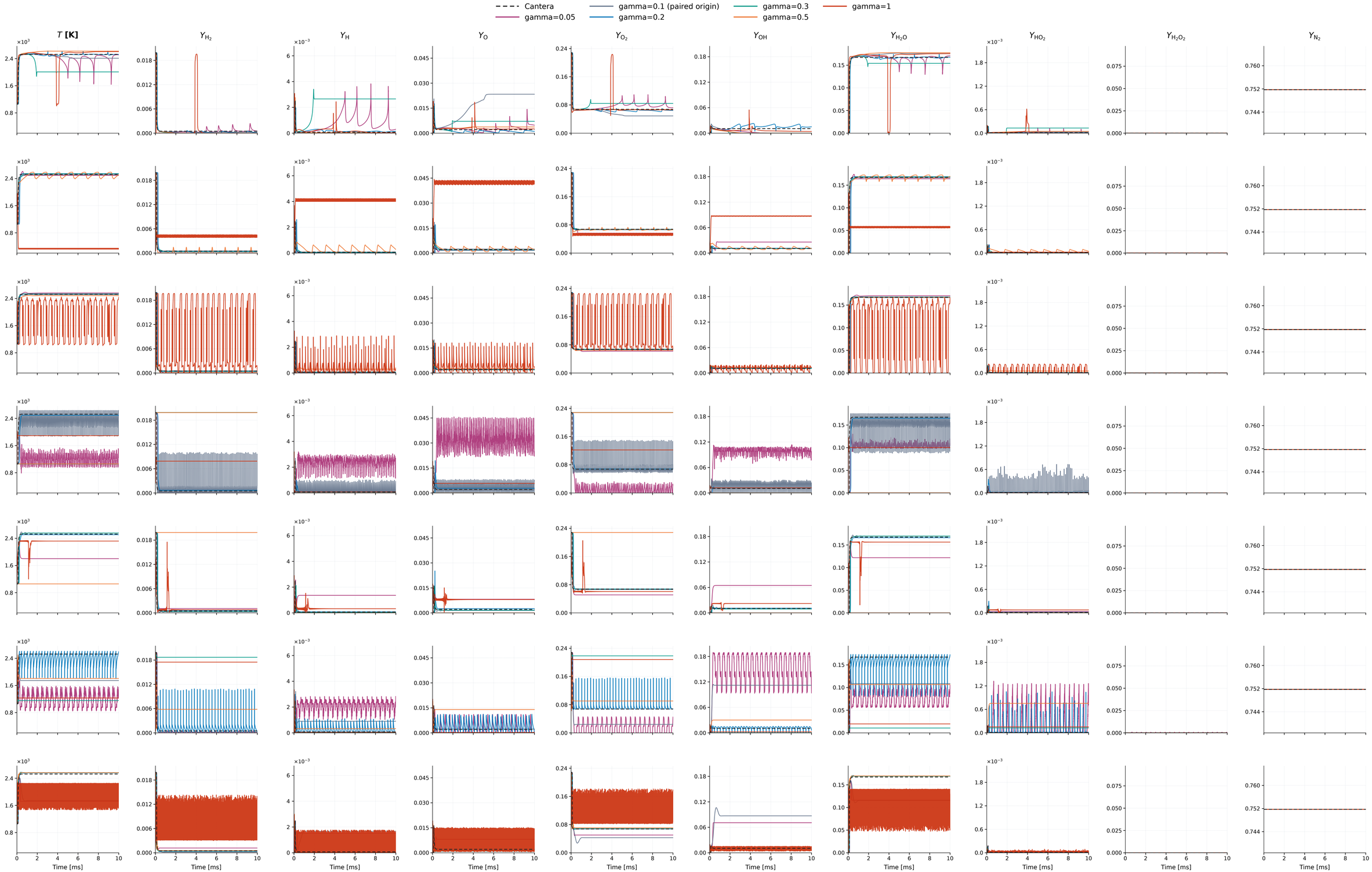
Repaired | Cantera vs NN NRMSE | T0=1057.2K, P0=1.276atm, phi0=0.692

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.125	0.222	5.11	0.767	0.244	4.16	0.168	0.948	1.35	0	Complete
1 ANN10	0.1	0.0388	0.602	1.01	5.37	0.194	0.94	0.0738	23.8	70.6	0	Complete
1 ANN10	0.2	0.101	1.29	1.03	9.39	0.391	0.988	0.146	11.2	75	0	Complete
1 ANN10	0.3	0.0485	0.683	1.91	0.623	0.17	0.9	0.0868	1.18	1.55	0	Complete
1 ANN10	0.5	0.187	2.85	1.25	1.69	0.721	0.987	0.326	7.04	31	0	Complete
1 ANN10	1	0.58	8.82	0.994	0.995	2.23	0.999	1	0.963	1.38	0	Complete
2 P-only DeepONet10	0.05	0.271	1.46	0.972	6.96	0.265	4.51	0.314	0.64	0.483	0	Complete
2 P-only DeepONet10	0.1	0.0559	0.481	0.884	2.2	0.082	0.668	0.0653	0.708	0.673	0	Complete
2 P-only DeepONet10	0.2	0.049	0.427	0.973	0.968	0.101	0.719	0.0643	0.883	0.718	0	Complete
2 P-only DeepONet10	0.3	0.0399	0.415	0.744	5.73	0.194	0.64	0.0484	88.4	221	0	Complete
2 P-only DeepONet10	0.5	0.0572	0.829	0.937	0.775	0.19	0.297	0.0969	1.01	0.602	0	Complete
2 P-only DeepONet10	1	0.143	1.59	1.2	1.78	0.316	0.716	0.209	0.82	25.7	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.162	0.539	2.27	0.95	0.219	4.2	0.206	0.776	0.519	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0414	0.337	0.684	1.95	0.039	0.63	0.0488	0.591	0.652	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0427	0.659	1.08	0.677	0.167	0.251	0.073	1.83	2.94	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.048	0.524	0.697	3.64	0.0615	0.564	0.0408	18.5	61.2	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0385	0.516	0.867	6.96	0.253	0.781	0.0702	40.5	133	0	Complete
3 ANN10 + P-only DeepONet10	1	0.555	7.94	0.784	1.42	1.94	0.437	0.938	9.04	25.5	0	Complete
4 T/P DeepONet10	0.05	0.58	8.83	1	1	2.23	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.1	0.58	8.83	1	1	2.23	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.2	0.58	8.83	1	1	2.23	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.3	0.581	8.62	1.02	0.971	2.19	0.974	0.985	9.34	24.8	0	Complete
4 T/P DeepONet10	0.5	0.54	0.7	21.6	20.2	0.727	4.61	0.301	70.8	100	0	Complete
4 T/P DeepONet10	1	0.261	3.13	2.34	3.09	0.698	0.141	0.379	0.884	47.4	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.295	0.854	8.19	3.68	0.327	5.41	0.303	1.83	1.35	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.193	2.1	4.47	1.52	0.533	2.34	0.274	1.53	0.959	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0505	0.827	1.28	0.479	0.214	0.164	0.0906	0.791	0.953	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0158	0.282	0.916	0.255	0.0747	0.16	0.0296	0.424	0.667	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.564	8.52	0.736	0.986	2.16	0.997	0.968	11.1	14.1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.515	5.35	6.22	8.55	1.12	0.155	0.664	0.657	98	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.485	3.83	13.5	4.86	1.07	7.98	0.587	7.18	3.99e+03	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.58	8.83	1	1	2.23	1	1	0.998	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.58	8.83	1	1	2.23	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.58	8.83	1	1	2.23	1	1	0.999	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.58	8.83	1	1	2.23	1	1	1	0.999	0	Complete
6 T/P/ϕ DeepONet10	1	0.0752	0.766	4.16	1.91	0.214	0.619	0.0849	0.717	4.23	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.64	5.92	24.4	0.998	1.63	0.908	0.881	63.7	6.04e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.251	0.954	10.2	0.724	0.265	4.43	0.27	3.71	3.61	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.58	8.83	1	1	2.23	1	1	0.996	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.58	8.83	1	1	2.23	1	1	0.982	0.995	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.58	8.83	1	1	2.23	1	1	0.999	0.922	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.257	3.27	1.43	3.23	0.729	0.271	0.382	0.554	12	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 17

T_0=1057.2 K P_0=1.276 atm ϕ_0=0.692 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 17

Repaired | Cantera vs NN NRMSE | T0=1057.2K, P0=1.276atm, phi0=0.692

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0634	0.3	7.3	0.739	0.166	0.806	0.0439	0.857	1.4	0	Complete
1 ANN10	0.1	0.0338	0.131	0.57	7.32	0.201	0.899	0.0453	1.01	11.8	0	Complete
1 ANN10	0.2	0.0353	0.644	1.17	0.674	0.169	0.564	0.0676	0.908	1.15	0	Complete
1 ANN10	0.3	0.187	0.338	19.7	2.01	0.223	0.915	0.0772	7.24	27.8	0	Complete
1 ANN10	0.5	0.0344	0.203	0.378	0.617	0.0128	0.894	0.0565	1.99	18.5	0	Complete
1 ANN10	1	0.101	1.49	1.67	0.619	0.367	0.51	0.174	4.06	64.1	0	Complete
2 P-only DeepONet10	0.05	0.0376	0.674	1.1	0.478	0.223	1.38	0.0709	1.19	1.39	0	Complete
2 P-only DeepONet10	0.1	0.0349	0.622	0.935	0.453	0.161	0.122	0.0665	0.876	0.891	0	Complete
2 P-only DeepONet10	0.2	0.0628	1.02	1.12	0.505	0.263	0.178	0.112	1.07	1.1	0	Complete
2 P-only DeepONet10	0.3	0.0248	0.479	0.89	0.433	0.128	0.0982	0.0502	1.02	1.34	0	Complete
2 P-only DeepONet10	0.5	0.0382	0.5	2.69	0.627	0.129	0.327	0.0529	3.28	26.2	0	Complete
2 P-only DeepONet10	1	0.867	1.84	33.7	17.7	0.287	7.1	0.66	1.01	218	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0263	0.493	1.01	0.362	0.138	0.265	0.0506	1.02	1.1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0192	0.377	0.759	0.392	0.0962	0.0738	0.0391	0.603	0.582	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0422	0.731	0.997	0.454	0.189	0.156	0.0788	0.964	0.998	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0206	0.406	0.816	0.369	0.108	0.0809	0.0424	0.752	0.787	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0196	0.261	1.98	0.348	0.0668	0.256	0.0208	2.71	22	0	Complete
3 ANN10 + P-only DeepONet10	1	0.366	5.25	4.62	1.91	1.31	0.575	0.608	7.82	79.8	0	Complete
4 T/P DeepONet10	0.05	0.526	0.921	19.1	13.5	0.825	8.02	0.382	1.16	59.7	0	Complete
4 T/P DeepONet10	0.1	0.0962	0.936	3.55	1.36	0.254	1.11	0.122	5.27	6.4	0	Complete
4 T/P DeepONet10	0.2	0.0609	0.993	0.918	0.659	0.254	0.177	0.109	1.98	1.22	0	Complete
4 T/P DeepONet10	0.3	0.58	8.83	1	1	2.23	1	1	0.974	0.973	0	Complete
4 T/P DeepONet10	0.5	0.579	8.81	0.999	1	2.23	0.999	0.998	0.993	0.991	0	Complete
4 T/P DeepONet10	1	0.252	3.41	1.01	1.66	0.773	0.181	0.405	0.665	45.7	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.284	0.5	10.7	2.7	0.251	4.96	0.273	1.66	1.46	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0154	0.241	0.669	0.374	0.0596	0.0874	0.0255	0.376	0.4	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0523	0.882	0.854	0.606	0.226	0.149	0.096	1.42	1.51	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0432	0.736	0.961	0.47	0.19	0.193	0.08	0.811	0.666	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.58	8.82	0.999	0.999	2.23	1	1	0.674	0.968	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0933	0.741	2.41	2.64	0.208	1.04	0.104	4.36	19.7	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.527	0.633	17.4	1.52	0.784	13.4	0.548	27.5	296	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.307	0.474	6.9	0.339	0.612	9.35	0.352	1.02	1.02	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0813	0.713	2.97	1.7	0.176	0.324	0.0936	5.15	2.14	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.541	8.27	0.926	0.957	2.09	0.982	0.936	7.69	4.14	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.287	2.56	2.35	5.25	0.381	1.8	0.372	49.6	10.4	0	Complete
6 T/P/ϕ DeepONet10	1	0.512	7.75	1.13	0.972	1.96	0.781	0.883	9.48	5.94	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.316	0.488	12	2.65	0.255	5.53	0.305	1.69	1.33	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.304	0.288	13	0.524	0.353	7.04	0.307	0.493	0.934	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.014	0.164	0.654	0.306	0.0393	0.137	0.0196	0.234	0.273	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0239	0.369	0.972	0.534	0.0952	0.17	0.0397	0.595	0.516	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0461	0.741	1.18	0.525	0.197	0.497	0.0812	0.955	0.989	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.237	2.55	6.09	3.14	0.62	0.359	0.322	1.13	28.9	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 19

T_0=1416.7 K P_0=2.395 atm ϕ_0=0.972 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 19

Repaired | Cantera vs NN NRMSE | T0=1416.7K, P0=2.395atm, phi0=0.972

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.187	0.893	0.897	0.893	0.765	3.84	0.33	3.8	6.82	0	Complete
1 ANN10	0.1	0.0202	0.46	2.09	0.424	0.308	0.863	0.0617	2.85	10.1	0	Complete
1 ANN10	0.2	7.68e-03	0.12	0.0413	0.459	0.325	0.662	0.0176	0.208	0.163	0	Complete
1 ANN10	0.3	0.0239	0.15	0.207	0.427	0.121	0.689	0.0548	1.32	15.9	0	Complete
1 ANN10	0.5	0.0686	0.159	0.952	0.915	0.221	1.05	0.0832	1.16	14.7	0	Complete
1 ANN10	1	0.125	0.931	3.32	0.4	0.932	1.54	0.0419	16.3	156	0	Complete
2 P-only DeepONet10	0.05	0.182	0.86	0.728	0.902	0.36	2.74	0.278	1.1	0.55	0	Complete
2 P-only DeepONet10	0.1	0.0172	0.175	0.571	0.375	0.123	0.283	9.45e-03	0.377	0.408	0	Complete
2 P-only DeepONet10	0.2	0.276	2.4	0.802	5.48	1.3	0.508	0.385	5.61	28.1	0	Complete
2 P-only DeepONet10	0.3	0.0593	0.391	1.71	0.23	0.252	0.508	0.0166	0.648	7.02	0	Complete
2 P-only DeepONet10	0.5	0.272	0.928	5.79	1.57	0.842	1.72	0.119	0.963	98.6	0	Complete
2 P-only DeepONet10	1	0.215	0.74	4.41	2.76	0.853	0.969	0.0711	0.995	965	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.572	3	0.956	1.19	0.228	7.75	0.886	1.75	2.85	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0887	0.31	1.53	0.755	0.449	1.4	0.0793	0.577	2.32	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.136	1.47	0.839	2.53	0.928	0.212	0.221	4.99	31.5	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.159	1.65	0.926	3.04	0.962	0.316	0.252	12.3	96.8	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.053	0.937	2.21	0.408	0.941	1.03	0.0258	0.766	97	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0744	1.41	0.855	0.194	1.5	0.305	0.181	2.61	1.38	0	Complete
4 T/P DeepONet10	0.05	0.0834	0.316	0.986	0.991	0.67	0.998	0.0667	0.965	0.904	0	Complete
4 T/P DeepONet10	0.1	0.119	0.48	1.18	0.634	0.391	0.853	0.139	1.65	2.09	0	Complete
4 T/P DeepONet10	0.2	0.106	1	0.6	2.01	0.47	0.396	0.164	3.07	2.5	0	Complete
4 T/P DeepONet10	0.3	1.92e-03	0.0197	0.0204	0.0199	0.0202	3.67e-03	3.04e-03	0.143	0.123	0	Complete
4 T/P DeepONet10	0.5	0.253	0.976	6	1.81	0.636	1.03	0.08	26.7	28.9	0	Complete
4 T/P DeepONet10	1	0.236	3.13	0.484	0.282	3.58	0.496	0.453	0.422	25.5	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0606	0.736	0.981	0.986	1.16	0.995	0.071	0.738	0.599	0	Complete
5 ANN10 + T/P DeepONet10	0.1	6.22e-03	0.0107	0.0845	0.0118	5.45e-03	0.0597	6.41e-03	0.0428	0.0523	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0438	0.53	0.521	0.938	0.236	0.19	0.0794	1.73	1.87	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.88e-03	0.0268	0.0332	0.0339	0.0155	8.59e-03	3.83e-03	0.177	0.18	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0214	0.853	0.959	0.0588	0.871	0.248	0.0932	0.941	0.948	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0323	0.874	1.77	0.202	0.875	0.531	0.0603	0.627	1.58	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.426	0.415	4.81	0.814	0.292	3.86	0.422	4.16	1.34	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.116	0.49	0.955	0.801	0.53	0.986	0.135	1.23	2.94	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0127	0.171	0.35	0.11	0.0264	0.299	0.0339	0.584	0.945	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.189	1.2	0.223	2.04	0.274	1.6	0.274	0.154	1.8	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.206	0.922	4.67	0.703	0.891	1.64	0.0848	29.8	55.9	0	Complete
6 T/P/ϕ DeepONet10	1	0.0699	0.93	2.52	1.23	0.945	0.446	0.0523	0.662	2.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0911	0.443	0.978	0.988	0.73	0.996	0.0894	0.871	0.721	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.102	0.246	0.844	0.477	0.0187	0.919	0.115	0.841	0.815	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0107	0.166	0.386	0.0877	0.0345	0.307	0.0325	0.583	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.109	0.874	0.762	1.28	0.103	1.2	0.184	0.45	0.667	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.182	2.27	0.952	0.262	2.1	0.772	0.374	0.982	0.913	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0454	0.966	2.22	0.864	0.961	0.409	0.0661	0.732	2.14	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 19

T_0=1416.7 K P_0=2.395 atm ϕ_0=0.972 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 19

Repaired | Cantera vs NN NRMSE | T0=1416.7K, P0=2.395atm, phi0=0.972

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0566	0.561	0.837	1.26	0.0534	0.642	0.101	0.549	0.337	0	Complete
1 ANN10	0.1	0.0154	0.0923	0.478	2.45	0.212	0.914	0.0395	0.779	6.08	0	Complete
1 ANN10	0.2	0.135	1.72	0.906	0.987	1.46	0.278	0.261	0.123	0.671	0	Complete
1 ANN10	0.3	0.122	1.51	0.888	0.18	1.22	0.789	0.258	0.915	2.09	0	Complete
1 ANN10	0.5	0.0372	0.509	0.474	0.431	0.551	0.363	0.072	0.759	6.11	0	Complete
1 ANN10	1	0.13	0.952	3.42	0.285	0.961	1.28	0.0461	24.1	319	0	Complete
2 P-only DeepONet10	0.05	5.51e-03	0.0703	0.161	0.0724	0.0857	0.131	2.11e-03	0.0304	0.0889	0	Complete
2 P-only DeepONet10	0.1	2.57e-03	0.0253	6.28e-03	0.0381	0.0218	2.18e-03	3.85e-03	0.0278	0.0392	0	Complete
2 P-only DeepONet10	0.2	9.52e-04	0.0101	6.85e-03	0.0193	0.0121	9.49e-03	1.17e-03	0.0332	0.0665	0	Complete
2 P-only DeepONet10	0.3	0.0958	0.836	0.669	2.66	0.15	0.349	0.133	29.7	147	0	Complete
2 P-only DeepONet10	0.5	0.338	0.9	7.26	1.89	0.487	1.51	0.158	3.51	177	0	Complete
2 P-only DeepONet10	1	0.0162	0.244	0.147	0.0723	0.225	0.0177	0.0352	0.973	1.07	0	Complete
3 ANN10 + P-only DeepONet10	0.05	5.33e-03	0.0607	0.134	0.0706	0.084	0.14	2.56e-03	0.0174	0.0624	0	Complete
3 ANN10 + P-only DeepONet10	0.1	4.14e-04	7.23e-03	0.0103	0.0149	9.09e-03	0.0152	4.83e-04	0.0418	0.0489	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.08e-03	9.69e-03	6.20e-03	0.0103	6.65e-03	6.49e-03	1.73e-03	0.0449	0.142	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.04	0.59	0.726	1.4	0.24	0.0337	0.0738	13.7	69.7	0	Complete
3 ANN10 + P-only DeepONet10	0.5	2.94e-03	0.951	1.51	0.653	0.899	0.631	0.0703	1.69	148	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0155	0.255	0.17	0.0671	0.239	5.68e-03	0.0354	0.962	0.667	0	Complete
4 T/P DeepONet10	0.05	0.17	2.28	0.95	1.11	2.73	0.973	0.325	27	111	0	Complete
4 T/P DeepONet10	0.1	5.16e-03	0.0181	0.0477	0.0309	0.0139	0.0271	5.72e-03	0.0733	0.0654	0	Complete
4 T/P DeepONet10	0.2	1.36e-03	0.015	0.0178	0.023	0.0136	3.52e-03	2.08e-03	0.0425	0.0334	0	Complete
4 T/P DeepONet10	0.3	2.97e-04	0.0257	0.0576	0.0293	0.0134	3.88e-03	2.20e-03	0.0305	0.0349	0	Complete
4 T/P DeepONet10	0.5	0.0152	0.186	0.657	0.0203	0.095	0.0516	7.64e-03	0.0878	0.437	0	Complete
4 T/P DeepONet10	1	0.0891	0.956	2.46	0.897	0.995	0.627	0.0809	2.85	10.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.149	0.615	0.98	0.938	0.316	0.993	0.182	0.95	0.808	0	Complete
5 ANN10 + T/P DeepONet10	0.1	4.58e-03	9.00e-03	0.0428	6.00e-03	3.07e-03	0.048	5.32e-03	0.0966	0.106	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.25e-03	0.0237	9.97e-03	0.0398	0.0219	7.62e-03	3.27e-03	0.0621	0.0655	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.24e-04	0.0179	0.0424	0.018	9.80e-03	4.18e-03	1.38e-03	8.52e-03	0.0196	0	Complete
5 ANN10 + T/P DeepONet10	0.5	5.37e-03	0.243	0.59	0.0979	0.149	0.037	0.0192	0.0501	0.549	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0844	1.11	2.16	0.841	1.19	0.528	0.12	3.86	11.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.156	0.7	0.978	0.948	0.418	0.996	0.195	1.7	2.47	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.477	0.364	6.87	0.501	0.875	4.89	0.42	1.07	1.4	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0349	0.0508	0.92	0.355	0.137	0.173	8.92e-03	0.0288	1.2	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0761	1.32	0.986	0.178	1.22	0.0805	0.185	0.0486	0.467	0	Complete
6 T/P/ϕ DeepONet10	0.5	4.11e-03	0.14	0.258	0.104	0.0679	0.0515	0.0176	0.201	0.106	0	Complete
6 T/P/ϕ DeepONet10	1	0.0417	0.808	1.7	0.964	0.892	0.363	0.0631	1.28	2.02	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.154	0.686	0.977	0.883	0.407	0.993	0.193	0.968	0.97	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.115	0.428	0.828	0.365	0.212	0.92	0.144	0.893	0.844	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	1.75e-03	0.28	0.83	0.0338	0.0255	0.281	0.0362	0.0564	0.786	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0932	1.43	0.983	0.4	1.25	0.191	0.208	0.0445	0.362	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.143	1.41	0.89	0.408	0.779	1.35	0.273	0.701	0.173	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0166	0.855	1.47	0.729	0.925	0.311	0.0755	1.49	1.48	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 23

T_0=1226.9 K P_0=1.917 atm ϕ_0=1.176 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 23

Repaired | Cantera vs NN NRMSE | T0=1226.9K, P0=1.917atm, phi0=1.176

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0418	0.295	0.93	0.847	0.32	0.934	0.0863	0.699	1.17	0	Complete
1 ANN10	0.1	7.38e-03	0.178	0.916	1.14	0.195	0.921	0.0477	3.78	16.5	0	Complete
1 ANN10	0.2	0.0197	0.193	0.489	0.325	0.199	0.657	0.0592	1.18	5.83	0	Complete
1 ANN10	0.3	0.0418	0.153	0.194	0.829	0.345	0.76	0.0694	4.26	47.6	0	Complete
1 ANN10	0.5	0.0385	0.227	0.348	0.631	0.511	0.407	0.0657	2.71	116	0	Complete
1 ANN10	1	0.204	1.24	1.91	0.867	4.78	0.398	0.314	26.8	137	0	Complete
2 P-only DeepONet10	0.05	0.647	2.01	0.371	0.649	1.64	10.9	0.93	18.5	4.79	0	Complete
2 P-only DeepONet10	0.1	4.44e-03	0.023	0.0375	0.0603	0.0564	0.061	7.10e-03	0.106	0.0969	0	Complete
2 P-only DeepONet10	0.2	4.11e-03	0.0354	0.0633	0.0697	0.131	0.0331	7.71e-03	0.331	0.343	0	Complete
2 P-only DeepONet10	0.3	0.48	2.3	0.932	14.8	4.94	1.54	0.628	1.75	25.5	0	Complete
2 P-only DeepONet10	0.5	0.317	1.08	0.479	7.25	1.93	2.25	0.38	1.06	29	0	Complete
2 P-only DeepONet10	1	0.0154	0.476	2.05	0.761	0.334	0.841	0.0875	0.599	202	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.634	2.08	0.788	0.783	1.68	11	0.938	17.5	4.07	0	Complete
3 ANN10 + P-only DeepONet10	0.1	3.10e-03	0.0252	0.0486	0.0655	0.0487	0.0522	6.76e-03	0.0912	0.0584	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.51e-03	0.0225	0.0403	0.0459	0.0791	0.0219	4.91e-03	0.174	0.208	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.335	1.66	0.869	8.12	3.53	1.9	0.479	2.2	37.1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0508	0.653	2.71	1.31	0.58	0.9	0.12	0.825	29.4	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0122	0.504	2.13	0.76	0.374	0.898	0.0941	0.897	208	0	Complete
4 T/P DeepONet10	0.05	0.437	0.156	4.8	2.42	0.364	3.87	0.349	3.65	0.885	0	Complete
4 T/P DeepONet10	0.1	0.0995	0.157	0.883	0.97	0.523	0.989	0.0947	5.11	8.14	0	Complete
4 T/P DeepONet10	0.2	0.435	1.83	0.484	14.1	3.24	1.92	0.537	12.7	1.29	0	Complete
4 T/P DeepONet10	0.3	0.402	1.65	0.677	14.4	2.23	2.13	0.491	45.2	28.9	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.256 ms
4 T/P DeepONet10	1	0.25	1.68	0.245	0.968	6.62	0.379	0.419	0.876	44.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0232	0.449	0.98	0.976	2.04	0.995	0.0477	0.509	0.0996	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0964	0.059	0.937	0.959	0.113	0.983	0.083	1.37	1.48	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.309	1.43	0.832	8.49	2.44	2.1	0.427	7.72	2.51	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.325	1.51	0.892	9.33	2.56	2.1	0.446	17.7	14.5	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.233	0.849	2.99	3.66	2.65	1.53	0.243	18	22.9	0	Complete
5 ANN10 + T/P DeepONet10	1	0.285	0.737	6.31	2.08	0.769	0.817	0.0617	5.6	27.7	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.543	0.148	5.6	0.598	0.25	6.18	0.462	9.47	5.67	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.437	0.0251	5.39	0.224	0.238	4.31	0.341	6.4	8.33	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0115	0.0612	0.109	0.16	0.207	0.0943	0.0165	0.476	0.262	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.179	0.569	0.135	2.19	0.537	2.4	0.241	4.46	7.32	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.722	0.415	9.96	8.84	0.697	4.02	0.384	45.6	317	0	Complete
6 T/P/ϕ DeepONet10	1	0.245	0.967	3.8	3.03	3.63	0.669	0.24	5.43	13.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0631	0.181	0.973	0.976	0.985	0.995	0.0247	0.605	0.149	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.554	1.31	0.625	0.852	0.364	10.3	0.753	1.99	4.67	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0156	0.0786	0.0928	0.177	0.155	0.244	0.0262	0.62	0.76	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.133	0.533	0.44	1.6	0.413	2.11	0.205	2.32	5.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.101	0.771	0.657	1.12	3.01	0.315	0.181	1.01	0.834	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.219	0.552	4.73	2.28	0.715	0.631	0.049	3.2	30.6	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 23

T_0=1226.9 K P_0=1.917 atm ϕ_0=1.176 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 23

Repaired | Cantera vs NN NRMSE | T0=1226.9K, P0=1.917atm, phi0=1.176

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.117	0.434	0.781	0.835	0.499	3.4	0.217	0.647	0.8	0	Complete
1 ANN10	0.1	0.0172	0.135	0.334	3.81	0.2	0.881	0.0242	2.65	24.1	0	Complete
1 ANN10	0.2	0.0221	0.0168	0.0747	0.511	0.247	0.712	0.0318	0.317	0.0929	0	Complete
1 ANN10	0.3	0.0231	0.256	0.594	0.286	0.372	0.509	0.0663	0.306	0.828	0	Complete
1 ANN10	0.5	0.049	0.323	0.478	0.604	0.546	0.691	0.0952	0.613	0.783	0	Complete
1 ANN10	1	0.149	1.01	0.641	0.461	3.92	0.217	0.255	12.9	178	0	Complete
2 P-only DeepONet10	0.05	4.15e-03	8.84e-03	0.0417	0.0253	0.0284	0.053	4.23e-03	0.0625	0.0482	0	Complete
2 P-only DeepONet10	0.1	9.19e-04	0.012	0.0367	0.0321	0.0263	4.13e-03	2.08e-03	0.0513	0.0682	0	Complete
2 P-only DeepONet10	0.2	2.21e-03	7.00e-03	0.0315	0.0178	0.0264	9.81e-03	2.11e-03	0.0641	0.12	0	Complete
2 P-only DeepONet10	0.3	1.68e-03	7.90e-03	9.69e-03	0.0352	0.026	4.96e-03	2.20e-03	0.0675	0.0499	0	Complete
2 P-only DeepONet10	0.5	0.468	0.486	7.78	3.95	0.113	2.19	0.218	11.6	133	0	Complete
2 P-only DeepONet10	1	0.847	0.384	11.4	15.9	0.471	2.77	0.4	0.984	1.38e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	8.50e-03	0.0633	0.266	0.142	0.177	0.151	2.35e-03	0.081	0.231	0	Complete
3 ANN10 + P-only DeepONet10	0.1	8.10e-04	0.0202	0.049	0.07	0.047	8.16e-03	3.38e-03	0.0677	0.13	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.91e-04	7.27e-03	0.0193	0.0133	0.0252	3.14e-03	1.46e-03	0.0558	0.0943	0	Complete
3 ANN10 + P-only DeepONet10	0.3	9.44e-04	3.46e-03	6.11e-03	0.0158	0.0108	4.39e-03	1.13e-03	0.0232	0.055	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0474	0.46	1.15	0.957	0.537	0.964	0.12	1.88	43.1	0	Complete
3 ANN10 + P-only DeepONet10	1	4.09e-03	0.0533	0.0849	0.0941	0.202	0.0166	0.011	0.79	0.949	0	Complete
4 T/P DeepONet10	0.05	0.108	0.749	0.931	0.874	3.09	0.972	0.175	10.9	68.1	0	Complete
4 T/P DeepONet10	0.1	0.0787	0.0715	0.856	0.567	0.285	0.559	0.0668	0.972	1.75	0	Complete
4 T/P DeepONet10	0.2	0.0135	0.0403	0.0987	0.215	0.136	0.0496	0.0144	0.183	0.104	0	Complete
4 T/P DeepONet10	0.3	1.58e-03	0.0178	0.0305	0.0363	0.0598	7.67e-03	3.74e-03	0.122	0.0554	0	Complete
4 T/P DeepONet10	0.5	0.0668	0.686	0.953	0.774	1.95	0.256	0.157	0.829	0.585	0	Complete
4 T/P DeepONet10	1	0.324	1.43	3.84	3.5	5.59	0.889	0.379	13.9	63.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0928	0.234	0.95	0.832	0.969	0.982	0.0892	0.752	3.89	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0702	0.0579	0.677	0.317	0.123	0.755	0.0655	0.591	0.708	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0784	0.476	0.341	2.34	1.17	0.179	0.12	2.95	2.83	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.01e-03	0.0114	0.019	0.0259	0.0363	5.43e-03	2.44e-03	0.0638	0.0377	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.067	0.678	0.936	0.663	1.92	0.305	0.158	0.763	0.689	0	Complete
5 ANN10 + T/P DeepONet10	1	0.312	1.56	3.3	2.55	5.97	0.773	0.41	15.1	73	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.53	0.295	7.14	1.01	0.58	5.57	0.374	0.703	3.44	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.507	0.774	8.97	1	0.677	3.88	0.25	6.4	329	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.137	0.867	0.878	1.15	3.49	0.431	0.216	40.1	56.4	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0326	0.536	0.981	0.0348	1.55	0.137	0.113	0.36	2.04	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.144	0.639	0.735	1.6	0.541	2.49	0.237	0.85	1.12	0	Complete
6 T/P/ϕ DeepONet10	1	0.875	0.978	14.5	22.6	0.786	0.983	0.22	2.17	1.38e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.423	0.0295	4.71	0.529	0.315	5.01	0.355	1.64	1.84	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0832	0.0711	0.781	0.171	0.107	0.948	0.0807	0.6	0.533	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0305	0.105	0.863	0.302	0.0773	0.14	4.87e-03	0.113	0.646	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0745	0.699	0.978	0.398	1.87	0.591	0.173	0.766	2.12	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0796	0.115	0.801	0.659	0.333	0.756	0.0726	0.759	0.922	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.251	0.568	5.21	2.3	0.657	0.826	0.0663	4.78	40.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 24

T_0=1522.6 K P_0=1.432 atm ϕ_0=0.599 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 24

Repaired | Cantera vs NN NRMSE | T0=1522.6K, P0=1.432atm, phi0=0.599

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0355	0.352	5.92	0.86	0.202	0.948	8.57e-03	0.756	0.837	0	Complete
1 ANN10	0.1	0.119	0.283	0.848	0.784	0.347	4.13	0.232	0.856	0.883	0	Complete
1 ANN10	0.2	0.0427	0.522	0.876	0.896	0.204	2.31	0.104	0.945	0.965	0	Complete
1 ANN10	0.3	0.055	0.299	0.847	0.838	0.152	2.19	0.13	0.917	0.921	0	Complete
1 ANN10	0.5	0.0622	0.773	0.832	0.229	6.90e-03	0.954	0.111	4.03	17.2	0	Complete
1 ANN10	1	0.483	16.3	0.793	3.51	1.14	0.468	0.882	1.34	106	0	Complete
2 P-only DeepONet10	0.05	0.209	0.829	4.12	0.665	0.17	3.07	0.283	0.235	0.492	0	Complete
2 P-only DeepONet10	0.1	0.0833	0.776	0.87	0.91	0.0575	0.952	0.112	0.979	0.977	0	Complete
2 P-only DeepONet10	0.2	0.0722	0.782	0.871	0.96	0.0303	0.611	0.0912	0.989	0.986	0	Complete
2 P-only DeepONet10	0.3	0.0516	0.786	2.22	0.672	0.105	1.36	0.0669	0.992	0.981	0	Complete
2 P-only DeepONet10	0.5	0.0975	0.787	5.81	0.515	0.0322	0.486	0.0569	0.969	0.992	0	Complete
2 P-only DeepONet10	1	0.106	1.34	1.54	4.37	0.0631	0.809	0.0476	20.6	6.57	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.186	0.221	4.77	0.247	0.155	3.04	0.255	0.95	0.239	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0285	0.523	0.747	1.1	0.0119	0.54	0.0472	0.613	0.588	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0548	0.768	0.856	0.819	0.0132	0.214	0.0657	0.985	0.991	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0195	0.781	0.409	0.737	0.125	1.29	0.0405	0.988	0.978	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0595	0.788	3.43	0.188	0.0674	0.629	0.0367	0.987	0.987	0	Complete
3 ANN10 + P-only DeepONet10	1	0.154	3.81	2.04	1.24	0.284	0.978	0.251	3.3	25.8	0	Complete
4 T/P DeepONet10	0.05	0.0578	1.13	0.835	0.942	0.183	0.993	0.0695	2.77	2.4	0	Complete
4 T/P DeepONet10	0.1	0.0641	0.364	0.778	0.643	0.0711	0.883	0.0834	0.751	0.701	0	Complete
4 T/P DeepONet10	0.2	0.0682	0.975	0.709	1.26	6.19e-03	0.276	0.0793	2.22	2.03	0	Complete
4 T/P DeepONet10	0.3	0.0116	0.197	0.126	0.224	4.51e-03	0.0438	0.0143	0.0907	0.137	0	Complete
4 T/P DeepONet10	0.5	0.52	0.425	10.7	12.2	0.839	3.48	0.322	68.4	92.1	0	Complete
4 T/P DeepONet10	1	0.131	4.55	0.238	0.789	0.316	0.0129	0.253	0.738	28.1	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0589	0.298	0.868	0.951	0.146	0.991	0.0558	0.713	0.429	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0529	0.15	0.684	0.415	0.0762	0.874	0.0697	0.769	0.74	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0155	0.0811	0.389	0.0851	6.73e-03	0.171	0.0195	0.358	0.4	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.42e-03	0.0891	0.0727	0.0985	1.47e-03	6.30e-03	4.58e-03	0.0965	0.115	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0462	2.77	0.765	0.484	0.278	0.511	0.123	0.934	0.854	0	Complete
5 ANN10 + T/P DeepONet10	1	0.138	5.27	0.162	0.526	0.396	0.146	0.282	0.775	26.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.442	1.3	9.72	4.44	0.376	4.6	0.472	0.77	0.747	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0549	0.129	0.803	0.335	0.0778	0.94	0.0736	0.694	0.523	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0331	0.331	0.649	0.358	2.85e-03	0.227	0.0401	0.226	0.203	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0117	0.126	0.0727	0.109	0.0115	0.17	0.0175	0.129	0.227	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.686	5.16	3.14	19.3	0.916	3.48	0.535	32	6.36	0	Complete
6 T/P/ϕ DeepONet10	1	0.125	3.33	5.01	1.49	0.297	0.333	0.179	1.01	13.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.349	1.14	5.37	2.62	0.4	5.03	0.437	1.15	0.331	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0423	0.12	0.738	0.056	0.0771	0.921	0.0597	0.703	0.586	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0167	0.149	0.328	0.142	4.44e-03	0.15	0.0212	0.157	0.189	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	4.48e-03	0.0306	0.0842	0.0205	9.92e-03	0.118	8.22e-03	0.104	0.216	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0268	3.2	0.832	0.758	0.344	0.762	0.12	0.99	0.99	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0955	2.42	3.33	1.08	0.204	0.24	0.136	0.66	12.3	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 24

T_0=1522.6 K P_0=1.432 atm ϕ_0=0.599 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 24

Repaired | Cantera vs NN NRMSE | T0=1522.6K, P0=1.432atm, phi0=0.599

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	5.91e-03	0.0694	0.482	1.54	0.0425	0.81	0.0508	0.376	0.385	0	Complete
1 ANN10	0.1	0.165	0.429	7.43	0.826	0.245	3.06	0.215	1.1	3	0	Complete
1 ANN10	0.2	0.0144	0.408	0.446	0.497	0.0909	0.952	0.0375	0.405	0.962	0	Complete
1 ANN10	0.3	7.65e-03	0.232	0.407	0.169	0.0151	0.284	0.0221	0.129	0.286	0	Complete
1 ANN10	0.5	0.0227	1.1	0.292	0.0831	0.0832	0.106	0.0542	0.0905	0.934	0	Complete
1 ANN10	1	0.208	6.93	1.84	0.586	0.511	1.07	0.421	12.6	86.8	0	Complete
2 P-only DeepONet10	0.05	0.0197	0.77	0.864	0.126	0.192	1.38	0.0346	0.976	0.982	0	Complete
2 P-only DeepONet10	0.1	2.81e-03	0.0446	0.111	0.0365	3.91e-03	0.026	2.63e-03	0.0436	0.1	0	Complete
2 P-only DeepONet10	0.2	1.18e-03	7.76e-03	0.0491	0.0337	5.53e-03	0.0351	1.68e-03	0.086	0.104	0	Complete
2 P-only DeepONet10	0.3	0.0505	0.503	0.108	1.92	0.0763	0.0772	0.0333	17.1	64.1	0	Complete
2 P-only DeepONet10	0.5	0.37	0.398	7.88	5.76	0.46	3.49	0.331	17.5	193	0	Complete
2 P-only DeepONet10	1	0.767	2.07	18.4	11.6	0.582	4.79	0.63	0.602	79.8	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0168	0.737	0.85	0.266	0.175	1.06	0.024	0.947	0.958	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.45e-03	0.0615	0.0562	0.0354	3.17e-03	9.18e-03	2.59e-03	0.0324	0.0713	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.10e-03	0.0105	0.0155	0.0312	2.46e-03	0.0144	1.20e-03	0.0168	0.0306	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0178	0.271	0.176	0.687	0.0244	0.038	0.015	5.32	22.8	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0735	0.758	2.39	0.291	0.154	1.38	0.0729	2.84	71.2	0	Complete
3 ANN10 + P-only DeepONet10	1	0.442	9.16	7.33	2.37	0.611	1.21	0.669	2.18	222	0	Complete
4 T/P DeepONet10	0.05	0.322	0.704	9.31	2.55	0.187	3	0.335	0.957	0.25	0	Complete
4 T/P DeepONet10	0.1	0.0346	1.39	0.0866	1.72	0.0907	0.888	0.0236	1.81	1.59	0	Complete
4 T/P DeepONet10	0.2	9.64e-03	0.125	0.13	0.16	4.21e-03	0.0542	0.0114	0.201	0.261	0	Complete
4 T/P DeepONet10	0.3	0.0171	0.374	0.0766	0.363	9.42e-03	0.0218	0.0223	0.206	0.258	0	Complete
4 T/P DeepONet10	0.5	0.105	0.614	6.11	0.771	0.0167	0.312	0.0605	0.542	1.53	0	Complete
4 T/P DeepONet10	1	0.166	5.74	0.509	0.823	0.413	0.0268	0.321	0.425	26.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.264	0.543	6.59	1.85	0.213	3.06	0.297	0.646	0.158	0	Complete
5 ANN10 + T/P DeepONet10	0.1	7.11e-03	0.735	0.0572	0.82	0.0782	0.756	7.05e-03	0.912	1.02	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.61e-03	0.0822	0.0472	0.108	3.10e-03	0.0326	6.78e-03	0.192	0.279	0	Complete
5 ANN10 + T/P DeepONet10	0.3	9.30e-03	0.278	0.114	0.247	7.79e-03	0.0127	0.0133	0.199	0.305	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0672	2.64	0.876	0.508	0.152	0.134	0.144	0.446	0.847	0	Complete
5 ANN10 + T/P DeepONet10	1	0.249	8.61	0.603	0.85	0.618	0.208	0.494	0.321	55.3	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.126	0.485	0.874	0.153	0.4	4.07	0.215	0.99	0.995	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.336	0.334	3.68	0.352	0.641	7.58	0.491	0.991	0.992	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0549	1.32	0.964	0.96	0.11	0.247	0.0756	6.38	3.01	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0316	0.203	0.877	0.62	8.81e-03	0.0539	0.0248	0.761	0.457	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.025	0.189	0.79	0.289	0.03	0.246	0.0237	0.941	0.972	0	Complete
6 T/P/ϕ DeepONet10	1	0.755	1.07	19.9	18.7	0.972	2.23	0.36	0.991	7.08e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.341	0.422	8.82	2.19	0.291	4.07	0.38	0.353	0.114	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.32	0.411	4.35	0.148	0.576	7.02	0.46	0.977	0.981	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0232	0.172	0.797	0.401	7.04e-03	0.0225	0.0179	0.263	0.584	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	6.16e-03	0.224	0.868	0.417	3.85e-03	0.257	0.0177	0.681	0.512	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0202	0.261	0.781	0.256	0.0326	0.202	0.0171	0.907	0.933	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.195	5.4	2.76	1.53	0.381	0.515	0.335	1.18	39.5	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 29

T_0=1127.4 K P_0=2.860 atm ϕ_0=1.850 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 29

Repaired | Cantera vs NN NRMSE | T0=1127.4K, P0=2.860atm, phi0=1.850

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0344	0.016	0.691	0.728	0.0534	0.939	0.0181	0.139	0.191	0	Complete
1 ANN10	0.1	0.014	0.0421	0.76	0.852	0.175	1.14	0.0239	0.419	0.531	0	Complete
1 ANN10	0.2	0.0202	0.0482	0.877	1.04	0.279	0.864	0.0244	0.749	2.92	0	Complete
1 ANN10	0.3	0.0172	0.0444	0.613	11.9	0.344	0.912	0.0231	1.52	9.44	0	Complete
1 ANN10	0.5	0.0209	0.0186	0.361	1.39	0.189	0.846	0.0192	0.457	0.607	0	Complete
1 ANN10	1	0.17	0.067	1.65	7.41	0.562	8.5	0.168	18.5	152	0	Complete
2 P-only DeepONet10	0.05	0.0231	0.0449	0.616	1.65	0.386	0.965	0.0317	1.12	3.56	0	Complete
2 P-only DeepONet10	0.1	0.0441	0.0433	0.931	0.985	0.41	0.923	0.0326	1.21	2.23	0	Complete
2 P-only DeepONet10	0.2	0.0217	0.059	0.216	0.711	0.66	0.13	0.0442	1.92	4.67	0	Complete
2 P-only DeepONet10	0.3	0.444	0.624	0.857	137	3.9	10.6	0.606	3.44	17.8	0	Complete
2 P-only DeepONet10	0.5	0.647	0.765	1.77	24.2	7.48	17.6	0.871	1	309	0	Complete
2 P-only DeepONet10	1	0.582	1.21	1	1	14.4	1	1	43.3	23.8	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0244	0.0511	0.646	1.76	0.269	2.29	0.0468	0.707	1.11	0	Complete
3 ANN10 + P-only DeepONet10	0.1	6.97e-03	0.0254	0.152	0.517	0.265	0.103	0.0174	0.71	1.1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	8.84e-03	0.0305	0.156	0.556	0.325	0.0802	0.0213	0.931	1.56	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.99e-03	0.0103	0.0898	0.696	0.0827	0.165	7.04e-03	0.332	0.679	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.111	0.146	0.617	29.6	0.295	7.24	0.172	0.946	84.8	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0266	0.0399	0.2	7.41	0.388	1.06	0.0368	1.25	2.96	0	Complete
4 T/P DeepONet10	0.05	0.0457	0.0512	1	1.09	0.475	0.999	0.0351	1.27	3.02	0	Complete
4 T/P DeepONet10	0.1	0.0427	0.0603	0.97	1.05	0.574	0.985	0.0386	1.5	7.5	0	Complete
4 T/P DeepONet10	0.2	0.0194	0.0528	0.227	0.656	0.588	0.114	0.0392	1.33	1.6	0	Complete
4 T/P DeepONet10	0.3	0.0105	0.0196	0.32	0.442	0.164	0.0612	0.0108	1.03	2.98	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.664 ms
4 T/P DeepONet10	1	0.0721	0.125	0.905	7.53	1.46	0.476	0.103	1.08	12.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0356	0.0559	0.989	1.04	0.443	0.99	0.0235	0.951	1.68	0	Complete
5 ANN10 + T/P DeepONet10	0.1	9.88e-03	0.0308	0.177	0.55	0.329	0.102	0.0217	0.858	1.49	0	Complete
5 ANN10 + T/P DeepONet10	0.2	7.16e-03	0.0258	0.143	0.512	0.272	0.0593	0.0178	0.997	2.94	0	Complete
5 ANN10 + T/P DeepONet10	0.3	6.54e-03	0.0197	0.228	0.44	0.18	0.117	0.012	1.57	6.55	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.115	0.17	1.04	15.1	1.8	3.63	0.161	4.59	16.1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.176	0.328	0.301	5.85	4.03	0.474	0.282	1.01	27.9	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0395	0.0445	0.99	0.977	0.313	0.994	0.0196	0.737	1.97	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0363	0.0412	0.961	0.88	0.237	0.991	9.57e-03	0.539	3.57	0	Complete
6 T/P/ϕ DeepONet10	0.2	8.73e-03	0.0321	0.152	0.791	0.34	0.0881	0.0224	2.47	11.2	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.345	0.102	3.3	24.4	0.414	15.1	0.324	3.71	18.2	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.207	0.192	3.3	20.4	1.93	5.31	0.19	4.6	12.7	0	Complete
6 T/P/ϕ DeepONet10	1	0.0353	0.0754	0.341	2.1	0.878	0.166	0.06	1.01	9.59	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.425	0.301	1.04	0.867	0.215	31.2	0.567	0.558	2.57	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0178	9.10e-03	0.343	0.267	0.0662	0.5	0.0105	0.261	0.376	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	9.46e-03	0.0317	0.165	0.567	0.338	0.0871	0.0222	1.19	1.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0757	0.0993	2.52	0.923	0.238	0.646	0.0192	1.82	5.27	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0201	0.0442	0.319	0.92	0.487	0.431	0.0328	1.98	3.68	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0537	0.0982	0.27	0.801	1.24	0.205	0.0846	1	12.4	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 29

T_0=1127.4 K P_0=2.860 atm ϕ_0=1.850 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

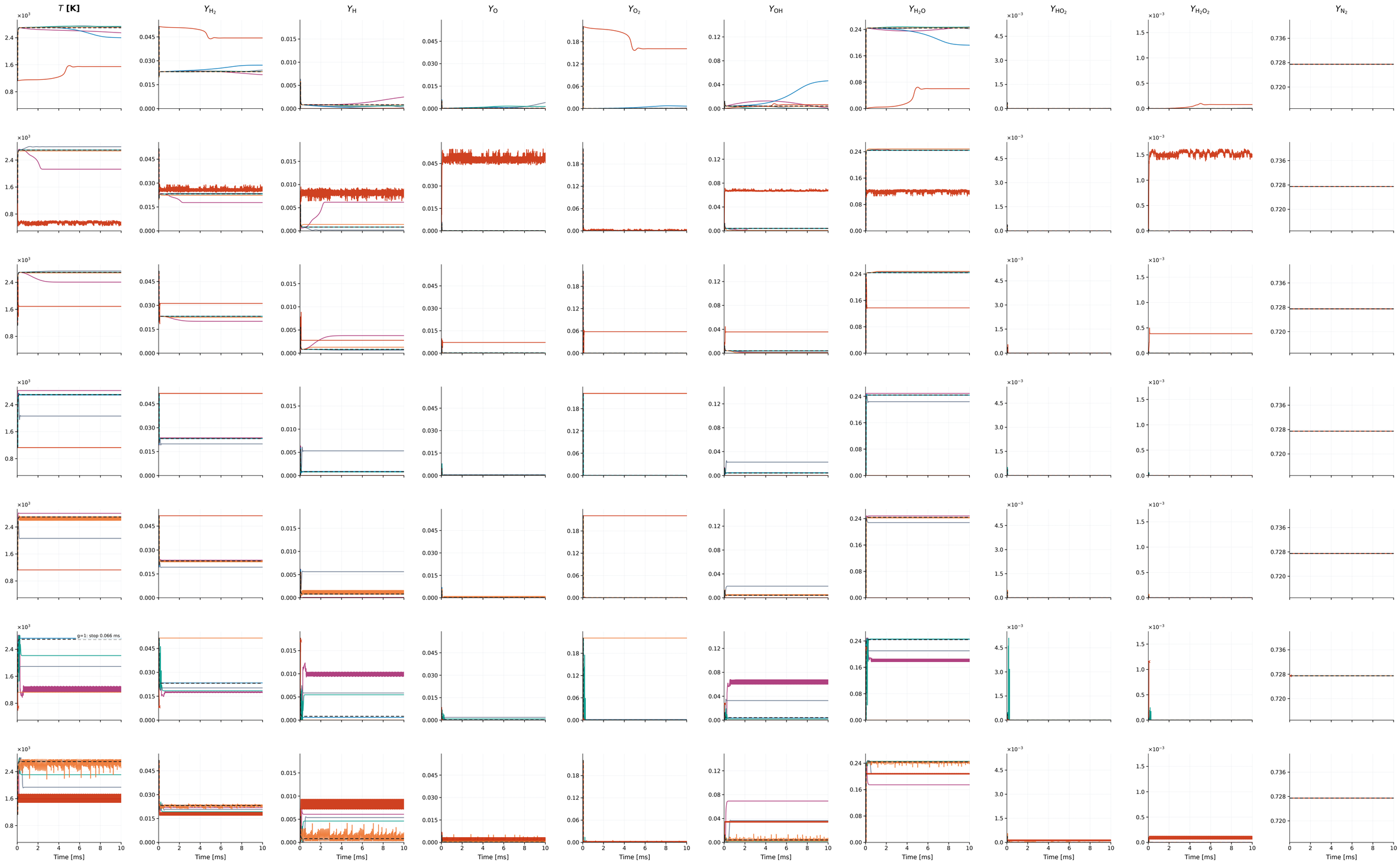
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 29

Repaired | Cantera vs NN NRMSE | T0=1127.4K, P0=2.860atm, phi0=1.850

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0337	0.032	0.841	0.57	0.049	1.36	0.0242	0.129	0.125	0	Complete
1 ANN10	0.1	0.0122	0.0184	0.375	5.39	0.0863	0.644	7.15e-03	0.416	3.57	0	Complete
1 ANN10	0.2	0.056	0.104	0.736	2.02	0.286	5.25	0.116	0.137	0.14	0	Complete
1 ANN10	0.3	0.0133	8.04e-03	0.256	4.64	0.0406	0.874	0.012	0.182	1.76	0	Complete
1 ANN10	0.5	5.17e-03	0.0162	0.103	0.35	0.168	0.125	0.0113	0.432	0.556	0	Complete
1 ANN10	1	0.493	1.03	0.969	0.689	12.1	0.52	0.857	1	77.3	0	Complete
2 P-only DeepONet10	0.05	0.189	0.208	5.77	0.662	0.0272	0.219	4.88e-03	0.143	4.61	0	Complete
2 P-only DeepONet10	0.1	0.0354	0.0247	0.744	0.867	0.195	0.778	0.0202	0.476	0.607	0	Complete
2 P-only DeepONet10	0.2	7.01e-03	0.0247	0.133	0.489	0.261	0.055	0.0172	0.676	0.867	0	Complete
2 P-only DeepONet10	0.3	6.68e-03	0.0229	0.123	0.474	0.243	0.0776	0.016	0.687	0.92	0	Complete
2 P-only DeepONet10	0.5	0.0147	0.0485	0.691	0.965	0.364	0.856	0.0294	0.932	1.04	0	Complete
2 P-only DeepONet10	1	0.8	0.132	8.83	219	0.674	15	0.511	0.98	1.93e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0964	0.121	3.18	0.701	0.0631	0.568	0.0124	0.159	0.319	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0137	0.016	0.202	0.761	0.166	0.491	0.015	0.413	0.517	0	Complete
3 ANN10 + P-only DeepONet10	0.2	5.69e-03	0.0194	0.112	0.393	0.205	0.0547	0.0134	0.511	0.68	0	Complete
3 ANN10 + P-only DeepONet10	0.3	6.28e-03	0.0224	0.121	0.459	0.237	0.0646	0.0155	0.633	0.827	0	Complete
3 ANN10 + P-only DeepONet10	0.5	9.59e-03	0.0391	0.553	0.902	0.273	0.809	0.024	0.671	0.86	0	Complete
3 ANN10 + P-only DeepONet10	1	0.373	0.345	2.35	32	3.77	7.48	0.436	1.87	487	0	Complete
4 T/P DeepONet10	0.05	0.0458	0.0333	0.98	0.995	0.259	0.998	0.026	0.71	1.6	0	Complete
4 T/P DeepONet10	0.1	0.235	0.151	5.34	1.88	0.605	4.31	0.0917	1.59	2.7	0	Complete
4 T/P DeepONet10	0.2	0.0178	0.0505	0.21	0.774	0.558	0.174	0.0372	1.57	1.77	0	Complete
4 T/P DeepONet10	0.3	0.0152	0.0453	0.198	0.863	0.495	0.109	0.0329	1.57	2.03	0	Complete
4 T/P DeepONet10	0.5	0.582	1.21	1	1	14.4	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.582	1.21	1	1	14.4	1	1	0.999	1.06	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0407	0.0231	0.874	0.848	0.0791	0.994	0.0179	0.17	0.868	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.234	0.169	5.71	0.562	0.324	3.62	0.0691	0.96	2.3	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0113	0.0365	0.174	0.673	0.393	0.132	0.026	1.11	1.39	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0121	0.0383	0.174	0.678	0.414	0.0877	0.0274	1.2	1.48	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0307	0.063	0.542	1.89	0.689	0.21	0.0464	1.78	2.72	0	Complete
5 ANN10 + T/P DeepONet10	1	0.582	1.21	1	0.996	14.4	0.999	1	1	1.04	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.541	0.24	10.7	0.987	0.275	13.8	0.251	0.838	6.74	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.298	0.134	5.99	7.62	0.546	6.76	0.143	1.41	2.43	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0158	0.0376	0.325	0.689	0.396	0.153	0.0262	1.12	1.46	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.179	0.212	5.46	3.18	0.842	0.602	0.0577	11.2	8.18	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.582	1.21	1	1	14.4	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.407	0.0598	6.24	1.87	0.142	15.4	0.283	0.744	3.55	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.275	0.108	5.31	2.78	0.291	7.52	0.143	0.812	1.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0102	0.0303	0.236	0.521	0.314	0.0696	0.0206	0.844	1.17	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.144	0.172	4.47	2.59	0.57	0.603	0.039	1.5	1.62	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0451	0.0663	0.993	4.74	0.671	0.382	0.0459	2.37	3.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.413	0.235	9.11	9.18	0.516	7.23	0.152	11.2	122	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 33

T_0=1254.9 K P_0=4.275 atm ϕ_0=0.886 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 33

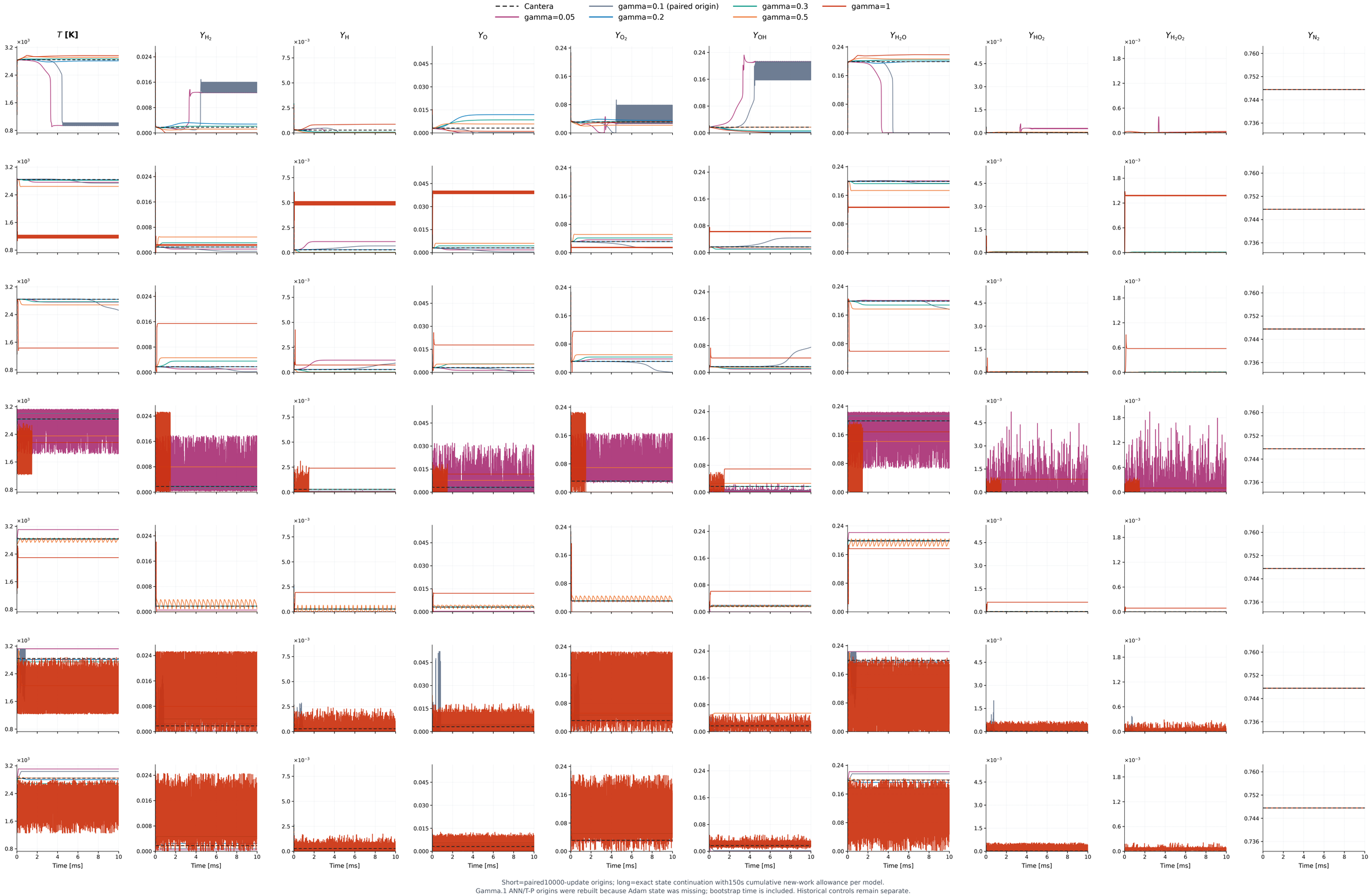
Repaired | Cantera vs NN NRMSE | T0=1254.9K, P0=4.275atm, phi0=0.886

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.131	0.963	2.44	1.36	0.338	2.54	0.187	4.99	16.6	0	Complete
1 ANN10	0.1	0.337	3.31	4.41	0.762	0.647	5.16	0.511	1.05	3.44	0	Complete
1 ANN10	0.2	0.412	6.07	0.92	0.931	2.1	3.69	0.691	0.838	0.91	0	Complete
1 ANN10	0.3	0.0126	0.406	0.685	0.282	0.0981	0.0981	0.0317	0.411	0.838	0	Complete
1 ANN10	0.5	0.0538	0.883	1.27	0.746	0.0733	0.827	0.0997	0.945	0.988	0	Complete
1 ANN10	1	0.0337	0.875	2.15	0.303	0.0753	0.924	0.0918	0.935	20.1	0	Complete
2 P-only DeepONet10	0.05	0.118	0.971	0.825	1.59	0.286	2.08	0.168	6.29	7.15	0	Complete
2 P-only DeepONet10	0.1	0.25	0.209	5.59	0.962	0.757	4.56	0.261	1.22	7.02	0	Complete
2 P-only DeepONet10	0.2	0.172	2.25	0.85	4.59	0.371	0.821	0.227	16.5	138	0	Complete
2 P-only DeepONet10	0.3	0.239	0.339	8.39	1.15	0.221	2.09	0.174	8.86	215	0	Complete
2 P-only DeepONet10	0.5	0.0276	1.26	0.985	0.201	0.659	0.532	0.0747	0.527	0.593	0	Complete
2 P-only DeepONet10	1	0.418	6.93	4.26	2.14	3.44	1.09	0.647	1.89	358	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.197	1.36	0.717	0.861	0.84	4.68	0.301	17.9	23.8	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.226	0.12	3.12	0.96	0.879	4.92	0.272	0.393	4.91	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0393	0.778	0.665	0.803	0.216	0.167	0.0677	7.02	46.8	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.177	0.0536	5	0.162	0.165	2.08	0.163	2.33	113	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0293	1.25	0.977	0.296	0.639	0.503	0.0755	0.57	0.588	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0102	0.683	0.233	0.983	0.672	0.958	0.0202	0.476	104	0	Complete
4 T/P DeepONet10	0.05	0.222	5.3	0.959	0.955	2.82	0.998	0.424	0.941	1.07	0	Complete
4 T/P DeepONet10	0.1	0.0711	0.348	0.77	0.654	0.11	0.973	0.0838	0.965	0.987	0	Complete
4 T/P DeepONet10	0.2	7.40e-03	0.16	0.143	0.105	0.0833	0.0314	0.0131	0.295	0.19	0	Complete
4 T/P DeepONet10	0.3	5.62e-03	0.158	0.134	0.0829	0.0864	0.0245	0.0122	0.288	0.155	0	Complete
4 T/P DeepONet10	0.5	0.0176	0.9	1.07	1.61	0.69	0.644	0.0382	13.2	13.5	0	Complete
4 T/P DeepONet10	1	0.465	9.47	0.893	1.57	4.59	0.716	0.796	0.998	70.7	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0356	1.8	0.983	0.971	1.15	0.998	0.105	0.636	1.51	0	Complete
5 ANN10 + T/P DeepONet10	0.1	4.03e-03	0.0807	0.101	0.0391	0.0435	0.0296	6.75e-03	0.2	0.127	0	Complete
5 ANN10 + T/P DeepONet10	0.2	4.74e-03	0.118	0.132	0.0768	0.0632	0.0232	9.27e-03	0.305	0.201	0	Complete
5 ANN10 + T/P DeepONet10	0.3	5.49e-03	0.14	0.134	0.0715	0.0757	0.0237	0.011	0.296	0.154	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0416	1.01	1.04	0.484	0.598	0.459	0.0745	0.91	0.926	0	Complete
5 ANN10 + T/P DeepONet10	1	0.14	2.28	2.75	0.794	1.12	0.633	0.211	3.56	14.1	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.208	0.737	5.08	0.812	0.42	2.57	0.211	4.12	3.85	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0834	0.494	0.929	0.803	0.0847	0.987	0.0992	0.873	0.938	0	Complete
6 T/P/ϕ DeepONet10	0.2	5.55e-03	0.156	0.233	0.089	0.083	0.0572	0.0121	0.278	0.153	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.25	2.55	0.142	4.35	0.282	2.18	0.325	0.323	2.53	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0175	0.304	0.315	0.228	0.132	0.126	0.0278	0.493	0.359	0	Complete
6 T/P/ϕ DeepONet10	1	0.271	4.89	3.5	1.51	2.46	0.711	0.436	7.89	9.84	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0489	1.32	0.979	0.988	0.865	0.997	0.0854	0.773	0.721	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	6.72e-03	0.0854	0.0955	0.0382	0.0465	0.124	0.0105	0.257	0.27	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.52e-03	0.128	0.227	0.091	0.0638	0.0645	0.0102	0.29	0.148	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.065	0.845	0.835	1.41	0.076	0.754	0.0982	0.679	0.187	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.321	4.53	0.782	0.74	1.59	2.58	0.511	0.98	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.255	4.39	3.12	0.969	2.15	0.776	0.407	6.78	18.3	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 33

T_0=1254.9 K P_0=4.275 atm ϕ_0=0.886 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 33

Repaired | Cantera vs NN NRMSE | T0=1254.9K, P0=4.275atm, phi0=0.886

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.548	4.53	0.807	0.89	0.318	9.39	0.818	10.8	17.9	0	Complete
1 ANN10	0.1	0.488	4.89	0.798	0.785	0.768	7.22	0.747	0.471	0.457	0	Complete
1 ANN10	0.2	0.0131	0.526	0.713	2.4	0.153	0.717	9.29e-03	0.761	2.45	0	Complete
1 ANN10	0.3	8.82e-03	0.151	0.647	1.48	0.0334	0.544	0.0202	0.818	1.72	0	Complete
1 ANN10	0.5	0.0247	0.322	0.843	0.833	0.298	0.0123	0.0392	1.48	5.99	0	Complete
1 ANN10	1	0.0385	0.863	1.84	0.609	0.0901	0.805	0.0891	0.873	13.1	0	Complete
2 P-only DeepONet10	0.05	0.0255	0.316	2.71	0.399	0.154	0.305	7.49e-03	0.821	1.01	0	Complete
2 P-only DeepONet10	0.1	0.0225	0.503	0.963	0.618	0.363	0.994	0.0176	0.329	0.542	0	Complete
2 P-only DeepONet10	0.2	2.02e-03	0.0463	0.0877	0.0341	0.0221	0.0362	3.54e-03	0.0515	0.106	0	Complete
2 P-only DeepONet10	0.3	7.87e-03	0.653	0.812	0.401	0.316	0.378	0.0304	2.51	8.8	0	Complete
2 P-only DeepONet10	0.5	0.0688	1.58	0.904	0.917	0.628	0.05	0.126	1.78	2.61	0	Complete
2 P-only DeepONet10	1	0.578	0.361	16.2	11.4	0.505	2.67	0.363	2.04	837	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0258	0.342	3	0.522	0.197	0.399	0.0103	0.571	0.722	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0385	0.408	0.926	0.509	0.407	1.31	0.0389	0.39	0.476	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.37e-03	0.0424	0.0682	0.0333	0.0206	0.0308	2.99e-03	0.058	0.0964	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0239	0.84	0.785	0.716	0.358	0.294	0.0511	1.86	6.13	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0563	1.41	0.916	0.708	0.578	0.0972	0.109	1.42	2.32	0	Complete
3 ANN10 + P-only DeepONet10	1	0.493	6.87	1.67	4.65	2.63	1.51	0.699	1.73	348	0	Complete
4 T/P DeepONet10	0.05	0.167	3.72	0.96	1.96	1.91	0.945	0.299	25.3	164	0	Complete
4 T/P DeepONet10	0.1	0.056	0.341	0.647	0.343	0.012	0.672	0.0691	0.796	0.834	0	Complete
4 T/P DeepONet10	0.2	1.40e-03	0.0274	0.086	0.0366	8.04e-03	3.98e-03	1.65e-03	0.0735	0.0475	0	Complete
4 T/P DeepONet10	0.3	1.98e-03	0.0331	0.0882	0.0381	0.018	6.59e-03	2.64e-03	0.113	0.064	0	Complete
4 T/P DeepONet10	0.5	0.172	3.13	0.954	1.43	1.19	0.528	0.289	0.628	1.16	0	Complete
4 T/P DeepONet10	1	0.25	2.36	6.93	2.62	1.41	2.83	0.242	37	58.5	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0923	0.616	0.97	0.863	0.0349	0.994	0.112	0.927	0.914	0	Complete
5 ANN10 + T/P DeepONet10	0.1	7.49e-03	0.0265	0.0906	0.041	0.0217	0.152	9.89e-03	0.196	0.225	0	Complete
5 ANN10 + T/P DeepONet10	0.2	9.79e-04	0.021	0.0283	0.0333	9.85e-03	5.34e-03	1.54e-03	0.0604	0.0437	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.07e-03	0.0287	0.0659	0.0353	0.0115	6.35e-03	1.87e-03	0.0678	0.0258	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0213	0.672	0.798	0.291	0.269	0.116	0.0479	0.538	0.299	0	Complete
5 ANN10 + T/P DeepONet10	1	0.192	0.99	5.82	2.83	0.989	2.57	0.119	29.3	53.7	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0985	0.726	0.974	0.924	0.0822	0.996	0.122	0.935	0.959	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0348	0.58	0.429	0.903	0.311	0.267	0.0509	2.53	7.32	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0117	0.325	0.731	0.222	0.0259	0.239	0.0303	0.0971	0.519	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0346	1.12	0.989	0.467	0.484	0.201	0.0778	0.517	0.369	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.272	3.08	0.365	3.39	0.654	2.17	0.375	0.739	2.34	0	Complete
6 T/P/ϕ DeepONet10	1	0.298	5.9	1.88	1.59	2.94	0.781	0.512	10.2	30.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0933	0.666	0.964	0.732	0.072	0.994	0.117	0.934	0.971	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0674	0.43	0.744	0.246	0.0251	0.851	0.0865	0.8	0.802	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0136	0.341	0.657	0.293	0.0407	0.207	0.0313	0.0788	0.434	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0558	1.45	0.981	0.755	0.599	0.143	0.109	0.421	0.314	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	5.58e-03	0.158	0.196	0.0639	0.0708	0.0326	0.0118	0.263	0.154	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.268	5.25	1.33	1.13	2.56	0.722	0.464	9.3	32.8	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 41

T_0=1339.8 K P_0=1.702 atm ϕ_0=1.947 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 41

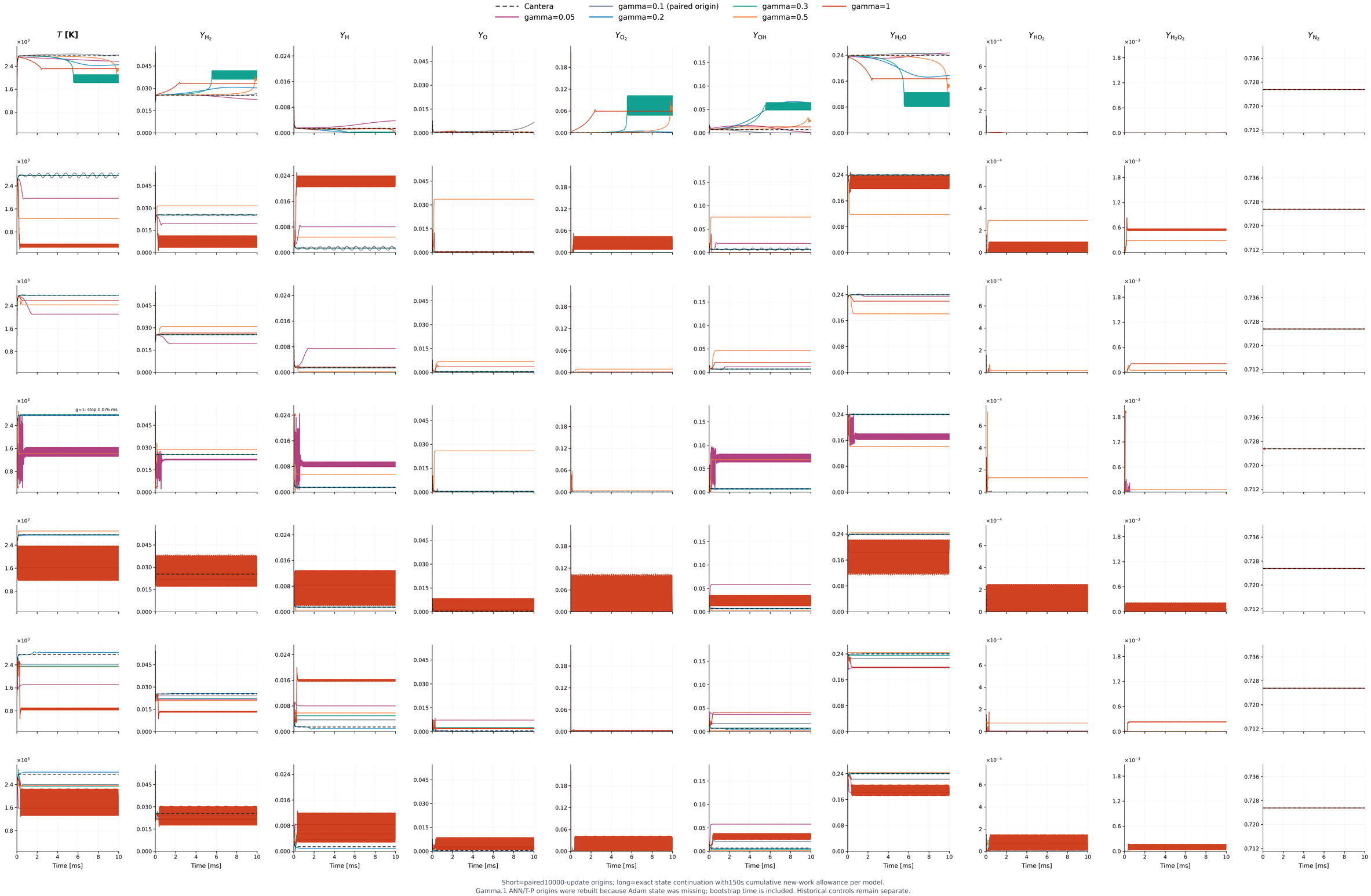
Repaired | Cantera vs NN NRMSE | T0=1339.8K, P0=1.702atm, phi0=1.947

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0231	0.0292	0.355	0.41	0.0344	0.843	0.0282	0.124	0.549	0	Complete
1 ANN10	0.1	0.0198	0.0563	0.663	2.67	0.11	0.561	0.0247	0.105	0.395	0	Complete
1 ANN10	0.2	0.0733	0.163	0.776	3.29	0.97	4.3	0.179	0.153	0.689	0	Complete
1 ANN10	0.3	0.0235	5.70e-03	0.214	2.23	0.0495	0.902	0.0238	13.3	334	0	Complete
1 ANN10	0.5	0.0242	1.63e-03	0.209	1.13	0.0515	0.761	0.0226	0.227	0.736	0	Complete
1 ANN10	1	0.159	0.0589	0.754	2.45	0.442	5.56	0.18	55.8	786	0	Complete
2 P-only DeepONet10	0.05	0.0713	0.134	0.956	13.1	0.368	2.77	0.126	0.29	1.03	0	Complete
2 P-only DeepONet10	0.1	7.70e-03	0.0238	0.278	0.634	0.0551	0.266	0.0114	0.201	0.58	0	Complete
2 P-only DeepONet10	0.2	0.0132	5.61e-03	0.153	0.4	0.0728	0.162	7.35e-03	0.178	0.145	0	Complete
2 P-only DeepONet10	0.3	6.46e-03	3.71e-03	0.0642	0.325	0.0676	0.0878	5.01e-03	0.206	1.01	0	Complete
2 P-only DeepONet10	0.5	3.13e-03	0.0151	0.125	0.554	0.104	0.252	0.0116	0.354	0.805	0	Complete
2 P-only DeepONet10	1	0.0225	0.0292	0.322	0.865	0.497	0.198	0.0194	1.06	4.68	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0392	0.113	0.947	10.8	0.353	1.6	0.0827	0.187	0.453	0	Complete
3 ANN10 + P-only DeepONet10	0.1	3.57e-03	0.0154	0.165	0.504	0.0385	0.2	8.75e-03	0.12	0.515	0	Complete
3 ANN10 + P-only DeepONet10	0.2	6.81e-03	2.71e-03	0.0655	0.261	0.0518	0.113	5.03e-03	0.12	0.186	0	Complete
3 ANN10 + P-only DeepONet10	0.3	3.31e-03	1.04e-03	0.0226	0.135	0.0151	0.0754	3.16e-03	0.0821	0.176	0	Complete
3 ANN10 + P-only DeepONet10	0.5	3.79e-03	8.03e-03	0.0388	0.394	0.0374	0.202	8.66e-03	0.0739	0.727	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0169	0.026	0.234	0.865	0.474	0.155	0.0185	1.4	101	0	Complete
4 T/P DeepONet10	0.05	0.487	0.196	5.69	1.87	0.13	7.73	0.247	4.94	25	0	Complete
4 T/P DeepONet10	0.1	0.0222	0.0121	0.282	0.374	0.124	0.272	0.0111	0.343	0.368	0	Complete
4 T/P DeepONet10	0.2	4.03e-03	0.0108	0.0682	0.166	0.198	0.0546	7.46e-03	0.524	0.371	0	Complete
4 T/P DeepONet10	0.3	0.661	0.298	2.94	87.5	2.3	8.52	0.577	64.6	4.07	0	Complete
4 T/P DeepONet10	0.5	0.558	0.303	1.76	93.4	0.5	8.45	0.515	148	790	0	Complete
4 T/P DeepONet10	1	0.0847	0.0624	0.581	1.82	2.01	0.211	0.0914	0.999	56.7	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.37	0.0152	2.8	0.54	0.0972	10.5	0.327	1.47	8.51	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0142	7.68e-03	0.176	0.181	0.0887	0.198	7.60e-03	0.231	0.241	0	Complete
5 ANN10 + T/P DeepONet10	0.2	1.75e-03	6.91e-03	0.0399	0.0973	0.125	0.0338	4.67e-03	0.35	0.269	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.36	0.418	0.606	56.9	1.93	9.23	0.507	41.1	80	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.331	0.298	0.207	50.1	0.276	9.6	0.426	78.6	322	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0659	0.0619	0.4	1.45	1.79	0.118	0.0784	1.05	52.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.456	0.157	5.08	3.35	0.19	7.78	0.252	4.32	17.8	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0223	0.0151	0.283	0.384	0.21	0.282	0.0129	0.451	0.382	0	Complete
6 T/P/ϕ DeepONet10	0.2	7.03e-03	0.0221	0.226	0.207	0.259	0.232	0.0123	0.571	0.43	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.598	0.379	8.23	1.63	0.254	6.12	0.192	70.4	1.85e+03	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0306	0.0199	0.288	0.741	0.366	0.642	0.0272	0.96	0.814	0	Complete
6 T/P/ϕ DeepONet10	1	0.067	0.0501	0.954	2.29	0.471	0.467	0.0255	1.15	31.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.459	0.377	0.464	0.806	0.14	22.9	0.707	1.55	20.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	9.30e-03	8.53e-03	0.119	0.175	0.142	0.124	6.83e-03	0.323	0.217	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	9.21e-03	0.0238	0.285	0.194	0.166	0.292	0.0116	0.411	0.427	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.244	0.135	0.667	6.69	0.202	9.36	0.31	11.9	49.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0277	0.0159	0.241	0.711	0.296	0.623	0.0252	0.756	0.726	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.057	0.0465	0.823	1.99	0.519	0.369	0.0239	1.25	34.6	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 41

T_0=1339.8 K P_0=1.702 atm ϕ_0=1.947 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 41

Repaired | Cantera vs NN NRMSE | T0=1339.8K, P0=1.702atm, phi0=1.947

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0468	0.0526	0.835	0.634	0.0315	0.862	0.0258	0.0772	0.244	0	Complete
1 ANN10	0.1	0.0154	0.0101	0.169	3.18	0.051	0.753	0.0161	0.488	11.1	0	Complete
1 ANN10	0.2	0.0791	0.147	0.727	0.548	0.318	5.7	0.188	0.112	0.643	0	Complete
1 ANN10	0.3	0.186	0.341	0.641	0.568	5.32	5.37	0.37	0.107	0.71	0	Complete
1 ANN10	0.5	0.0424	0.0816	0.129	0.287	1.44	0.914	0.0839	0.0652	3.89	0	Complete
1 ANN10	1	0.147	0.284	0.156	0.747	6.27	0.757	0.274	0.881	5	0	Complete
2 P-only DeepONet10	0.05	0.28	0.227	4.4	0.13	0.0369	1.76	0.0545	0.566	4.13	0	Complete
2 P-only DeepONet10	0.1	0.0207	0.0126	0.279	0.464	0.0494	0.234	9.69e-03	0.19	0.789	0	Complete
2 P-only DeepONet10	0.2	6.58e-04	2.44e-03	0.0127	0.0689	0.0404	0.0311	1.86e-03	0.135	0.262	0	Complete
2 P-only DeepONet10	0.3	2.50e-03	2.57e-03	0.0296	0.0544	0.0459	0.0417	2.25e-03	0.158	0.15	0	Complete
2 P-only DeepONet10	0.5	0.538	0.238	2.3	61.6	1.11	9.74	0.504	83.7	1.94e+03	0	Complete
2 P-only DeepONet10	1	0.875	0.687	13.9	1.76	3.43	1.03	0.111	16.3	3.73e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.224	0.215	3.84	0.531	0.0352	0.654	0.018	0.227	3.15	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.51e-03	3.11e-03	0.0341	0.0901	0.0243	0.0437	2.01e-03	0.0812	0.14	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.15e-04	1.43e-03	8.27e-03	0.0796	0.0161	0.0309	1.40e-03	0.0569	0.173	0	Complete
3 ANN10 + P-only DeepONet10	0.3	8.20e-04	9.56e-04	9.84e-03	0.0371	0.0157	0.0205	8.63e-04	0.0868	0.224	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.119	0.21	0.821	11.7	0.976	5.53	0.241	4.73	348	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0664	0.0439	0.174	5.63	0.225	1.98	0.081	1.12	1.38e+03	0	Complete
4 T/P DeepONet10	0.05	0.452	0.17	5.03	0.95	0.0775	8.72	0.267	0.314	99.6	0	Complete
4 T/P DeepONet10	0.1	3.53e-03	1.59e-03	0.0412	0.0613	0.0175	0.0546	2.05e-03	0.144	0.116	0	Complete
4 T/P DeepONet10	0.2	6.51e-03	2.59e-03	0.079	0.172	0.0158	0.0751	3.19e-03	0.137	0.162	0	Complete
4 T/P DeepONet10	0.3	5.28e-03	2.91e-03	0.0408	0.199	0.0522	0.117	5.06e-03	0.212	0.19	0	Complete
4 T/P DeepONet10	0.5	0.483	0.126	2.78	47.3	0.42	8.76	0.407	37.4	437	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
5 ANN10 + T/P DeepONet10	0.05	0.415	0.143	4.62	2.03	0.0403	7.32	0.233	1.03	8.26	0	Complete
5 ANN10 + T/P DeepONet10	0.1	1.92e-03	1.55e-03	0.0285	0.0266	0.0148	0.017	8.45e-04	0.0714	0.0815	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.65e-03	1.56e-03	0.062	0.178	0.0138	0.0774	3.31e-03	0.0852	0.169	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.47e-03	2.28e-03	8.41e-03	0.1	0.0266	0.0605	2.63e-03	0.114	0.0957	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0464	0.0263	0.623	0.793	0.0578	0.538	0.0179	0.358	0.782	0	Complete
5 ANN10 + T/P DeepONet10	1	0.345	0.212	3.95	6.97	4.64	2.51	0.242	27.1	603	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.38	0.141	4.46	12.6	0.345	4.24	0.177	1.47	1.95	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.122	0.0512	1.5	2.39	0.129	1.49	0.0573	0.374	0.718	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0237	0.0126	0.317	0.427	0.027	0.233	9.18e-03	0.138	0.249	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.146	0.12	2.36	4.01	0.16	0.0697	0.0176	0.679	0.595	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.156	0.179	2.96	2.55	0.339	0.742	0.0214	22.3	35.9	0	Complete
6 T/P/ϕ DeepONet10	1	0.677	0.468	9.72	2.83	0.597	4.84	0.169	2.45	1.55e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.427	0.139	4.7	4.96	0.0509	7.23	0.239	1.03	3.95	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.129	0.046	1.49	1.65	0.0783	2.01	0.0697	0.218	0.246	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0253	0.019	0.391	0.399	0.0198	0.118	5.32e-03	0.0879	0.138	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.15	0.152	2.67	3.15	0.0635	0.408	3.94e-03	0.749	0.299	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.143	0.161	2.69	3.21	0.239	0.738	0.0181	19.9	20.9	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.401	0.211	5.11	7.61	1.99	3.85	0.19	24.7	666	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 46

T_0=1002.9 K P_0=1.808 atm ϕ_0=0.789 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 46

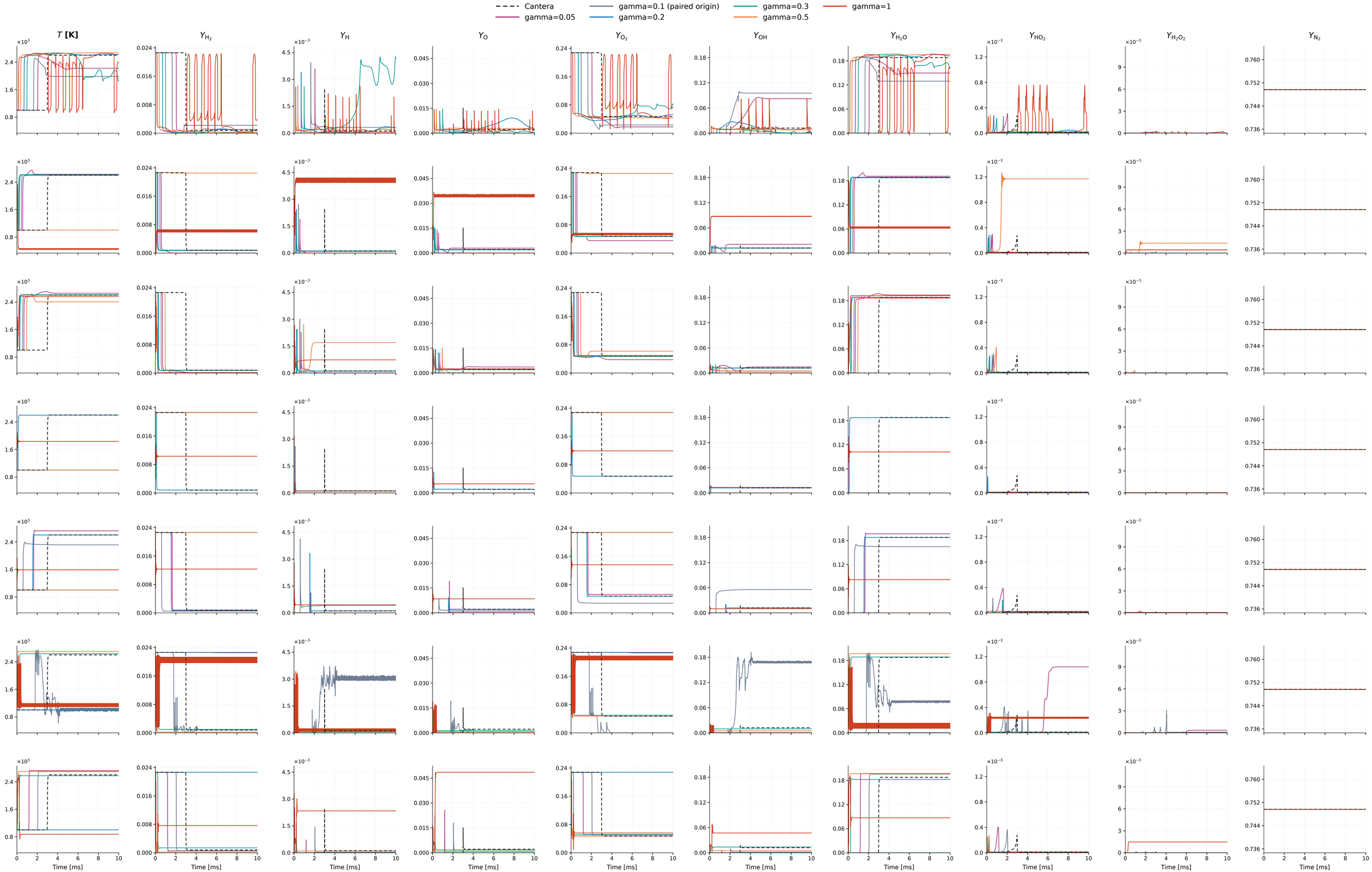
Repaired | Cantera vs NN NRMSE | T0=1002.9K, P0=1.808atm, phi0=0.789

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.237	0.496	0.87	1.31	0.421	4.89	0.383	1.12	1.54	0	Complete
1 ANN10	0.1	0.291	0.712	2.26	0.994	0.607	5.99	0.49	1.16	1.1	0	Complete
1 ANN10	0.2	0.656	1.44	0.97	6.18	1.02	0.588	0.995	2.08	5.57	0	Complete
1 ANN10	0.3	0.349	0.86	1.35	0.952	0.669	1.75	0.586	1.09	1.01	0	Complete
1 ANN10	0.5	0.596	1.47	0.997	1	1.15	1	1	1	1	0	Complete
1 ANN10	1	0.596	1.47	0.959	0.983	1.15	0.962	1	0.998	0.995	0	Complete
2 P-only DeepONet10	0.05	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
2 P-only DeepONet10	0.1	0.336	0.813	8.18	0.971	0.655	4.14	0.537	1.28	0.979	0	Complete
2 P-only DeepONet10	0.2	0.392	0.917	1.16	1.07	0.7	0.719	0.639	1.07	1	0	Complete
2 P-only DeepONet10	0.3	0.339	0.956	3.34	0.865	0.792	3.3	0.59	1.02	0.983	0	Complete
2 P-only DeepONet10	0.5	0.331	0.974	12.7	4.38	0.818	3.85	0.573	1	1.42	0	Complete
2 P-only DeepONet10	1	0.524	1.05	8.14	9.73	0.841	4.89	0.753	2.84	20.8	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.291	0.862	2.42	1.05	0.744	4.23	0.515	1.12	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.379	0.902	1.2	0.949	0.696	0.608	0.622	1.09	1.01	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.377	0.938	1.08	0.929	0.744	1.15	0.626	1.05	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.359	0.97	2.64	0.821	0.8	2.62	0.618	1.02	1.22	0	Complete
3 ANN10 + P-only DeepONet10	1	0.363	0.976	4.1	1.03	0.761	0.983	0.639	1.04	3.69	0	Complete
4 T/P DeepONet10	0.05	0.596	1.47	1	1	1.15	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.1	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
4 T/P DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
4 T/P DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.5	0.596	1.47	1	1	1.15	1	1	0.979	0.999	0	Complete
4 T/P DeepONet10	1	0.366	0.849	2.24	3.72	0.651	0.549	0.584	1.04	2.92	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.625	1.43	1.71	4.9	1.06	0.998	0.974	12.5	1.06	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.651	1.4	7.19	0.992	1.14	0.999	1	21	101	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.315	0.942	6.38	8.62	0.871	3.97	0.563	17.3	2.32	0	Complete
5 ANN10 + T/P DeepONet10	1	0.485	1.01	3.62	6.64	0.744	0.552	0.712	0.981	5.97	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.346	0.801	0.964	1.06	0.609	1.01	0.563	1	0.986	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	1	0.343	0.984	6.62	3.14	0.785	1.37	0.628	0.864	0.966	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.339	0.783	0.985	1.1	0.596	1.01	0.55	1	0.991	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.493	1.01	3.45	6.81	0.734	0.651	0.72	0.869	3.36	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 46

T_0=1002.9 K P_0=1.808 atm ϕ_0=0.789 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 46

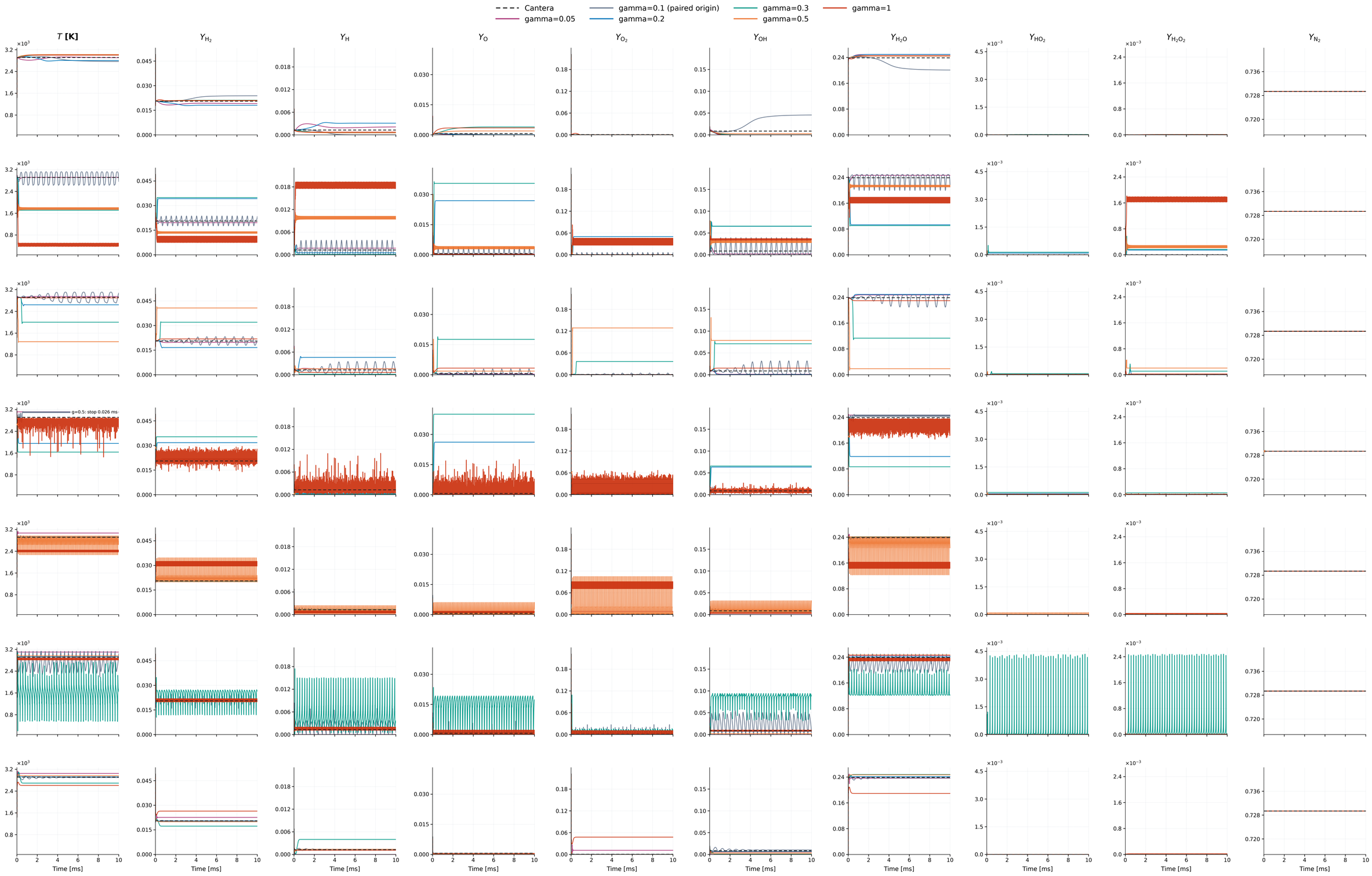
Repaired | Cantera vs NN NRMSE | T0=1002.9K, P0=1.808atm, phi0=0.789

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.247	0.544	1.53	0.997	0.463	5.45	0.404	1.4	1.87	0	Complete
1 ANN10	0.1	0.317	0.657	2.07	1	0.576	7.18	0.496	1.04	1.02	0	Complete
1 ANN10	0.2	0.34	0.864	1.45	1.75	0.686	1.3	0.572	1.2	1.05	0	Complete
1 ANN10	0.3	0.362	0.805	16.9	0.775	0.634	0.882	0.552	1.09	1.03	0	Complete
1 ANN10	0.5	0.369	0.901	0.983	0.863	0.704	0.573	0.618	1.07	1.03	0	Complete
1 ANN10	1	0.476	1.15	1.77	1.01	0.887	0.993	0.795	6.17	4.7	0	Complete
2 P-only DeepONet10	0.05	0.358	0.889	1.15	0.819	0.709	1.09	0.599	1.22	1.07	0	Complete
2 P-only DeepONet10	0.1	0.358	0.887	0.995	0.694	0.695	0.619	0.601	1.02	0.997	0	Complete
2 P-only DeepONet10	0.2	0.374	0.924	0.944	0.675	0.723	0.622	0.627	1.02	0.998	0	Complete
2 P-only DeepONet10	0.3	0.381	0.939	0.923	0.674	0.734	0.619	0.638	0.956	0.99	0	Complete
2 P-only DeepONet10	0.5	0.595	1.47	1	0.964	1.14	1	1	33.7	85.8	0	Complete
2 P-only DeepONet10	1	0.812	0.814	32.2	17.3	0.727	7.73	0.701	0.876	29.2	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.345	0.853	1.29	0.896	0.67	0.672	0.579	1.24	1.1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.35	0.867	1.04	0.651	0.679	0.598	0.588	1.02	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.37	0.914	0.957	0.67	0.716	0.621	0.62	1.02	1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.374	0.923	0.953	0.668	0.722	0.614	0.627	0.966	0.991	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.305	0.817	11.9	1.04	0.599	0.994	0.538	1.54	2.34	0	Complete
3 ANN10 + P-only DeepONet10	1	0.369	0.967	5.26	0.902	0.726	0.692	0.651	0.95	1.17	0	Complete
4 T/P DeepONet10	0.05	0.596	1.47	1	1	1.15	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.1	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
4 T/P DeepONet10	0.2	0.38	0.942	0.972	0.722	0.738	0.641	0.639	0.962	0.988	0	Complete
4 T/P DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
4 T/P DeepONet10	0.5	0.596	1.47	0.999	1	1.15	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.35	0.843	0.95	2.16	0.647	0.711	0.578	0.871	2.82	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.283	0.648	1.02	0.775	0.491	0.998	0.46	2.19	2.72	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.307	0.87	2.84	0.59	0.753	4.38	0.529	1.07	1	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.269	0.666	1.02	0.525	0.522	0.461	0.452	0.894	0.985	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.596	1.47	0.999	1	1.15	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.403	0.901	3.11	3.68	0.686	0.567	0.629	0.88	4.14	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.595	1.47	1	0.999	1.15	1	0.999	20.7	15.5	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.619	0.6	20.7	1.15	0.592	13.1	0.674	1.85	13.2	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.381	0.915	1.03	0.664	0.712	0.541	0.628	0.958	0.989	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.413	0.983	1.05	0.806	0.75	0.668	0.682	0.971	0.998	0	Complete
6 T/P/ϕ DeepONet10	1	0.55	1.36	1.99	1.26	1.07	1.02	0.922	6.99	1.09	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.329	0.755	0.959	0.802	0.572	0.995	0.535	1.9	1.38	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.233	0.543	0.866	0.515	0.415	0.844	0.382	1.43	1.46	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.596	1.47	1	1	1.15	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.368	0.9	1.04	0.657	0.711	0.723	0.611	0.965	0.986	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.411	0.979	0.921	0.754	0.748	0.657	0.679	0.971	0.998	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.65	0.823	18.1	24.3	0.719	3.8	0.63	1.01	99.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 48

T_0=1440.4 K P_0=4.726 atm ϕ_0=1.757 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 48

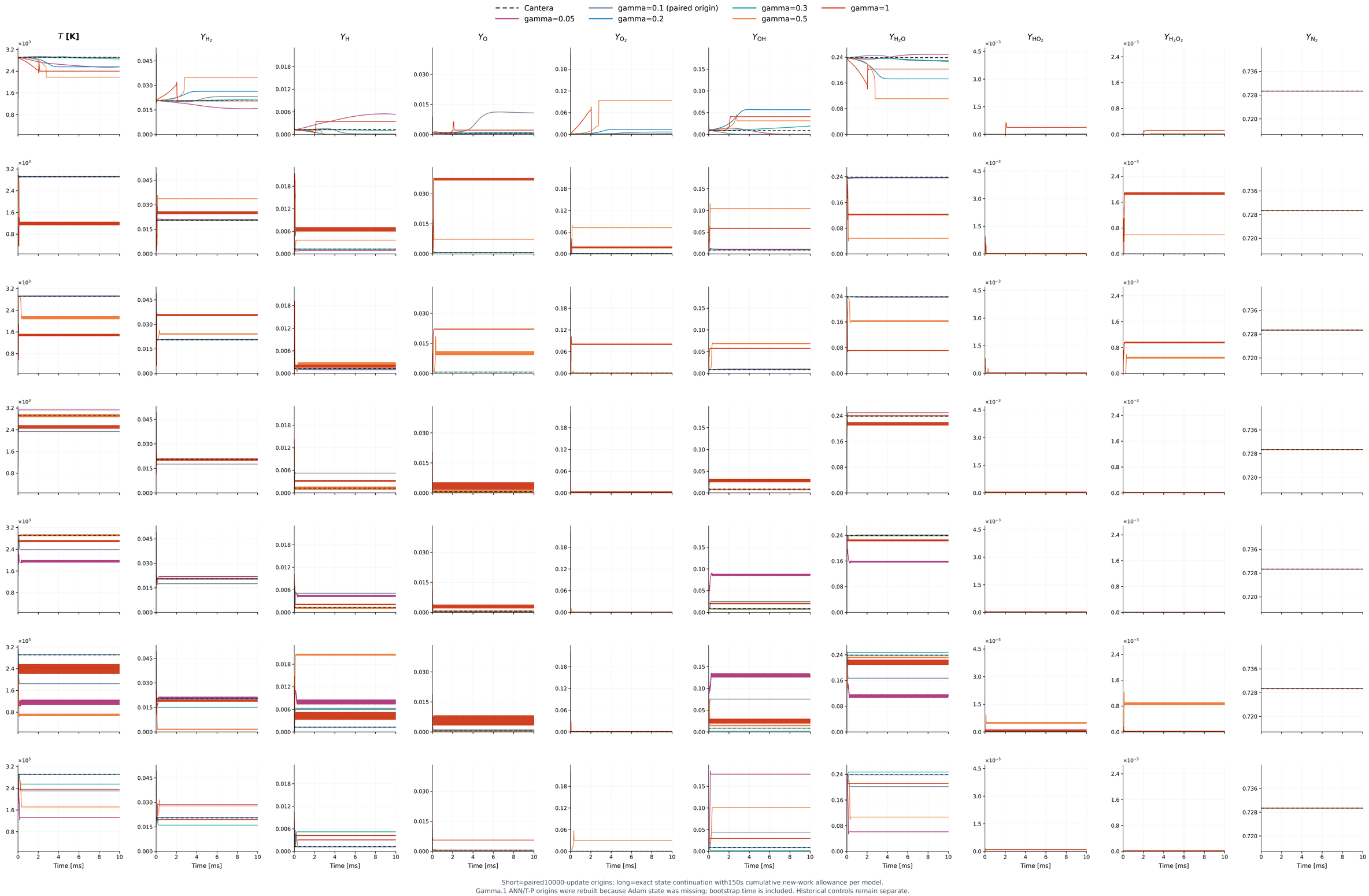
Repaired | Cantera vs NN NRMSE | T0=1440.4K, P0=4.726atm, phi0=1.757

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0139	0.0743	0.725	0.9	0.104	0.965	0.042	0.699	0.861	0	Complete
1 ANN10	0.1	0.0335	0.12	0.791	0.656	0.136	3.07	0.114	0.729	0.959	0	Complete
1 ANN10	0.2	0.034	0.109	1.27	0.83	0.139	0.958	0.0422	0.597	0.809	0	Complete
1 ANN10	0.3	0.0321	0.026	0.525	4.56	0.0836	0.917	0.0233	8.68	29.7	0	Complete
1 ANN10	0.5	0.0357	0.0213	0.538	1.9	0.144	0.638	0.0219	0.714	0.97	0	Complete
1 ANN10	1	0.0288	0.0171	0.405	4.26	0.209	0.956	0.0268	0.7	23	0	Complete
2 P-only DeepONet10	0.05	5.21e-03	0.0443	0.375	0.436	0.0547	0.752	0.0309	1.14	0.384	0	Complete
2 P-only DeepONet10	0.1	0.0671	0.081	1.01	1.78	0.433	1.47	0.0693	3.85	14.9	0	Complete
2 P-only DeepONet10	0.2	0.397	0.659	0.572	40.3	10.9	6.47	0.605	50.1	478	0	Complete
2 P-only DeepONet10	0.3	0.411	0.693	0.899	53.7	9.46	6.52	0.617	76.2	547	0	Complete
2 P-only DeepONet10	0.5	0.393	0.337	6.63	4.71	0.33	2.52	0.111	3.37	844	0	Complete
2 P-only DeepONet10	1	0.845	0.533	13.3	1.11	8.3	3.06	0.297	1.08	5.61e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0101	0.0393	0.254	0.436	0.0488	0.845	0.0345	0.499	0.486	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0439	0.0643	0.768	1.17	0.273	1.02	0.0473	2.47	10.9	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0916	0.188	2.51	0.952	0.147	0.931	0.0418	5.96	49.2	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.306	0.545	0.866	25.7	7.74	7	0.513	31.1	362	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.555	0.974	0.235	2.24	28.3	8.03	0.916	4.21	682	0	Complete
3 ANN10 + P-only DeepONet10	1	6.43e-03	0.0619	0.494	4.08	0.15	0.738	0.0382	10.1	81.7	0	Complete
4 T/P DeepONet10	0.05	0.0691	0.0444	0.996	0.976	0.433	0.999	0.0335	1.32	0.739	0	Complete
4 T/P DeepONet10	0.1	0.0629	0.052	0.956	0.949	0.777	0.981	0.0263	2.41	20.9	0	Complete
4 T/P DeepONet10	0.2	0.327	0.54	0.727	39.2	6.64	6.25	0.501	28.8	16.9	0	Complete
4 T/P DeepONet10	0.3	0.437	0.715	0.858	60.8	9.34	6.54	0.636	64.2	180	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
4 T/P DeepONet10	1	0.0747	0.14	0.89	3.95	4.74	0.334	0.0981	5.68	32.3	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.053	0.0826	0.992	0.97	1.8	0.996	4.61e-03	0.652	0.0821	0	Complete
5 ANN10 + T/P DeepONet10	0.1	8.31e-03	6.00e-03	0.119	0.148	0.0272	0.17	5.92e-03	0.228	0.189	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.51e-03	7.64e-03	0.0233	0.325	0.0754	0.163	8.59e-03	0.295	0.291	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.36e-03	0.0159	0.118	0.317	0.1	0.201	0.0104	0.208	0.0578	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0687	0.178	0.538	3.74	5.4	1.24	0.133	22.4	35.7	0	Complete
5 ANN10 + T/P DeepONet10	1	0.174	0.52	0.517	1.06	18.2	0.617	0.364	0.593	102	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0645	0.0568	0.997	0.991	0.891	0.999	0.0258	1.76	0.885	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.117	0.0453	1.13	0.732	0.597	2.77	0.11	4.81	10.6	0	Complete
6 T/P/ϕ DeepONet10	0.2	5.87e-03	9.75e-03	0.037	0.287	0.234	0.159	9.09e-03	0.721	0.233	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.474	0.266	3.99	22.7	2.21	8.81	0.448	496	2.01e+03	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0173	0.031	0.0958	0.522	0.312	0.752	0.0338	0.809	0.22	0	Complete
6 T/P/ϕ DeepONet10	1	0.0233	0.0463	0.277	1.28	1.64	0.0915	0.0321	2.34	20	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.046	0.0996	0.992	0.97	2.42	0.996	0.0107	0.616	0.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	9.28e-03	3.38e-03	0.0871	0.137	0.0452	0.251	9.19e-03	0.321	0.404	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.27e-03	0.0119	0.0569	0.324	0.077	0.214	0.0105	0.236	0.264	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0745	0.156	2.07	0.472	0.154	0.854	0.0374	1.54	0.187	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0179	0.0292	0.0851	0.506	0.178	0.752	0.0334	0.731	0.272	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.104	0.285	0.131	0.19	10.4	0.374	0.206	2.57	72.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 48

T_0=1440.4 K P_0=4.726 atm ϕ_0=1.757 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 48

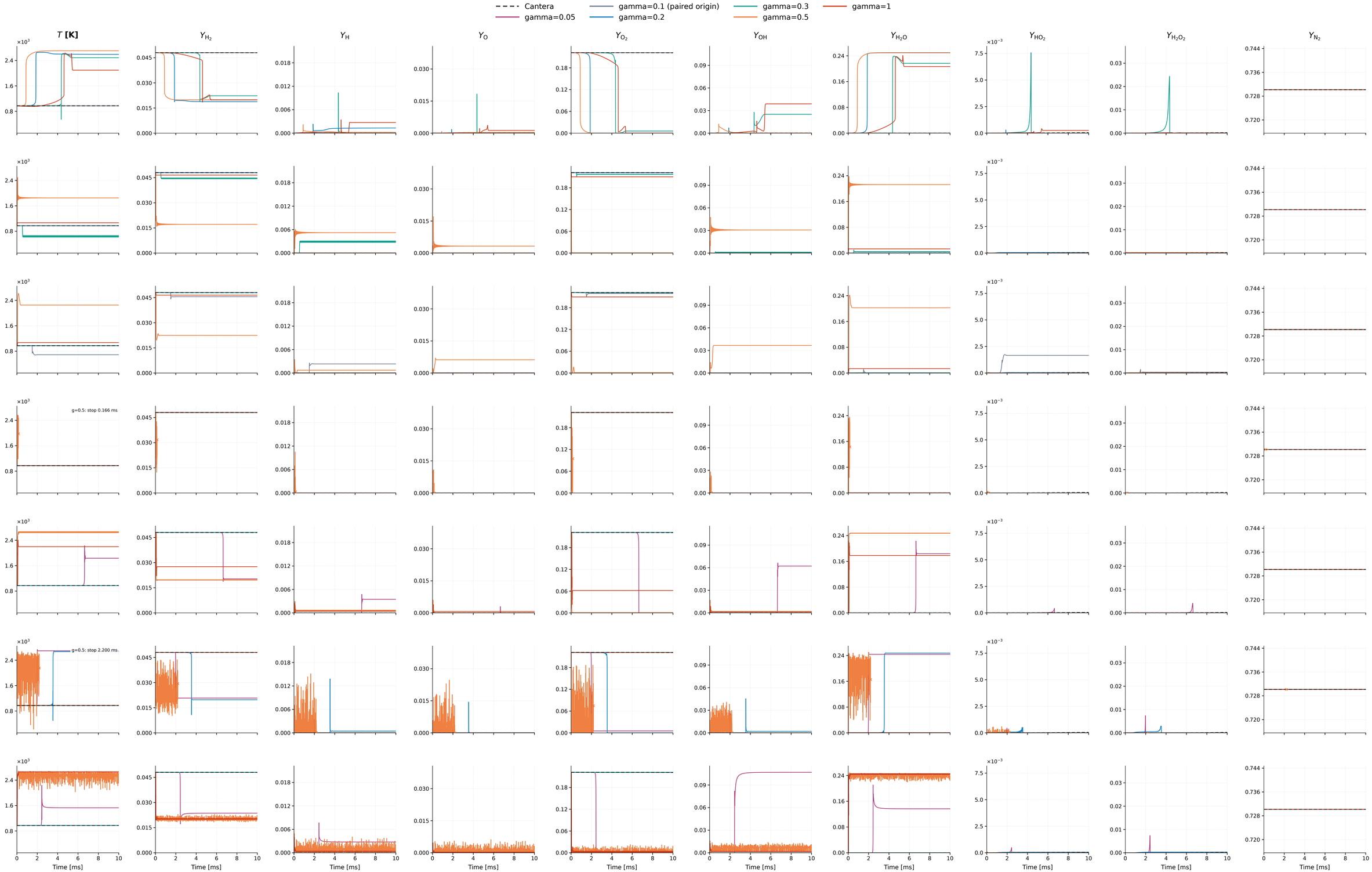
Repaired | Cantera vs NN NRMSE | T0=1440.4K, P0=4.726atm, phi0=1.757

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0956	0.167	2.37	0.863	0.129	0.708	0.0301	0.595	0.638	0	Complete
1 ANN10	0.1	5.53e-03	0.0924	0.652	11.7	0.879	0.902	0.0291	9.36	22.4	0	Complete
1 ANN10	0.2	0.101	0.24	0.867	0.306	2.41	4.74	0.234	0.343	0.217	0	Complete
1 ANN10	0.3	0.0101	0.0249	0.111	0.25	0.133	0.613	0.0257	0.453	0.961	0	Complete
1 ANN10	0.5	0.215	0.59	0.841	0.849	17.6	2.3	0.457	0.567	56	0	Complete
1 ANN10	1	0.162	0.12	1.5	2.22	3.95	3.29	0.159	183	360	0	Complete
2 P-only DeepONet10	0.05	7.17e-03	0.0256	0.284	0.235	0.081	0.237	0.0108	0.288	0.152	0	Complete
2 P-only DeepONet10	0.1	3.44e-03	3.96e-03	0.0469	0.154	0.0733	0.0606	3.11e-03	0.255	0.146	0	Complete
2 P-only DeepONet10	0.2	8.88e-04	3.28e-03	0.0173	0.107	0.0297	0.0426	2.52e-03	0.0911	0.263	0	Complete
2 P-only DeepONet10	0.3	1.54e-03	3.94e-03	0.0161	0.139	0.0459	0.0594	3.41e-03	0.335	0.98	0	Complete
2 P-only DeepONet10	0.5	0.59	0.639	1.83	10.5	15.9	10.9	0.792	18.5	1.94e+03	0	Complete
2 P-only DeepONet10	1	0.591	0.236	4.24	56.5	3.9	5.78	0.485	22.9	6.13e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	5.88e-03	0.0196	0.227	0.0996	0.0263	0.179	7.67e-03	0.129	0.224	0	Complete
3 ANN10 + P-only DeepONet10	0.1	6.10e-04	3.70e-03	0.0279	0.0839	0.035	0.0418	2.31e-03	0.0671	0.129	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.46e-04	4.03e-03	0.029	0.106	0.0273	0.0423	2.53e-03	0.0874	0.202	0	Complete
3 ANN10 + P-only DeepONet10	0.3	4.70e-04	4.14e-03	0.0289	0.123	0.0334	0.0423	2.62e-03	0.208	0.606	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.268	0.174	1.05	15.1	0.275	6.8	0.316	13.7	1.59e+03	0	Complete
3 ANN10 + P-only DeepONet10	1	0.489	0.733	0.757	33.2	17.8	5.62	0.698	22.2	3.15e+03	0	Complete
4 T/P DeepONet10	0.05	0.0751	0.03	0.999	1	0.162	0.999	0.0445	0.762	7.4	0	Complete
4 T/P DeepONet10	0.1	0.198	0.142	3.1	0.875	0.114	2.28	0.0838	10.2	19.4	0	Complete
4 T/P DeepONet10	0.2	0.0108	6.82e-03	0.0981	0.455	0.105	0.207	0.0106	0.693	0.42	0	Complete
4 T/P DeepONet10	0.3	5.46e-03	0.0108	0.107	0.304	0.0944	0.134	6.82e-03	0.723	0.225	0	Complete
4 T/P DeepONet10	0.5	0.0138	0.0166	0.228	0.574	0.289	0.155	7.75e-03	1.39	1.9	0	Complete
4 T/P DeepONet10	1	0.139	0.0234	1.51	4.65	0.308	2.21	0.0991	16.5	39.6	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.328	0.0662	2.44	0.901	0.141	8.92	0.339	1.98	10.4	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.18	0.143	2.94	0.868	0.119	1.88	0.0676	6.94	15.5	0	Complete
5 ANN10 + T/P DeepONet10	0.2	4.66e-03	9.47e-03	0.0325	0.346	0.0816	0.176	9.16e-03	0.411	0.406	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.68e-03	0.0105	0.0742	0.224	0.0843	0.14	7.25e-03	0.407	0.22	0	Complete
5 ANN10 + T/P DeepONet10	0.5	6.73e-03	0.0113	0.124	0.282	0.222	0.0671	5.28e-03	0.772	0.767	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0731	7.64e-03	0.642	3.64	0.168	1.36	0.0611	14.1	15.1	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.594	0.0337	5.23	0.989	0.165	13.8	0.525	1.8	50.4	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.363	0.0312	3.62	0.824	0.175	7.65	0.299	10.6	15.8	0	Complete
6 T/P/ϕ DeepONet10	0.2	4.15e-03	5.10e-03	0.0741	0.188	0.0726	0.0327	8.44e-04	0.145	0.0416	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.163	0.265	3.87	0.209	0.162	0.794	0.0334	4.53	1.7	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.761	0.918	15	0.808	0.373	0.908	0.0408	259	2.85e+03	0	Complete
6 T/P/ϕ DeepONet10	1	0.172	0.0663	2.21	7.42	0.172	1.85	0.0912	33.2	46.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.543	0.392	1.37	0.867	0.148	19.3	0.74	2.98	16.9	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.207	0.048	2.35	0.244	0.064	4.04	0.154	5.22	6.59	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.84e-04	5.52e-03	0.0478	0.0668	0.0495	0.074	3.32e-03	0.269	0.0778	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.123	0.214	3.04	0.0871	0.156	0.805	0.0342	2.34	1.36	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.407	0.352	1.49	0.249	6.57	10.5	0.545	1.91	83.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.19	0.0528	2.26	7.8	0.159	2.43	0.115	54.8	83.8	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 52

T_0=972.8 K P_0=4.350 atm ϕ_0=1.718 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 52

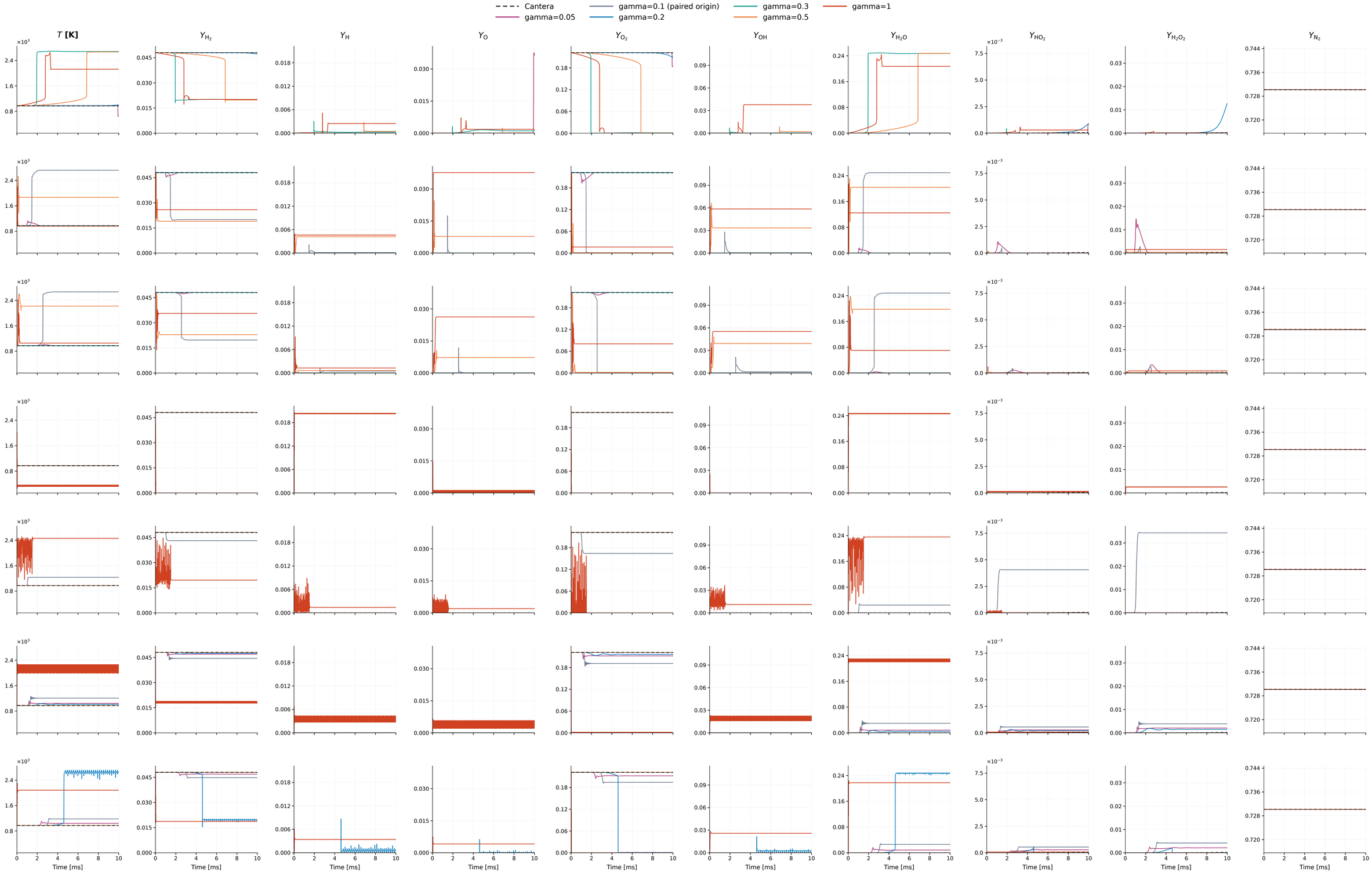
Repaired | Cantera vs NN NRMSE | T0=972.8K, P0=4.350atm, phi0=1.718

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	7.11e-04	2.65e-04	1	11.8	6.78e-04	1	1	0.999	1	0	Complete
1 ANN10	0.1	7.11e-04	2.64e-04	1	29.9	6.75e-04	1	1	0.984	0.996	0	Complete
1 ANN10	0.2	1.51	0.547	9.45e+05	5.37e+04	0.902	3.32e+05	3.23e+03	1.02	0.99	0	Complete
1 ANN10	0.3	1.18	0.406	1.96e+05	2.91e+05	0.734	1.49e+07	2.36e+03	24.5	33.8	0	Complete
1 ANN10	0.5	1.68	0.557	2.31e+05	2.46e+05	0.951	1.49e+06	3.39e+03	0.999	1	0	Complete
1 ANN10	1	0.911	0.427	1.61e+06	8.27e+05	0.729	2.19e+07	2.22e+03	7.38	1.06	0	Complete
2 P-only DeepONet10	0.05	7.11e-04	2.65e-04	1	0.738	6.77e-04	0.939	1	0.997	1	0	Complete
2 P-only DeepONet10	0.1	7.11e-04	2.65e-04	1	0.99	6.78e-04	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.2	1.87e-03	6.48e-04	1.18	4.57	1.49e-03	3.48	2.91	1.51	1.75	0	Complete
2 P-only DeepONet10	0.3	0.338	0.0709	2.52e+06	1.22e+03	0.0206	9.13e+05	58.1	1	0.998	0	Complete
2 P-only DeepONet10	0.5	0.901	0.641	4.53e+06	2.80e+06	0.995	2.55e+07	3.05e+03	1.01	2.6	0	Complete
2 P-only DeepONet10	1	0.0965	0.0308	2.70e+04	3.47e+04	0.0519	9.49e+03	186	0.963	0.972	0	Complete
3 ANN10 + P-only DeepONet10	0.05	7.11e-04	2.65e-04	1	0.723	6.77e-04	0.927	0.999	0.995	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.268	0.0469	1.86e+06	1.13e+04	0.0101	4.81e+04	11	69.8	4.01	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.16e-03	4.06e-04	1.15	2.52	9.52e-04	2.91	1.79	1.12	1.16	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.89e-03	1.39e-03	8.81e+03	129	1.97e-03	3.55e+03	7.63	0.945	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.32	0.531	6.33e+05	5.23e+06	0.991	3.00e+07	2.91e+03	0.89	1.47	0	Complete
3 ANN10 + P-only DeepONet10	1	0.1	0.032	2.79e+04	3.39e+04	0.0539	7.01e+03	193	0.98	0.971	0	Complete
4 T/P DeepONet10	0.05	7.11e-04	2.65e-04	0.998	0.997	6.78e-04	0.999	1	0.997	1	0	Complete
4 T/P DeepONet10	0.1	7.11e-04	2.65e-04	1	0.976	6.78e-04	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.2	7.11e-04	2.65e-04	1	0.979	6.78e-04	1	1	0.997	1	0	Complete
4 T/P DeepONet10	0.3	7.11e-04	2.65e-04	0.979	1.01	6.78e-04	0.994	1	1	1	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
4 T/P DeepONet10	1	2.54e-03	1.16e-03	5.92e+03	136	1.71e-03	2.25e+04	6.42	1	0.999	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.513	0.334	1.75e+06	9.00e+04	0.579	2.99e+07	1.53e+03	1.71	4.9	0	Complete
5 ANN10 + T/P DeepONet10	0.1	7.11e-04	2.65e-04	1	0.967	6.78e-04	1	1	0.998	1	0	Complete
5 ANN10 + T/P DeepONet10	0.2	7.11e-04	2.65e-04	1	0.962	6.77e-04	1	1	0.995	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	7.11e-04	2.65e-04	0.998	1.01	6.78e-04	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	1.72	0.589	5.14e+05	1.37e+05	0.996	1.57e+06	3.54e+03	0.999	0.999	0	Complete
5 ANN10 + T/P DeepONet10	1	1.25	0.424	3.77e+05	5.42e+05	0.718	1.09e+06	2.55e+03	0.986	0.839	0	Complete
6 T/P/ϕ DeepONet10	0.05	1.58	0.508	4.89e+04	1.47e+04	0.873	4.06e+05	3.13e+03	1.47	4.37	0	Complete
6 T/P/ϕ DeepONet10	0.1	7.11e-04	2.65e-04	1	0.749	6.78e-04	0.997	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	1.4	0.473	3.77e+05	2.29e+05	0.803	1.57e+06	2.85e+03	2.68	5.32	0	Complete
6 T/P/ϕ DeepONet10	0.3	7.11e-04	2.65e-04	0.976	0.921	6.78e-04	0.999	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 2.200 ms
6 T/P/ϕ DeepONet10	1	3.52e-03	1.25e-03	3.33e+03	3.07e+03	1.93e-03	255	7.56	1	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.501	0.443	2.13e+06	3.40e+04	0.869	7.65e+07	1.72e+03	1.85	6.13	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	7.11e-04	2.65e-04	1	0.761	6.78e-04	0.997	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	1.70e-03	6.74e-04	15.8	0.997	1.84e-03	0.702	2.05	2.04	3.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	7.11e-04	2.65e-04	1	0.987	6.78e-04	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.62	0.585	9.52e+05	1.04e+06	0.988	5.94e+06	3.43e+03	0.885	0.994	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	1.73	0.578	3.28e+05	2.16e+05	0.985	1.54e+06	3.5e+03	0.962	0.974	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 52

T_0=972.8 K P_0=4.350 atm ϕ_0=1.718 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 52

Repaired | Cantera vs NN NRMSE | T0=972.8K, P0=4.350atm, phi0=1.718

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0312	2.61e-04	436	2.90e+06	0.0157	297	0.961	3.27	0.967	0	Complete
1 ANN10	0.1	7.11e-04	2.65e-04	1	19	6.76e-04	1	1	0.985	0.997	0	Complete
1 ANN10	0.2	5.11e-03	2.91e-03	1	2.88e+03	0.0106	1	0.881	8.21	32.7	0	Complete
1 ANN10	0.3	1.58	0.523	2.37e+05	8.77e+05	0.895	3.33e+05	3.19e+03	1.2	0.998	0	Complete
1 ANN10	0.5	0.98	0.331	2.34e+05	6.30e+04	0.563	9.64e+05	2e+03	0.978	1	0	Complete
1 ANN10	1	1.04	0.491	1.76e+06	1.33e+06	0.846	2.57e+07	2.55e+03	10.6	1.68	0	Complete
2 P-only DeepONet10	0.05	0.0296	0.0106	1.97e+03	26.8	0.0247	5.56	43.1	9.78	38.6	0	Complete
2 P-only DeepONet10	0.1	1.65	0.539	1.41e+05	2.89e+05	0.923	1.57e+06	3.3e+03	1.87	3.95	0	Complete
2 P-only DeepONet10	0.2	7.11e-04	2.65e-04	0.993	0.995	6.78e-04	0.999	1	0.997	1	0	Complete
2 P-only DeepONet10	0.3	7.11e-04	2.65e-04	0.989	0.952	6.77e-04	0.927	1	0.993	0.999	0	Complete
2 P-only DeepONet10	0.5	0.918	0.6	3.61e+06	6.63e+06	0.99	2.77e+07	2.91e+03	0.911	4.85	0	Complete
2 P-only DeepONet10	1	0.0521	0.46	4.00e+06	3.19e+07	0.922	4.85e+07	1.79e+03	0.92	20.3	0	Complete
3 ANN10 + P-only DeepONet10	0.05	8.90e-03	3.13e-03	55.4	11.5	7.29e-03	4.63	13	3.45	11	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.5	0.509	3.77e+05	2.18e+05	0.863	1.86e+06	3.07e+03	2.49	5.77	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.11e-04	2.65e-04	0.991	0.993	6.77e-04	0.999	1	0.995	1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	7.11e-04	2.65e-04	0.98	0.926	6.76e-04	0.916	1	0.987	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.28	0.519	5.47e+05	6.06e+06	0.983	3.20e+07	2.83e+03	0.75	2.18	0	Complete
3 ANN10 + P-only DeepONet10	1	0.13	0.262	1.14e+06	2.20e+07	0.635	4.53e+07	1.03e+03	1.14	11.9	0	Complete
4 T/P DeepONet10	0.05	7.11e-04	2.65e-04	0.995	0.996	6.77e-04	0.998	1	0.992	0.999	0	Complete
4 T/P DeepONet10	0.1	7.11e-04	2.65e-04	0.832	0.999	6.78e-04	0.974	1	1	1	0	Complete
4 T/P DeepONet10	0.2	7.11e-04	2.65e-04	1	0.999	6.78e-04	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.3	7.02e-04	2.62e-04	0.893	38.7	6.70e-04	18.7	0.983	0.967	1	0	Complete
4 T/P DeepONet10	0.5	7.09e-04	2.64e-04	8.74	17	6.76e-04	35.4	0.995	0.999	1	0	Complete
4 T/P DeepONet10	1	0.648	0.999	1.76e+07	6.75e+05	0.999	7.36e+05	3.53e+03	5.34	34.5	0	Complete
5 ANN10 + T/P DeepONet10	0.05	7.11e-04	2.65e-04	0.992	0.994	6.76e-04	0.997	0.999	0.987	0.999	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.252	0.0958	0.992	1.08e+03	0.246	2.02	330	176	444	0	Complete
5 ANN10 + T/P DeepONet10	0.2	7.11e-04	2.65e-04	1	0.998	6.78e-04	1	1	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	7.03e-04	2.62e-04	0.912	25.3	6.72e-04	9.72	0.985	0.983	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	7.09e-04	2.64e-04	10.7	19.4	6.76e-04	48.7	0.993	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	1	1.49	0.588	1.40e+06	1.90e+06	0.991	1.09e+07	3.33e+03	1.4	0.962	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0661	0.0209	95.6	190	0.0406	143	114	10.7	25.6	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.222	0.0691	3.72	1.08e+03	0.128	32.1	397	23	49.2	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0356	0.0115	228	71.5	0.0231	192	60	8.27	18.2	0	Complete
6 T/P/ϕ DeepONet10	0.3	7.11e-04	2.65e-04	0.999	0.975	6.78e-04	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	7.01e-04	2.59e-04	60.4	15.3	6.69e-04	22.2	0.972	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	1	1.2	0.62	3.07e+06	3.46e+06	0.999	1.57e+07	3.23e+03	1.9	0.915	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0626	0.0198	97.4	172	0.0384	145	108	9.75	24.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.175	0.0549	0.624	984	0.103	10	310	20	47.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	1.26	0.433	4.44e+05	1.33e+05	0.733	1.72e+06	2.6e+03	3.8	6.17	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	7.11e-04	2.65e-04	0.995	0.917	6.78e-04	0.98	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	7.00e-04	2.59e-04	68.6	13.5	6.68e-04	24.3	0.969	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	1.14	0.609	2.97e+06	3.52e+06	0.998	2.15e+07	3.12e+03	3.07	0.806	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 63

T_0=1101.2 K P_0=3.120 atm ϕ_0=0.820 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 63

Repaired | Cantera vs NN NRMSE | T0=1101.2K, P0=3.120atm, phi0=0.820

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.12	0.503	2.76	0.835	0.427	3.8	0.176	3.82	3.45	0	Complete
1 ANN10	0.1	0.193	0.991	5.35	0.894	0.405	4.98	0.281	1.02	1.56	0	Complete
1 ANN10	0.2	0.131	0.872	0.991	0.964	0.261	3.49	0.209	1.07	1.07	0	Complete
1 ANN10	0.3	0.0245	0.442	0.416	0.384	0.195	0.468	0.0438	0.772	0.894	0	Complete
1 ANN10	0.5	0.0589	0.612	0.879	0.667	0.179	0.897	0.088	1.07	1.06	0	Complete
1 ANN10	1	0.403	6.88	1.46	0.855	2.85	0.892	0.693	2.63	49	0	Complete
2 P-only DeepONet10	0.05	0.261	0.767	0.861	0.983	0.877	7.98	0.366	0.941	1.74	0	Complete
2 P-only DeepONet10	0.1	0.227	0.792	6.9	0.971	0.637	5.24	0.25	1.84	5.68	0	Complete
2 P-only DeepONet10	0.2	0.0346	0.331	1.44	0.272	0.137	0.393	0.0445	0.756	2.73	0	Complete
2 P-only DeepONet10	0.3	0.232	0.0478	12.3	1.1	0.0606	2.38	0.182	0.763	14	0	Complete
2 P-only DeepONet10	0.5	0.596	8.95	0.48	0.921	3.51	0.077	0.949	0.998	11.4	0	Complete
2 P-only DeepONet10	1	0.591	9.81	0.947	1	4.04	1	0.996	21.1	21.3	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0746	0.477	1.06	2.01	0.248	1.62	0.0961	1.87	1.76	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.169	0.435	1.84	0.971	0.763	5.63	0.219	0.993	2.23	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0327	0.0978	1.62	0.372	0.0418	0.534	0.0342	0.315	2.86	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0985	0.514	1.78	0.814	0.62	3.77	0.116	1	3.96	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.106	0.473	1.92	0.858	0.631	3.37	0.107	0.912	58	0	Complete
3 ANN10 + P-only DeepONet10	1	0.496	7.7	0.897	2.09	2.98	0.476	0.794	1.11	82.5	0	Complete
4 T/P DeepONet10	0.05	0.091	0.656	1.3	1.12	0.291	1.6	0.113	1.52	1.97	0	Complete
4 T/P DeepONet10	0.1	0.063	0.565	1.01	0.815	0.243	0.957	0.0805	1.16	1.18	0	Complete
4 T/P DeepONet10	0.2	0.0456	0.768	0.507	0.199	0.319	0.103	0.0773	0.824	0.905	0	Complete
4 T/P DeepONet10	0.3	0.0827	1.39	0.42	0.237	0.573	0.149	0.14	0.723	0.776	0	Complete
4 T/P DeepONet10	0.5	0.06	0.723	1.33	2.79	0.516	1.23	0.0608	11.1	9.85	0	Complete
4 T/P DeepONet10	1	0.105	0.776	7.07	1.19	0.313	0.875	0.0801	1.27	4.9	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.224	0.777	5.85	1.02	0.422	3.98	0.243	2.54	2.58	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.06	0.675	0.794	0.646	0.3	0.943	0.0835	1.12	1.23	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0448	0.767	0.44	0.183	0.318	0.0937	0.0768	0.821	0.966	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.017	0.33	0.367	0.17	0.141	0.0572	0.0315	0.6	0.79	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0654	0.946	2.27	0.56	0.422	0.289	0.096	1.12	1.6	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0737	0.738	5.03	0.546	0.3	0.765	0.0689	1.22	5.23	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.339	1.56	11.2	1.31	0.643	4.81	0.356	4.64	2.62	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.593	9.86	1	1	4.07	1	1	0.992	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.593	9.86	1	1	4.07	1	1	0.991	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.593	9.86	1	1	4.07	1	1	0.947	0.975	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.13	1.98	2.89	0.98	0.83	0.706	0.204	7.01	10.1	0	Complete
6 T/P/ϕ DeepONet10	1	0.0899	1.13	4.3	1.59	0.488	0.666	0.112	1.45	2.58	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.56	2.13	6.16	0.824	0.913	12.2	0.706	1.94	2.49	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0733	0.979	0.644	0.224	0.411	1.06	0.113	1.19	1.68	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0768	1.29	0.445	0.218	0.534	0.139	0.13	0.824	1.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0478	0.583	0.977	0.749	0.153	0.694	0.0765	0.799	0.769	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.149	2.39	1.38	0.994	0.977	0.338	0.247	1.22	1.81	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0729	1.02	3.27	0.98	0.437	0.527	0.101	1.33	3.47	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 63

T_0=1101.2 K P_0=3.120 atm ϕ_0=0.820 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 63

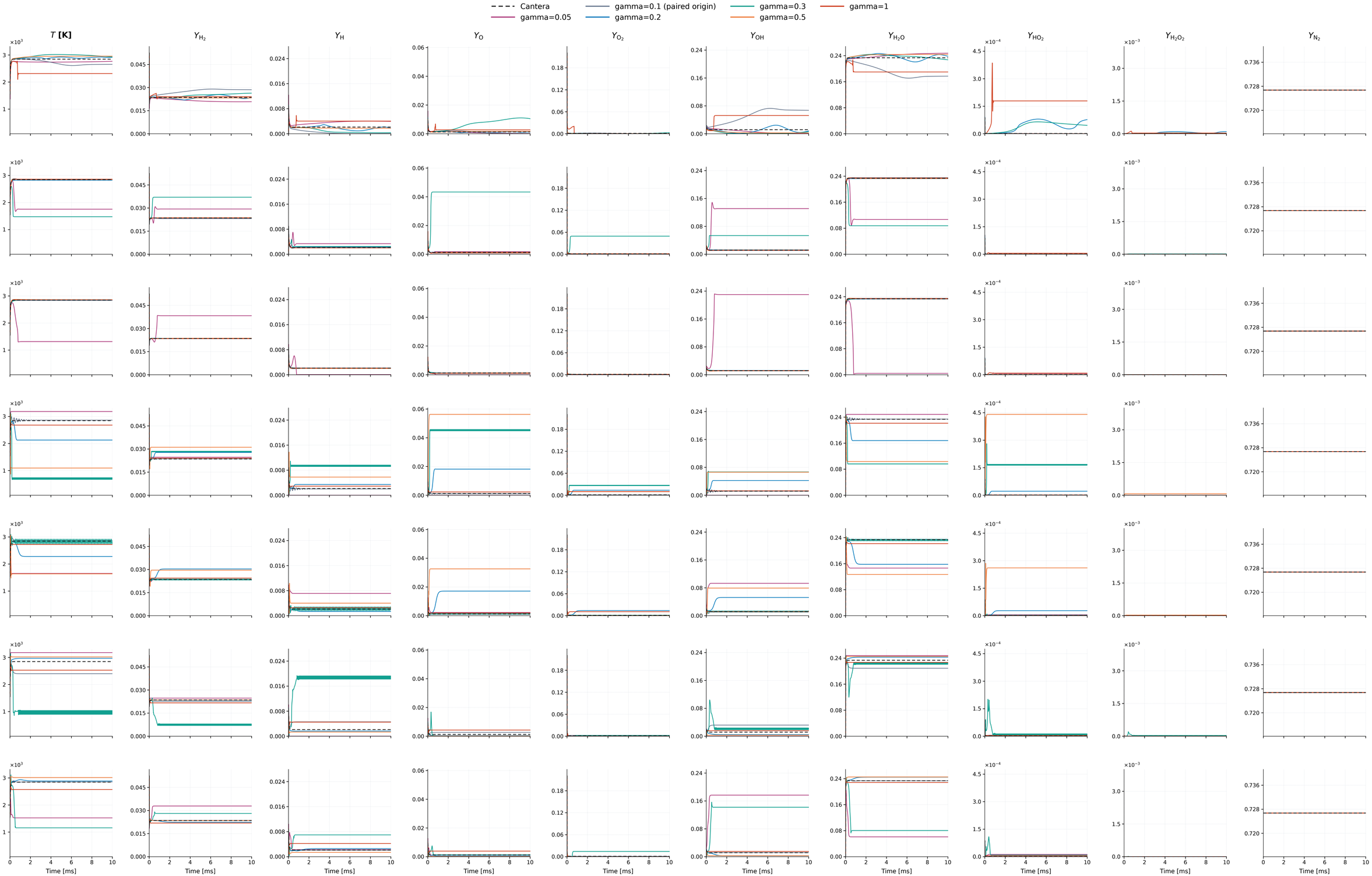
Repaired | Cantera vs NN NRMSE | T0=1101.2K, P0=3.120atm, phi0=0.820

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.211	0.786	0.903	0.909	0.589	6.21	0.306	0.81	0.889	0	Complete
1 ANN10	0.1	0.134	0.487	0.931	0.879	0.588	4.68	0.198	1.23	1.1	0	Complete
1 ANN10	0.2	0.0173	0.353	0.659	3.51	0.164	0.818	0.0452	1.06	3.11	0	Complete
1 ANN10	0.3	0.0204	0.446	0.536	0.713	0.202	0.468	0.037	0.709	1.12	0	Complete
1 ANN10	0.5	0.0255	0.509	0.444	0.201	0.246	0.301	0.0448	0.735	4.26	0	Complete
1 ANN10	1	0.236	3.7	1.28	0.614	1.46	0.69	0.393	7.54	92.4	0	Complete
2 P-only DeepONet10	0.05	0.376	0.766	6.2	0.898	0.744	8.75	0.454	1.29	3.19	0	Complete
2 P-only DeepONet10	0.1	9.90e-03	0.212	0.312	0.179	0.0874	0.0565	0.0195	0.427	0.721	0	Complete
2 P-only DeepONet10	0.2	0.0171	0.328	0.346	0.163	0.137	0.0558	0.0317	0.683	0.891	0	Complete
2 P-only DeepONet10	0.3	0.0175	0.343	0.384	0.174	0.144	0.129	0.0323	0.798	1	0	Complete
2 P-only DeepONet10	0.5	0.0579	1.2	1.01	0.604	0.47	0.365	0.106	0.993	0.916	0	Complete
2 P-only DeepONet10	1	0.781	1.3	29.7	20.4	0.499	4.12	0.502	0.998	599	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.617	7.84	1.47	6.11	2.52	2.32	0.891	8.17	5.6	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0121	0.256	0.362	0.199	0.105	0.0758	0.0236	0.526	0.809	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0176	0.334	0.348	0.159	0.139	0.0567	0.0323	0.69	0.891	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0185	0.36	0.395	0.189	0.149	0.0873	0.0342	0.744	0.928	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.121	0.455	2.55	0.883	0.607	3.49	0.124	0.878	55.9	0	Complete
3 ANN10 + P-only DeepONet10	1	0.56	5.18	6.78	7.18	1.5	2.41	0.691	0.632	341	0	Complete
4 T/P DeepONet10	0.05	0.364	0.583	13.4	2.76	0.218	4.61	0.334	3.67	1.41	0	Complete
4 T/P DeepONet10	0.1	0.0331	0.478	0.617	0.203	0.197	0.328	0.052	0.837	1.3	0	Complete
4 T/P DeepONet10	0.2	0.0235	0.429	0.373	0.183	0.178	0.0589	0.0421	0.837	1.05	0	Complete
4 T/P DeepONet10	0.3	0.0259	0.465	0.358	0.187	0.194	0.0621	0.0458	0.787	0.93	0	Complete
4 T/P DeepONet10	0.5	0.11	1.59	0.973	1.59	0.484	0.3	0.172	0.812	0.795	0	Complete
4 T/P DeepONet10	1	0.353	5.28	1.23	1.9	2.01	0.148	0.56	1.01	39.3	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.369	0.35	12.8	1.11	0.457	6.09	0.357	3.1	1.45	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.021	0.15	0.42	0.105	0.068	0.468	0.0283	0.501	0.678	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0226	0.413	0.372	0.175	0.172	0.0561	0.0404	0.784	0.931	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0241	0.437	0.379	0.176	0.182	0.0592	0.0429	0.777	0.917	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0254	0.528	0.888	0.388	0.201	0.112	0.0479	0.798	0.824	0	Complete
5 ANN10 + T/P DeepONet10	1	0.265	0.992	7.76	6.11	0.921	4.3	0.217	23.7	81.2	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0682	0.342	0.94	0.891	0.0432	0.992	0.0801	0.762	0.71	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.522	0.485	19.9	0.998	0.889	9.74	0.494	2.94	40.6	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0266	0.456	0.955	0.237	0.189	0.0756	0.0448	0.793	1.02	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0385	0.79	1.02	0.323	0.302	0.131	0.0731	0.826	0.955	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0577	0.698	0.408	0.703	0.256	0.628	0.0856	0.963	0.926	0	Complete
6 T/P/ϕ DeepONet10	1	0.594	9.87	1.13	1.04	4.06	1.01	1	19.9	19.5	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.387	0.254	13.8	0.835	0.568	7.24	0.388	3.58	3.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.35	0.242	14.8	0.517	0.3	5.47	0.331	3.09	2.49	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.024	0.453	0.846	0.194	0.179	0.149	0.0446	0.779	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0441	0.854	1.02	0.502	0.308	0.0847	0.081	0.825	0.928	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0287	0.506	0.437	0.178	0.212	0.0786	0.05	0.818	0.993	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.384	4.96	4.92	2.52	1.87	1.01	0.565	5.12	72.9	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 71

T_0=1549.8 K P_0=1.296 atm ϕ_0=1.887 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 71

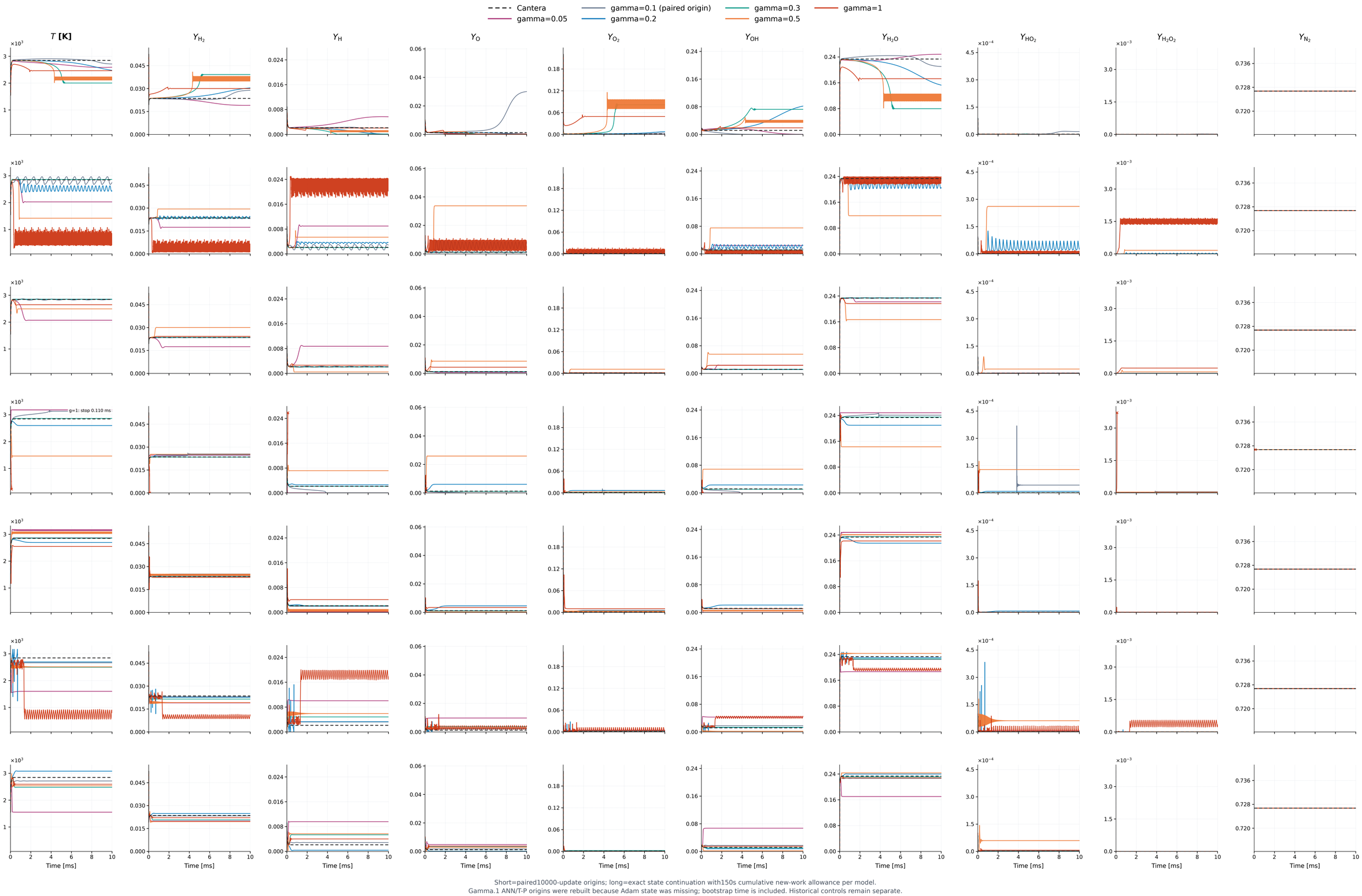
Repaired | Cantera vs NN NRMSE | T0=1549.8K, P0=1.296atm, phi0=1.887

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0362	0.0909	0.723	0.523	0.122	0.64	0.0403	0.302	0.53	0	Complete
1 ANN10	0.1	0.0558	0.186	0.837	0.628	0.121	3.87	0.201	0.472	0.784	0	Complete
1 ANN10	0.2	0.0188	0.0403	0.278	0.164	0.124	0.618	0.0345	24.1	322	0	Complete
1 ANN10	0.3	0.0466	0.0718	0.649	4.93	0.13	0.836	0.0257	22.3	139	0	Complete
1 ANN10	0.5	0.035	0.0107	0.153	0.29	0.17	0.695	0.0404	0.65	0.944	0	Complete
1 ANN10	1	0.181	0.0419	0.855	1.13	0.874	3.32	0.182	84.8	192	0	Complete
2 P-only DeepONet10	0.05	0.376	0.243	0.606	0.441	0.143	9.95	0.53	0.482	0.344	0	Complete
2 P-only DeepONet10	0.1	3.44e-03	0.0158	0.0982	0.329	0.111	0.0716	8.25e-03	0.225	0.264	0	Complete
2 P-only DeepONet10	0.2	7.41e-03	1.70e-03	0.0506	0.0947	0.022	0.0828	5.42e-03	0.175	0.322	0	Complete
2 P-only DeepONet10	0.3	0.475	0.564	0.194	31.4	6.97	3.51	0.616	0.289	83.4	0	Complete
2 P-only DeepONet10	0.5	2.29e-03	8.49e-03	0.0525	0.0929	0.0838	0.0272	4.08e-03	0.363	0.838	0	Complete
2 P-only DeepONet10	1	7.14e-03	0.0182	0.0712	0.366	0.426	0.056	0.0123	2.27	2.6	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.521	0.604	0.959	0.904	0.109	17.9	0.946	2.58	11	0	Complete
3 ANN10 + P-only DeepONet10	0.1	5.08e-03	4.33e-03	0.0315	0.201	0.0614	0.0453	5.09e-03	0.105	0.168	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.86e-03	1.80e-03	0.0182	0.0661	0.0253	0.0658	4.12e-03	0.161	0.302	0	Complete
3 ANN10 + P-only DeepONet10	0.3	4.21e-03	3.38e-03	0.0373	0.0952	0.0285	0.0308	2.99e-03	0.0682	0.327	0	Complete
3 ANN10 + P-only DeepONet10	0.5	4.69e-03	6.43e-03	0.058	0.0631	0.0445	0.0277	2.51e-03	0.279	0.693	0	Complete
3 ANN10 + P-only DeepONet10	1	2.63e-03	0.0129	0.0416	0.232	0.28	0.0195	8.33e-03	3.67	2.57	0	Complete
4 T/P DeepONet10	0.05	0.118	0.0491	0.983	0.947	0.151	0.994	0.064	0.514	1.5	0	Complete
4 T/P DeepONet10	0.1	0.0104	4.85e-03	0.0871	0.159	0.0387	0.0828	5.75e-03	0.171	0.158	0	Complete
4 T/P DeepONet10	0.2	0.246	0.174	0.582	12.5	1.79	2.53	0.273	9.85	5.97	0	Complete
4 T/P DeepONet10	0.3	0.748	0.201	3.33	33.2	3.74	4.66	0.583	80.5	2.83	0	Complete
4 T/P DeepONet10	0.5	0.614	0.322	1.69	41.5	1.26	4.62	0.554	214	372	0	Complete
4 T/P DeepONet10	1	0.0579	0.0235	0.347	0.779	1.3	0.0793	0.0519	0.261	30.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.418	7.23e-03	2.29	0.833	0.0416	6.87	0.37	1.96	2.06	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0118	5.22e-03	0.0984	0.155	0.0363	0.0862	6.41e-03	0.109	0.123	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.19	0.277	0.34	11.5	1.83	3.31	0.31	12.4	10.5	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0255	0.0145	0.233	0.309	0.0775	0.135	0.0111	0.711	1.18	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.424	0.254	0.877	23.6	0.125	5.73	0.454	126	155	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0437	0.0382	0.195	0.505	1.44	9.16e-03	0.0505	0.25	33.6	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.117	0.0512	0.984	0.949	0.101	0.995	0.0622	0.565	0.743	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.155	0.0382	1.12	1.08	0.189	1.7	0.103	2.11	2.8	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0426	7.42e-03	0.217	0.644	0.131	0.602	0.0399	0.581	0.803	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.637	0.658	7.39	1.27	0.201	1.31	0.0707	12.7	194	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0617	4.52e-03	0.373	0.834	0.168	0.753	0.0507	0.641	0.688	0	Complete
6 T/P/ϕ DeepONet10	1	0.111	0.0791	1.08	2.31	0.182	0.352	0.0284	1.51	8.03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.463	0.393	0.282	0.827	0.15	13.9	0.731	5.35	21.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	3.49e-03	3.32e-03	0.037	0.0504	0.0118	0.0382	2.23e-03	0.0589	0.104	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0154	0.0447	0.185	0.518	0.0908	0.727	0.045	0.561	0.858	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.581	0.19	2.18	0.548	1.9	10.9	0.641	7.54	9.33	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0595	5.77e-03	0.354	0.811	0.168	0.738	0.0496	0.64	0.633	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0955	0.0709	0.95	2	0.171	0.283	0.0226	1.94	10.9	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 71

T_0=1549.8 K P_0=1.296 atm ϕ_0=1.887 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 71

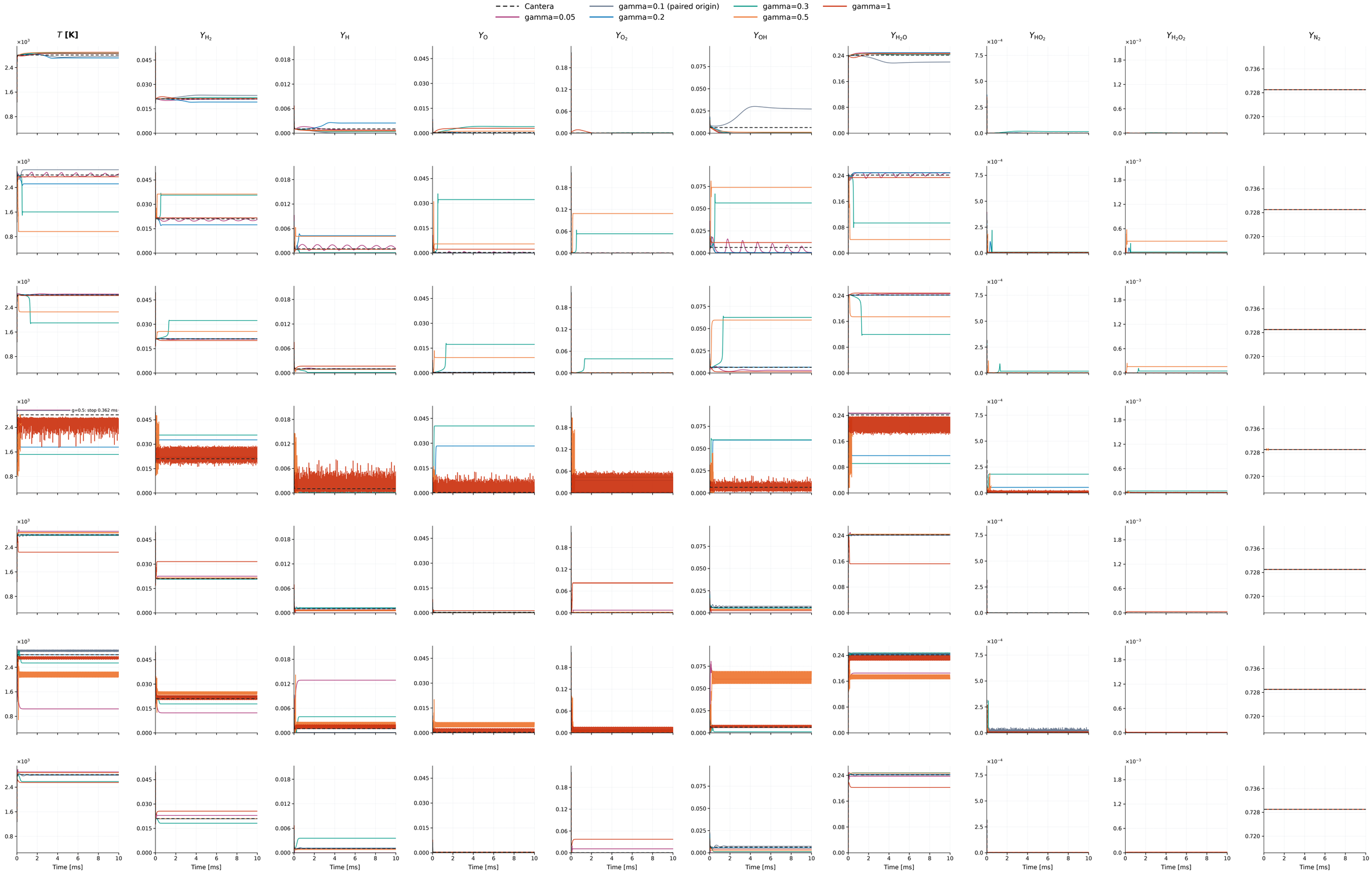
Repaired | Cantera vs NN NRMSE | T0=1549.8K, P0=1.296atm, phi0=1.887

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0624	0.114	1.01	0.703	0.112	0.574	0.0353	0.336	0.422	0	Complete
1 ANN10	0.1	0.0204	0.089	0.394	8.72	0.131	0.877	0.0464	3.52	43.2	0	Complete
1 ANN10	0.2	0.0533	0.138	0.501	0.272	0.344	2.76	0.156	0.333	0.677	0	Complete
1 ANN10	0.3	0.211	0.475	0.723	0.563	8.29	3.96	0.474	0.43	0.707	0	Complete
1 ANN10	0.5	0.178	0.408	0.396	0.65	8.93	1.8	0.383	0.528	5.37	0	Complete
1 ANN10	1	0.127	0.266	0.16	0.763	6.78	0.583	0.249	0.983	10.9	0	Complete
2 P-only DeepONet10	0.05	0.272	0.246	2.98	0.117	0.0481	1.03	0.0537	0.566	0.679	0	Complete
2 P-only DeepONet10	0.1	0.0359	0.0186	0.294	0.579	0.135	0.271	0.0222	0.448	1.41	0	Complete
2 P-only DeepONet10	0.2	0.113	0.0247	0.641	3.64	0.646	0.841	0.0894	19.4	104	0	Complete
2 P-only DeepONet10	0.3	6.57e-03	3.13e-03	0.0523	0.106	0.0535	0.0459	3.91e-03	0.143	0.194	0	Complete
2 P-only DeepONet10	0.5	0.48	0.238	1.43	23.6	1.36	5.17	0.472	122	863	0	Complete
2 P-only DeepONet10	1	0.752	0.77	8.71	4.8	0.749	0.6	0.0399	4.65	7.78e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.256	0.242	2.86	0.441	0.083	0.951	0.0455	0.432	1.2	0	Complete
3 ANN10 + P-only DeepONet10	0.1	5.66e-03	4.38e-03	0.0543	0.0781	0.0218	0.0396	3.04e-03	0.0875	0.252	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.85e-03	1.37e-03	0.012	0.0598	0.0212	0.0452	3.05e-03	0.0902	0.392	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.87e-03	2.29e-03	0.0245	0.0472	0.0426	0.019	1.90e-03	0.105	0.123	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.121	0.268	0.742	5.41	1.49	3.64	0.277	11.6	381	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0682	0.0295	0.238	2.32	0.174	0.971	0.0719	0.718	1.32e+03	0	Complete
4 T/P DeepONet10	0.05	0.119	0.0488	0.984	0.954	0.16	0.996	0.0644	0.497	34.8	0	Complete
4 T/P DeepONet10	0.1	0.0873	0.0673	0.823	0.824	0.643	0.832	0.0318	16.3	238	0	Complete
4 T/P DeepONet10	0.2	0.0854	0.0621	0.222	3.55	0.798	0.94	0.0989	3.48	5.09	0	Complete
4 T/P DeepONet10	0.3	6.09e-03	3.91e-03	0.0179	0.177	0.0407	0.0906	7.24e-03	0.14	0.239	0	Complete
4 T/P DeepONet10	0.5	0.484	0.0746	2.31	18.5	0.354	4.82	0.385	62	209	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.110 ms
5 ANN10 + T/P DeepONet10	0.05	0.109	0.0425	0.886	0.938	0.105	0.99	0.0619	0.419	7.18	0	Complete
5 ANN10 + T/P DeepONet10	0.1	7.98e-03	3.91e-03	0.0685	0.0415	0.0151	0.0693	4.11e-03	0.0457	0.0477	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0505	0.0614	0.0505	2.38	0.585	0.766	0.0752	2.85	5.28	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.24e-03	4.47e-03	5.02e-03	0.13	0.0306	0.0688	5.49e-03	0.107	0.184	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0738	0.0423	0.663	0.787	0.176	0.527	0.0322	0.611	0.8	0	Complete
5 ANN10 + T/P DeepONet10	1	0.109	0.0357	0.931	1.7	1.49	0.118	0.0573	2.84	81.8	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.436	0.192	3.63	6.42	0.519	2.65	0.199	2.65	0.864	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0638	0.0202	0.496	0.737	0.137	0.532	0.0373	0.342	0.252	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0554	0.0394	0.562	0.721	0.149	0.132	0.015	3.54	8.74	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.121	0.0856	1.23	1.84	0.286	0.162	0.0297	0.664	0.242	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.116	0.181	1.73	1.02	0.15	0.824	0.0419	29.8	23.1	0	Complete
6 T/P/ϕ DeepONet10	1	0.689	0.541	7	1.58	0.815	2.47	0.161	5.49	2.02e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.452	0.154	3.4	2.68	0.0587	4.7	0.269	1.32	2.41	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0459	0.0115	0.335	0.378	0.0717	0.484	0.0304	0.263	0.309	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0805	0.0532	0.79	0.784	0.117	0.338	0.0268	0.43	0.324	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.127	0.12	1.46	1.64	0.188	0.131	9.64e-03	0.605	0.0435	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.108	0.171	1.63	1.14	0.154	0.834	0.0418	28.5	16.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0878	0.0621	0.855	1.85	0.288	0.274	0.0247	2.79	4.06	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 75

T_0=1274.3 K P_0=3.589 atm ϕ_0=1.775 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 75

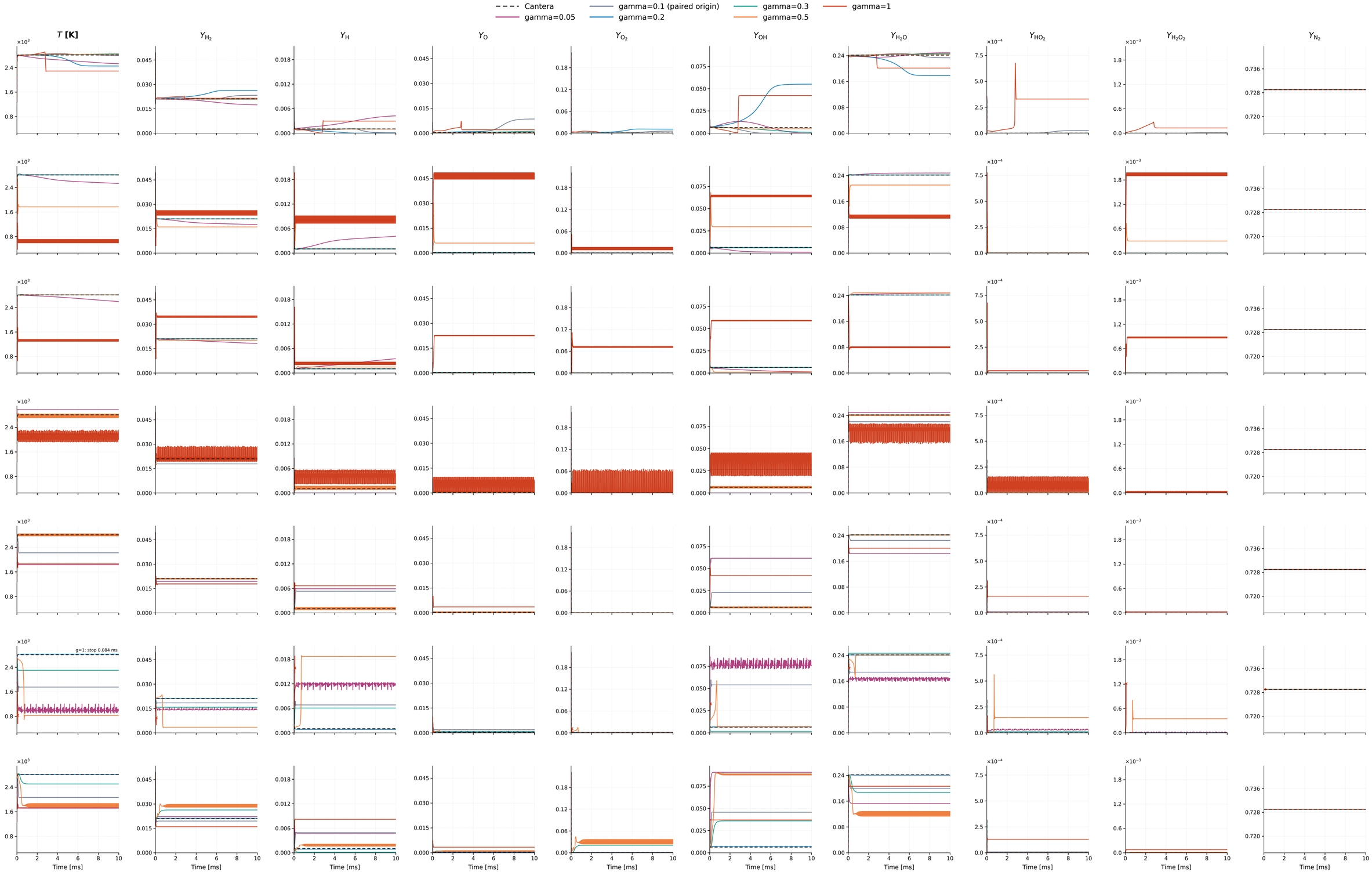
Repaired | Cantera vs NN NRMSE | T0=1274.3K, P0=3.589atm, phi0=1.775

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.026	0.02	0.278	0.833	0.0639	0.956	0.0294	0.314	0.819	0	Complete
1 ANN10	0.1	0.0175	0.091	0.854	0.584	0.0702	2.89	0.0787	0.238	0.827	0	Complete
1 ANN10	0.2	0.0289	0.0775	1.17	0.741	0.0681	0.926	0.0287	0.215	1.57	0	Complete
1 ANN10	0.3	0.0211	0.035	0.576	8.03	0.0761	0.896	9.88e-03	2.2	22.3	0	Complete
1 ANN10	0.5	0.0295	0.0115	0.463	1.88	0.0702	0.777	0.0201	0.205	0.75	0	Complete
1 ANN10	1	0.0158	0.0285	0.296	6.25	0.497	0.96	0.0206	0.73	8.44	0	Complete
2 P-only DeepONet10	0.05	0.0175	0.0409	0.549	0.745	0.197	0.768	0.0241	0.591	0.662	0	Complete
2 P-only DeepONet10	0.1	0.0594	0.0288	0.977	1.02	0.177	0.996	0.0321	0.562	1.01	0	Complete
2 P-only DeepONet10	0.2	0.101	0.17	3.01	0.937	0.148	0.893	0.0288	1.69	47.8	0	Complete
2 P-only DeepONet10	0.3	0.418	0.674	0.883	79.1	6.88	7.64	0.599	2.02	106	0	Complete
2 P-only DeepONet10	0.5	0.653	0.718	2.91	13.4	14.4	10.5	0.821	1.16	1.12e+03	0	Complete
2 P-only DeepONet10	1	0.0222	0.0412	0.151	5.08	0.556	0.869	0.0361	1.44	37	0	Complete
3 ANN10 + P-only DeepONet10	0.05	9.52e-03	0.012	0.0823	0.183	0.0711	0.53	0.0151	0.236	0.169	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.13e-03	4.35e-03	0.0615	0.111	0.0434	0.0345	1.81e-03	0.197	0.103	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.45e-03	5.98e-03	0.0665	0.228	0.0532	0.0452	2.82e-03	0.198	1.3	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.303	0.498	0.82	40.2	4.83	8.21	0.472	2.73	177	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.196	0.213	0.154	22.6	0.149	8.25	0.276	1.14	606	0	Complete
3 ANN10 + P-only DeepONet10	1	9.73e-03	0.0544	0.614	0.704	0.529	0.771	0.0292	1.1	14.4	0	Complete
4 T/P DeepONet10	0.05	0.0561	0.0412	0.998	1.01	0.384	0.999	0.0258	0.671	0.513	0	Complete
4 T/P DeepONet10	0.1	0.0506	0.0515	0.986	1.04	0.566	1	0.0182	1.02	24.1	0	Complete
4 T/P DeepONet10	0.2	0.369	0.543	0.631	70.1	4.5	8.19	0.514	8.19	13.1	0	Complete
4 T/P DeepONet10	0.3	0.457	0.687	0.887	101	5.73	8.32	0.619	27.8	181	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.362 ms
4 T/P DeepONet10	1	0.0909	0.149	1.18	6.42	3.21	0.48	0.112	1.63	37.2	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0403	0.0724	0.986	0.954	0.979	0.99	8.07e-03	0.51	0.404	0	Complete
5 ANN10 + T/P DeepONet10	0.1	8.99e-03	0.0152	0.151	0.212	0.281	0.185	0.0101	0.64	0.381	0	Complete
5 ANN10 + T/P DeepONet10	0.2	3.87e-03	0.0177	0.0692	0.261	0.345	0.0981	0.011	0.803	0.582	0	Complete
5 ANN10 + T/P DeepONet10	0.3	6.33e-03	0.0246	0.221	0.334	0.394	0.156	0.0134	0.913	0.706	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0245	0.0289	0.456	0.827	0.444	0.404	0.0156	0.956	0.815	0	Complete
5 ANN10 + T/P DeepONet10	1	0.201	0.497	0.307	2.46	10.9	0.604	0.369	1.04	114	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.625	0.41	11.2	0.947	0.317	8.49	0.235	1.71	44.3	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.047	0.0606	0.978	1.04	0.777	1	0.0174	2.54	23.2	0	Complete
6 T/P/ϕ DeepONet10	0.2	4.16e-03	0.0245	0.12	0.328	0.459	0.0979	0.0143	0.895	0.671	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0939	0.153	2.72	0.389	0.508	0.795	0.0282	1.31	0.856	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.23	0.167	1.05	11.6	0.933	8.65	0.281	1.24	5.79	0	Complete
6 T/P/ϕ DeepONet10	1	0.0378	0.0643	0.413	2.41	1.46	0.167	0.0506	1.23	29.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0318	0.0924	0.991	0.954	1.42	0.995	0.0205	0.425	0.909	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	7.59e-03	0.0153	0.124	0.207	0.286	0.173	0.01	0.641	0.386	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	3.25e-03	0.0191	0.101	0.247	0.348	0.131	0.0115	0.792	0.54	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0792	0.136	2.37	0.253	0.407	0.796	0.0272	0.971	0.713	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0239	0.024	0.265	0.76	0.46	0.63	0.025	0.974	0.771	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0903	0.217	0.115	0.41	4.94	0.349	0.163	1.21	68.6	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 75

T_0=1274.3 K P_0=3.589 atm ϕ_0=1.775 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 75

Repaired | Cantera vs NN NRMSE | T0=1274.3K, P0=3.589atm, phi0=1.775

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0671	0.0995	1.83	0.769	0.043	0.745	0.0209	0.0766	0.357	0	Complete
1 ANN10	0.1	8.63e-03	0.0533	0.479	10.8	0.346	0.789	0.0184	2.1	24.5	0	Complete
1 ANN10	0.2	0.0852	0.175	0.773	0.719	0.988	5.2	0.18	0.065	0.336	0	Complete
1 ANN10	0.3	7.17e-03	6.22e-03	0.0185	0.938	0.0258	0.485	0.0119	0.0839	1.27	0	Complete
1 ANN10	0.5	6.30e-03	3.60e-03	0.0503	0.0677	0.0537	0.19	7.03e-03	0.0674	0.62	0	Complete
1 ANN10	1	0.157	0.0371	1.57	4.61	0.436	4.78	0.141	44	509	0	Complete
2 P-only DeepONet10	0.05	0.0672	0.118	2.06	0.757	0.0518	0.7	0.0225	0.171	0.249	0	Complete
2 P-only DeepONet10	0.1	4.32e-03	1.64e-03	0.0659	0.125	0.011	0.0746	2.63e-03	0.0284	0.198	0	Complete
2 P-only DeepONet10	0.2	1.08e-03	8.07e-04	0.0193	0.0226	0.0148	0.012	6.20e-04	0.0429	0.152	0	Complete
2 P-only DeepONet10	0.3	1.77e-03	1.99e-03	0.0234	0.0863	0.034	0.0431	1.89e-03	0.149	0.454	0	Complete
2 P-only DeepONet10	0.5	0.368	0.233	6.65	14.8	0.0748	3.68	0.13	1.69	1.14e+03	0	Complete
2 P-only DeepONet10	1	0.767	0.171	6.97	117	1.53	9.04	0.527	4.01	7.36e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0419	0.0806	1.36	0.629	0.0383	0.576	0.0186	0.0814	0.235	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.43e-03	1.37e-03	0.0392	0.0607	0.012	0.0435	1.53e-03	0.0248	0.144	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.96e-04	1.11e-03	0.0164	0.0319	0.0167	0.0127	7.12e-04	0.0416	0.151	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.09e-03	1.40e-03	0.0137	0.0355	0.0264	0.0302	1.29e-03	0.0869	0.304	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.87e-03	0.0399	0.417	0.927	0.0489	0.837	0.0268	0.143	0.703	0	Complete
3 ANN10 + P-only DeepONet10	1	0.53	0.644	1.43	56.1	9.56	8.18	0.669	4.95	3.31e+03	0	Complete
4 T/P DeepONet10	0.05	0.0605	0.0272	0.994	0.962	0.0776	0.999	0.0315	0.356	6.07	0	Complete
4 T/P DeepONet10	0.1	0.221	0.149	4.11	0.482	0.1	3.11	0.0863	2.32	17.2	0	Complete
4 T/P DeepONet10	0.2	1.87e-03	5.71e-03	0.0368	0.108	0.104	0.0204	3.46e-03	0.361	0.296	0	Complete
4 T/P DeepONet10	0.3	3.25e-03	0.0138	0.0631	0.231	0.258	0.0788	8.60e-03	0.681	0.477	0	Complete
4 T/P DeepONet10	0.5	0.0178	0.0234	0.36	1.19	0.359	0.121	0.0115	0.982	2.24	0	Complete
4 T/P DeepONet10	1	0.257	0.1	3.03	14.3	2.26	5.16	0.199	12.6	91	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.347	0.0784	4.65	0.241	0.0458	8.61	0.24	1.7	25.3	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.208	0.158	4.05	0.751	0.0512	2.58	0.0699	1.56	14.4	0	Complete
5 ANN10 + T/P DeepONet10	0.2	9.32e-04	2.64e-03	0.0156	0.0756	0.0469	0.0247	1.75e-03	0.137	0.133	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.25e-03	6.47e-03	0.0344	0.151	0.109	0.0605	4.09e-03	0.309	0.3	0	Complete
5 ANN10 + T/P DeepONet10	0.5	9.47e-03	0.0192	0.202	0.528	0.326	0.0852	0.0103	0.735	0.997	0	Complete
5 ANN10 + T/P DeepONet10	1	0.337	0.149	5.33	8.41	0.533	5.59	0.172	24.9	123	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.644	0.317	10.4	0.951	0.129	11	0.309	4.66	48.3	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.374	0.123	5.51	3.81	0.196	7.42	0.221	1.64	5.43	0	Complete
6 T/P/ϕ DeepONet10	0.2	7.07e-03	0.0118	0.202	0.0852	0.0766	0.0556	2.90e-03	0.194	0.158	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.18	0.251	4.79	0.957	0.0598	0.719	0.02	1.31	1.47	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.681	0.804	16.3	0.978	0.422	0.904	0.0356	22.5	1.28e+03	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.084 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.388	0.0546	3.58	0.821	0.0684	13.1	0.365	0.407	6.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.262	0.0713	3.68	0.498	0.0351	6.1	0.173	1.27	6.23	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	2.74e-03	0.011	0.147	0.119	0.0364	0.149	4.99e-03	0.108	0.123	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.105	0.246	0.846	1.73	2.66	4.52	0.223	0.211	8.71	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.346	0.363	0.792	1.8	3.89	12.5	0.488	0.542	49.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.382	0.237	6.82	7.72	0.465	4.78	0.148	20.2	289	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Val IC 81

T_0=1039.6 K P_0=1.219 atm ϕ_0=1.500 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Val IC 81

Repaired | Cantera vs NN NRMSE | T0=1039.6K, P0=1.219atm, phi0=1.500

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0959	0.28	0.685	0.825	0.893	4.09	0.177	1.46	2.33	0	Complete
1 ANN10	0.1	0.0941	0.301	0.682	0.809	1.1	2.73	0.167	1.72	2.93	0	Complete
1 ANN10	0.2	0.0474	0.188	0.49	0.997	0.701	0.85	0.0918	3.43	62.5	0	Complete
1 ANN10	0.3	0.0665	0.252	0.862	0.773	0.93	0.846	0.122	3.2	73.1	0	Complete
1 ANN10	0.5	0.614	1.91	0.978	0.998	7.67	0.999	1	1	3.39	0	Complete
1 ANN10	1	0.613	1.91	0.999	0.941	7.67	0.999	1	0.999	18.5	0	Complete
2 P-only DeepONet10	0.05	0.727	0.988	1	1.05	0.737	28.4	0.996	2.2	1	0	Complete
2 P-only DeepONet10	0.1	0.0458	0.0988	0.558	1.21	0.37	0.429	0.052	0.815	0.89	0	Complete
2 P-only DeepONet10	0.2	0.0103	6.76e-03	0.157	0.251	0.019	0.0768	5.62e-03	0.563	0.818	0	Complete
2 P-only DeepONet10	0.3	0.531	1.27	0.916	39.8	3.55	2.55	0.708	1	8.14	0	Complete
2 P-only DeepONet10	0.5	0.423	0.706	0.394	27.6	1.44	6.64	0.506	0.999	3.7	0	Complete
2 P-only DeepONet10	1	0.491	0.301	7.3	8.24	1.02	3.77	0.249	0.998	11.9	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.681	1.15	0.864	0.982	1.8	21	0.963	14.1	4.23	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0304	0.0987	0.322	1.01	0.363	0.326	0.0482	0.925	1.03	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0204	0.102	0.265	0.561	0.375	0.1	0.0458	0.948	0.992	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0225	0.0994	0.264	0.657	0.371	0.167	0.0457	0.886	2.65	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.266	0.583	0.842	11.7	1.59	4.69	0.377	0.999	2.69	0	Complete
3 ANN10 + P-only DeepONet10	1	0.13	0.213	2.01	2.6	0.77	1.07	0.11	3.19	41.2	0	Complete
4 T/P DeepONet10	0.05	0.144	0.424	1	1	1.66	1	0.222	4.71	4.48e+03	0	Complete
4 T/P DeepONet10	0.1	0.613	1.91	1	1	7.67	1	1	0.999	0.999	0	Complete
4 T/P DeepONet10	0.2	0.68	0.315	6.34	26.9	1.25	7.32	0.481	3.92	6.19	0	Complete
4 T/P DeepONet10	0.3	0.632	1.54	0.899	26.4	3.06	0.999	0.971	5.78e+03	9.51e+04	0	Complete
4 T/P DeepONet10	0.5	0.613	1.91	1	1	7.67	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.137	0.222	2.02	4.01	0.807	1.22	0.111	1.11	11.4	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.419	0.329	2.94	2.64	1.12	10.7	0.407	2.07	4.04	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.603	1.88	0.948	0.917	7.53	0.996	0.983	18.8	66	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.404	0.737	0.862	37.3	1.31	6.24	0.489	4.83	4.29	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.468	0.71	0.504	50	0.733	7.58	0.51	25.5	12.2	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.613	1.91	1	1	7.67	1	1	1	0.988	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0936	0.188	1.31	2.54	0.695	0.884	0.0929	1.04	7.49	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.605	0.446	7.46	2.25	1.65	9.89	0.412	4.99	16.3	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.613	1.91	1	1	7.66	1	1	10.3	221	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.223	0.566	1.94	3.16	2.26	1.38	0.301	4	10.2	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.576	1.62	0.984	0.94	4.56	1	0.997	4.24e+03	9.82e+04	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.613	1.91	1	1	7.67	1	1	0.999	0.988	0	Complete
6 T/P/ϕ DeepONet10	1	0.612	1.89	0.924	0.818	7.59	0.987	0.992	19.2	150	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.193	0.63	0.943	1.36	2.5	0.995	0.32	5.47	20.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.362	0.55	3.6	0.737	2.19	6.67	0.366	1.77	7.33	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.614	1.9	0.907	0.942	7.63	0.998	0.995	21	61	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.611	1.89	0.893	0.983	7.59	0.997	0.991	15.5	34.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.613	1.91	1	1	7.67	1	1	0.999	0.958	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.62	1.9	0.858	0.958	7.67	0.999	0.999	0.996	30.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Val IC 81

T_0=1039.6 K P_0=1.219 atm ϕ_0=1.500 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Val IC 81

Repaired | Cantera vs NN NRMSE | T0=1039.6K, P0=1.219atm, phi0=1.500

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0511	0.162	1.21	0.796	0.497	0.644	0.0648	1.23	1.45	0	Complete
1 ANN10	0.1	0.0321	0.137	0.451	0.96	0.494	0.811	0.0659	1.25	1.54	0	Complete
1 ANN10	0.2	0.0489	0.191	0.464	0.736	0.722	0.513	0.0921	1.41	8.56	0	Complete
1 ANN10	0.3	0.0535	0.213	1.37	0.737	0.669	0.891	0.0904	1.43	5.52	0	Complete
1 ANN10	0.5	0.0237	0.0765	0.27	0.588	0.274	0.503	0.0388	0.734	5.7	0	Complete
1 ANN10	1	0.614	1.91	0.98	0.926	7.67	0.999	1	0.993	27.5	0	Complete
2 P-only DeepONet10	0.05	0.288	0.329	4.32	1.94	1.15	3.23	0.194	1.82	9.18	0	Complete
2 P-only DeepONet10	0.1	0.612	1.91	1	1	7.65	1	0.998	3.9	725	0	Complete
2 P-only DeepONet10	0.2	0.613	1.91	1	1	7.67	1	1	1	0.998	0	Complete
2 P-only DeepONet10	0.3	0.613	1.91	1	1	7.67	1	1	0.998	0.989	0	Complete
2 P-only DeepONet10	0.5	0.465	0.591	1.18	34.3	1.03	7.09	0.481	17.2	153	0	Complete
2 P-only DeepONet10	1	0.853	0.937	16.2	26.9	0.863	0.317	0.147	4.45	3.85e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.286	0.28	4.05	1.1	0.976	4.04	0.193	1.6	3.16	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0851	0.291	0.48	0.975	1.14	0.279	0.146	1.62	2.86	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.11	0.366	0.447	0.841	1.45	0.285	0.186	1.55	1.9	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0896	0.307	0.433	0.85	1.21	0.255	0.154	1.54	2.02	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.127	0.346	0.909	6.75	0.659	3.58	0.216	1.31	35.6	0	Complete
3 ANN10 + P-only DeepONet10	1	0.197	0.296	2.94	3.53	1.14	0.986	0.162	4.58	260	0	Complete
4 T/P DeepONet10	0.05	0.612	1.91	0.997	0.988	7.66	0.999	0.999	12	5.37	0	Complete
4 T/P DeepONet10	0.1	0.0406	0.132	0.458	0.794	0.495	0.441	0.0636	1.08	1.38	0	Complete
4 T/P DeepONet10	0.2	0.12	0.185	1.17	4.79	0.724	0.975	0.118	2.21	1.86	0	Complete
4 T/P DeepONet10	0.3	0.613	1.91	1	1	7.67	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.5	0.609	1.9	0.999	0.991	7.63	0.99	0.995	1	0.995	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
5 ANN10 + T/P DeepONet10	0.05	0.0973	0.213	0.97	1.12	0.828	1.28	0.118	2.3	6.68	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0247	0.0696	0.335	0.539	0.253	0.312	0.0334	0.586	0.684	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0311	0.103	0.358	0.782	0.379	0.373	0.0495	0.9	1.02	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.613	1.91	1	1	7.67	1	1	0.997	0.998	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.609	1.9	0.999	0.988	7.62	0.987	0.994	1	0.988	0	Complete
5 ANN10 + T/P DeepONet10	1	0.608	1.9	1.14	1.31	7.6	1.03	0.992	22.1	50.3	0	Complete
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.644 ms
6 T/P/ϕ DeepONet10	0.1	0.593	0.931	13.8	1.1	0.743	0.591	0.095	1.6	2.59	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.065	0.0769	1.06	1.53	0.256	0.412	0.0405	0.609	0.83	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.613	1.91	1	1	7.67	1	1	0.999	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.642	1.68	0.631	1.14	5.39	1.03	1	4.54e+03	3.97e+04	0	Complete
6 T/P/ϕ DeepONet10	1	0.607	1.9	0.994	1	7.6	0.972	0.991	1	2.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.418	0.202	5.45	1.03	0.596	7.08	0.261	1.41	2.45	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.397	0.309	6.64	0.779	0.388	4.49	0.166	0.947	3.61	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0185	0.0707	0.321	0.438	0.253	0.119	0.031	0.561	0.533	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.613	1.91	1	1	7.67	1	1	0.998	0.997	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.125	0.274	2.63	2.63	0.848	0.614	0.107	8.22	10.5	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.607	1.9	0.994	0.999	7.6	0.971	0.991	0.998	2.46	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.